

Math 113 Lectures, 2025

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Part I

Complex analysis

Chapter 1

Basic facts

In the first part of this lecture, we'll lay out the basic facts about complex numbers and some elementary functions. In the second part, we introduce the notion of a *holomorphic function*; to get an idea of their behavior, we'll list some of the basic theorems about them that we'll prove over the course of the next few weeks.

1.1 Complex numbers

A complex number is an expression of the form $x + iy$, where x and y are real numbers. The set of complex numbers forms a field $\mathbb{C}$, with the operation of addition being the obvious $(x + iy) + (u + iv) = (x + u) + i(y + v)$ and multiplication given by applying the distributive law and setting $i^2 = -1$; that is,

$$(x + iy)(u + iv) = (xu - yv) + i(xv + yu).$$

The main thing to check is that multiplicative inverses exist; that is, given x and y , not both zero, we can solve the system of equations

$$xu - yv = 1 \quad \text{and} \quad xv + yu = 0;$$

or, equivalently,

$$\begin{pmatrix} x & -y \\ y & x \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Since the matrix of coefficients has determinant $x^2 + y^2 > 0$, we can solve.

A better way to see this is by introducing two important notions. First, if $z = x + iy \in \mathbb{C}$, we define the *conjugate* $\bar{z}$ of z to be $\bar{z} = x - iy$. Second, we define the *absolute value* or *modulus* $|z|$ of z to be the length of the vector (x, y) in $\mathbb{R}^2$, so that we have

$$|z|^2 = x^2 + y^2 = z\bar{z}$$

This gives a direct way to show that inverses exist: if $z \neq 0$, we can write

$$z^{-1} = \frac{\bar{z}}{|z|^2}$$

As is probably becoming clear, one of the issues we'll be facing is knowing when a variable is real and when it's complex. Ahlfors generally adopts the convention that the letters z, w, ζ and ω refer to complex variables, while x, y, u and v are real. One convention of Ahlfors to keep in mind: we write $a > b$ *only* when a and b are real.

One more way of representing complex numbers is in what you might call polar coordinates: any complex number can be written as

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

with r and θ real. (Calling this $re^{i\theta}$ is just a convention we've adopted. In fact, we will define an exponential function e^z on $\mathbb{C}$, and this will turn out to be correct; but for now it's just a notational convention). Note that in these terms we have

$$|z| = r; \quad \bar{z} = re^{-i\theta}; \quad z^{-1} = \frac{1}{r}e^{-i\theta};$$

and in general if $z = re^{i\theta}$ and $w = se^{i\alpha}$, then

$$zw = rse^{i(\theta+\alpha)} \quad \text{and} \quad \frac{z}{w} = \frac{r}{s}e^{i(\theta-\alpha)}$$

Note that in the expression $z = re^{i\theta}$, the number θ is only defined modulo integral multiples of 2π . (Of course, this is just for $z \neq 0$; if $r = 0$, θ is completely indeterminate.) We have the option of specifying a range of allowable θ , or simply treating the value of θ as an element of $\mathbb{R}/2\pi\mathbb{Z}$. Either way, θ is called the *argument* of z .

1.1.1 Roots

Say we're given a complex number $z = x + iy$ and we want to find a square root of z ; that is, find a complex number $w = u + iv$ with

$$u^2 - v^2 = x \quad \text{and} \quad 2uv = y.$$

We can see by manipulating these two equations that solutions exist for all z , but there's an easier (and more insightful) way: if we write $z = re^{i\theta}$, then we can take

$$w = \sqrt{r}e^{i\theta/2}.$$

Note the ambiguity: we can replace $\theta/2$ by $\theta/2 + \pi$. Thus we have exactly two square roots of any nonzero complex number.

This ambiguity is important! It means that, even though there are two square roots of every nonzero complex number, *there is no way of choosing one consistently*: that is, there is no continuous function $w = w(z)$, defined in any neighborhood of the origin $z = 0$, such that $w(z)^2 \equiv z$.

Here's a really good exercise that will help you understand what's going on here. Consider the graph Γ of the function $z = w^2$ on the punctured plane $\mathbb{C}^* = \mathbb{C} \setminus 0$; that is,

$$\Gamma = \{(w, z) \in \mathbb{C}^* \times \mathbb{C}^* \mid z = w^2\}.$$

Now consider the projection map $\pi_2 : \Gamma \rightarrow \mathbb{C}^*$ sending (w, z) to z . The thing to see is that the map $\pi_2 : \Gamma \rightarrow \mathbb{C}^*$ *is the unique connected two-sheeted covering space of $\mathbb{C}^* \sim \mathbb{R}^2 \setminus \{(0, 0)\}$* ; this is why no lifting $\mathbb{C}^* \rightarrow \Gamma$ exists.

A similar story holds for n th roots: if $z = re^{i\theta}$ is any nonzero complex number, it has n th roots

$$z^{1/n} = r^{1/n}e^{i\theta/n}$$

where again the argument θ/n is only determined up to multiples of $2\pi/n$.

The picture here is analogous to the case $n = 2$; in particular, if we let

$$\Gamma_n = \{(w, z) \in \mathbb{C}^* \times \mathbb{C}^* \mid z = w^n\}$$

be the graph of the function $w \mapsto w^n$ over the punctured plane, then projection on the second factor expresses Γ_n as the unique connected n -sheeted

covering space of $\mathbb{C}^*$, so that there is no continuous function $w = w(z)$ defined in any neighborhood of the origin such that $w^n \equiv z$. On the other hand, we have

Exercise 1.1.1 Let $\Omega \subset \mathbb{C}^*$ be any connected open subset of the punctured plane. Show that if Ω is simply connected, then there *does* exist a continuous function $w = w(z)$ defined on Ω such that $w^n \equiv z$.

1.2 Holomorphic functions

For this, let $\Omega \subset \mathbb{C}$ be an open subset of the complex plane, and $f : \Omega \rightarrow \mathbb{C}$ any complex-valued function. We say that f is *holomorphic* (or *analytic*) if f has a derivative at every point $z \in \Omega$; that is, if for all $z \in \Omega$, the limit

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

exists; in other words, for all $\epsilon > 0$ there exists a δ such that

$$|h| < \delta \implies \left| f'(z) - \frac{f(z+h) - f(z)}{h} \right| < \epsilon.$$

This may seem like an innocuous condition, but it's not; the consequences are vastly different from asserting that f is differentiable as a map from $\Omega \subset \mathbb{R}^2$ to $\mathbb{R}^2$. The reason is simple: in order for the limit f' to exist, *it has to be independent of how h approaches 0*.

To understand what this says, write the argument of f as $z = x + iy$, and the value $w = f(z)$ as $w = u + iv$. Suppose first that $h = \epsilon$ approaches 0 along the real line, so that the limit

$$f'(z) = \lim_{\epsilon \rightarrow 0} \frac{f(z+\epsilon) - f(z)}{\epsilon} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}.$$

Now suppose $h = i\epsilon$ approaches zero along the imaginary axis $i\mathbb{R}$. We have

$$f'(z) = \lim_{\epsilon \rightarrow 0} \frac{f(z+i\epsilon) - f(z)}{i\epsilon} = \frac{1}{i} \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Comparing these, we see that we must have

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

These are called the *Cauchy-Riemann equations*, and they have enormous (and beautiful) consequences. We'll be spending much of the remainder of this semester exploring these consequences; let's take a moment now and give a preview of some of them. I should emphasize that (except for the first) these are things we have yet to prove; I'm mentioning them now so you have a sense of how different the behavior of holomorphic functions is from that of $\mathcal{C}^\infty$ functions.

1.2.1 Even $\mathcal{C}^\infty$ functions are not in general holomorphic

As we can see right away, even functions like $f(x + iy) = x$ or $f(x + iy) = y$, which are about as well-behaved as functions can be short of being constant, are not holomorphic.

1.2.2 The derivative of a holomorphic function is holomorphic

In the world of functions $f : \mathbb{R} \rightarrow \mathbb{R}$, a function can be 17 times differentiable, but not be 18 times differentiable—take, for example, $f(x) = x^{17+1/3}$. That can't happen in the world of holomorphic functions: even though the condition of being holomorphic requires only that the first derivative be defined, in fact we'll see that if a function satisfies that condition then its derivatives of all orders are defined and continuous.

1.2.3 The real and imaginary parts of a holomorphic function are harmonic

By the above, if f is holomorphic, then the derivative

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

is again holomorphic. This means it also must satisfy the Cauchy-Riemann equations; applying them, we deduce that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

and similarly for the imaginary part v of f . In other words, u and v are *harmonic* functions, meaning they satisfy Laplace's equation. This is a special class of real-valued functions on $\mathbb{C} = \mathbb{R}^2$, whose properties we'll be exploring as we analyze holomorphic functions.

To mention one of the consequences, note that if a harmonic function $u = u(x, y)$ is concave up in the real direction (i.e., $\frac{\partial^2 u}{\partial x^2} > 0$), then it must be concave down in the imaginary direction. Thus, for example, a harmonic function cannot have a local maximum or minimum. (Even more strikingly, we'll see that the value of a harmonic function at the center of a disc is equal to the average of its values on the boundary of the disc.)

1.2.4 Holomorphic functions are conformal

What this means is that f *does not distort angles*. Concretely, suppose $f : \Omega \rightarrow \mathbb{C}$ is a holomorphic function, and α and $\beta : (-\epsilon, \epsilon) \rightarrow \Omega$ are smooth arcs in Ω intersecting at $\alpha(0) = \beta(0) = z$. Then the angle formed by the tangent vectors $\alpha'(0)$ and $\beta'(0)$ is equal to the angle formed by the tangent vectors to the arcs $f \circ \alpha$ and $f \circ \beta$ at $f(z)$.

In other words, the map $df : T_z \Omega \rightarrow T_{f(z)} \mathbb{C}$ induced by f is angle-preserving. This is something we can see right away from the Cauchy-Riemann equations, which assert that the Jacobian matrix of f (viewed as a map from $\Omega \subset \mathbb{R}^2$ to $\mathbb{R}^2$) has the form

$$df = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

1.2.5 Holomorphic functions are open

A map $f : X \rightarrow Y$ of topological spaces is said to be *open* if the images $f(U)$ of open sets $U \subset X$ are open in Y . This is not a condition typically satisfied by continuous maps; even simple maps like the map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ sending

(x, y) to (x^2, y) , which folds the plane over and maps it to the right-hand half plane, is not open. But holomorphic functions $f : \Omega \rightarrow \mathbb{C}$, viewed as maps from an open set $\Omega \subset \mathbb{C}$ to $\mathbb{C}$ are always open. (You should check that the function $f(x + iy) = x^2 + iy$ is not holomorphic!)

1.2.6 Holomorphic functions are rigid

When I first encountered complex analysis, this was the point at which I realized we weren't in Kansas anymore. $\mathcal{C}^\infty$ functions are *flabby*, meaning that if $U, V \subset \Omega$ are open subsets of $\Omega \subset \mathbb{R}^2$ with disjoint closures, and $g : U \rightarrow \mathbb{R}$ and $h : V \rightarrow \mathbb{R}$ any $\mathcal{C}^\infty$ functions, we can find a $\mathcal{C}^\infty$ function $f : \Omega \rightarrow \mathbb{R}$ that agrees with both. (This is called “flabby” because it means we can alter the values of f on U without altering the values on V .) Since the condition of being holomorphic is simply that the first derivative exist, I assumed the same would be true of holomorphic functions.

I couldn't have been more wrong. In fact, holomorphic functions are *rigid*, meaning that if $\Omega \subset \mathbb{C}$ is connected, and $f : \Omega \rightarrow \mathbb{C}$ holomorphic, then f is *completely determined by its values on any open subset* $U \subset \Omega$: if I have a holomorphic function on the city of Cambridge, and its values on a subset the size of a dime are 0, then the function is identically zero.

Another reflection of the rigidity of holomorphic functions is the *Cauchy integral formula* (the centerpiece of this course, and the single most beautiful formula in mathematics), which gives an explicit formula for the value of a holomorphic function at any point of a region in terms of its values on the boundary.

Chapter 2

Holomorphic functions

2.1 Examples of holomorphic functions

Since we're going to be spending a lot of time studying holomorphic functions, we should start by introducing some examples of holomorphic functions. To begin with, some basic facts:

1. The constant function and the function $f(z) = z$ are holomorphic.
2. The sum of two holomorphic functions is holomorphic.
3. The product of two holomorphic functions is holomorphic.
4. The reciprocal of a holomorphic function is holomorphic (where the function is nonzero); hence the quotient of two holomorphic functions is holomorphic where the denominator is nonzero.
5. The composition of two holomorphic functions is holomorphic (where defined).
6. The inverse of a holomorphic function is holomorphic (where defined).

The first of these follows from direct examination; the others are proved exactly as you proved them in the one real variable case. In particular, the product, quotient and chain rules hold for holomorphic functions just as they do in that setting.

2.1.1 Polynomials

By the above, if $p \in \mathbb{C}[z]$ is any polynomial with complex coefficients, the function $p(z)$ is holomorphic.

Any polynomial $p(z)$ of degree n can be written

$$p(z) = \lambda(z - a_1) \dots (z - a_n),$$

with the a_i not necessarily distinct, or equivalently

$$p(z) = \lambda(z - a_1)^{m_1} \dots (z - a_k)^{m_k}$$

with the a_i distinct. In this setting, the a_i are called the *zeroes* of p , and m_i is called the *multiplicity* of the zero a_i .

2.1.2 Rational functions

Rational functions are defined to be quotients of polynomials; that is

$$r(z) = \frac{p(z)}{q(z)} = \lambda \frac{(z - a_1) \dots (z - a_m)}{(z - b_1) \dots (z - b_n)}$$

with $a_i \neq b_j$ for all i and j . Here the a_i are again called the *zeroes* of r , and the b_j are called the *poles* of r .

In order to better understand rational functions, we should introduce a basic notion: the *Riemann sphere*. This is just the sphere $S = S^2$, viewed as the one-point compactification of the complex plane $\mathbb{C}$; the added point is called “the point at infinity” and denoted ∞ . If $\Omega \subset S$ is an open subset and $f : \Omega \rightarrow \mathbb{C}$ a function, we’ll say that f is holomorphic at a point $z \neq \infty \in \Omega$ if it’s holomorphic in the usual sense; we’ll say that f is holomorphic at ∞ if the function $g(z) = f(1/z)$ is holomorphic at 0. Also, if $f : \Omega \rightarrow S$ is a map with ∞ a possible value, we’ll say that f is holomorphic at a point $z \in \Omega$ with $f(z) = \infty$ if the function $1/f$ is holomorphic at z .

With all that said, rational functions can be thought of as holomorphic maps $r : S \rightarrow S$. (Indeed, we will see in due course that they are *all* the holomorphic functions $S \rightarrow S$.) We can extend the notion of zero or pole to the point ∞ : if we set

$$s(z) = r\left(\frac{1}{z}\right),$$

we define the order of zero or pole of r at ∞ to be the order of zero or pole of $s(z)$ at 0. Explicitly, if we write our rational function r as

$$r(z) = \frac{p(z)}{q(z)} = \lambda \frac{(z - a_1) \dots (z - a_m)}{(z - b_1) \dots (z - b_n)}$$

as above, then if $m \geq n$, r will have a zero of order $m - n$ at ∞ ; and if $m \leq n$, it will have a pole of order $n - m$ at ∞ .

Exercise 2.1.1 Prove that a rational function $r(z)$ has the same number of zeroes as poles, counting multiplicity; and that similarly the functions $r(z) - \lambda$ all have the same number of zeroes.

At this point, we want to introduce the other elementary functions: exponential functions, logarithmic functions and trigonometric functions, and describe some of their behavior. In order to do this, we'll need to introduce *power series*; we'll take a break now and do that next.

2.2 Power series

2.2.1 lim sup and lim inf

Let $\{x_1, x_2, \dots\}$ be an infinite sequence of real numbers. Even if we allow ∞ as a possible limit (we say $\lim_{n \rightarrow \infty} x_n = \infty$ if for any $N \in \mathbb{R}$, there exists an $n_0 \in \mathbb{N}$ such that $x_n > N$ for all $n \geq n_0$), the sequence may not have a limit.

The idea of lim sup and lim inf is that they ascribe to an *arbitrary* sequence of real numbers something like a limit. Here's how: suppose $\{x_1, x_2, \dots\}$ is any sequence of real numbers. A basic property of the real numbers says that there is a least upper bound for the set $\{x_1, x_2, \dots\} \subset \mathbb{R}$, though we have to allow ∞ as a least upper bound if the sequence is not bounded from above.

Now suppose we start stripping terms away from the sequence, and considering the least upper bound of the remaining terms: that is, we set

$$y_k = \text{l.u.b.}\{x_{k+1}, x_{k+2}, \dots\}$$

The sequence $\{y_1, y_2, \dots\}$ is now a non-increasing sequence, and so it has a limit (of course, we have to allow $-\infty$ as a possible limit); we define

$$\limsup_{n \rightarrow \infty} x_n = \lim_{k \rightarrow \infty} y_k.$$

This is *always* defined (again allowing both ∞ and $-\infty$ as values).

From the definition, we see that $A = \limsup_{n \rightarrow \infty} x_n$ has the property that for any $\epsilon > 0$, there exists an n_0 such that $x_n \leq A + \epsilon$ for all $n \geq n_0$; moreover, it is the smallest number with this property.

The notion of $\liminf_{n \rightarrow \infty} x_n$ is defined in exactly the same way, but with the inequalities reversed: that is, we let z_k be the greatest lower bound for the set $\{x_{k+1}, x_{k+2}, \dots\} \subset \mathbb{R}$ and, noting that $z_1, z_2, \dots$ is a nondecreasing sequence, we define

$$\liminf_{n \rightarrow \infty} x_n = \lim_{k \rightarrow \infty} z_k,$$

where once more we allow $\pm\infty$ as values.

By way of an example, consider the sequence $x_n = (-1)^n$; in that case, it's easy to see that

$$\limsup_{n \rightarrow \infty} (-1)^n = 1 \quad \text{and} \quad \liminf_{n \rightarrow \infty} (-1)^n = -1.$$

In fact, we see that for any sequence $\{x_n\}$,

$$\limsup_{n \rightarrow \infty} x_n \quad \text{and} \quad \liminf_{n \rightarrow \infty} x_n \quad \text{always exist;}$$

and

$$\text{the actual limit } \lim_{n \rightarrow \infty} x_n \text{ exists if and only if } \limsup_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_n.$$

2.2.2 Convergence of power series

First off, by a *power series* we'll mean an expression of the form

$$\sum_{n=0}^{\infty} a_n z^n,$$

where $a_0, a_1, \dots$ is an arbitrary sequence of complex numbers. (More generally, we can consider expressions of the form $\sum_{n=0}^{\infty} a_n (z - z_0)^n$, but the treatment is exactly the same.) The question, naturally, is: for what values of z does this infinite sum converge, and what can we say about the function $f(z) = \sum a_n z^n$ defined in this way for such z ?

In fact, there's a very straightforward answer to this question. To express it, we begin by introducing the *radius of convergence* R of the power series: this is defined to be

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}}$$

(Note that since $\sqrt[n]{|a_n|}$ is nonnegative, the $\limsup$ is nonnegative or ∞ ; the radius of convergence R is correspondingly nonnegative or ∞ . In these terms, we have the basic Theorem 2 from Chapter 2 of Ahlfors:

Theorem 2.2.1 1. The series $\sum a_n z^n$ converges absolutely for all z with $|z| < R$;
 2. The series $\sum a_n z^n$ diverges for all z with $|z| > R$; and
 3. For any $\rho < R$, the series $\sum a_n z^n$ converges uniformly in the disc of radius ρ around the origin.

This is where Chapter 7 of Rudin kicks in. First of all, the first item tells us we can permute the terms in the sum without changing the value. In particular this means we can add, multiply and divide power series in the obvious way: we have

$$\begin{aligned} \sum a_n z^n + \sum b_n z^n &= \sum (a_n + b_n) z^n; \\ \left(\sum a_n z^n \right) \times \left(\sum b_n z^n \right) &= \sum \left(\sum_{i+j=n} a_i b_j \right) z^n; \end{aligned}$$

and if we write

$$\frac{\sum a_n z^n}{\sum b_n z^n} = \sum c_n z^n,$$

with $b_0 \neq 0$, we can rewrite this as

$$(b_0 + b_1z + b_2z^2 + \dots)(c_0 + c_1z + c_2z^2 + \dots) = (a_0 + a_1z + a_2z^2 + \dots)$$

and solve for the c_i in succession: we have

$$c_0 = \frac{a_0}{b_0}; \quad c_1 = \frac{a_1 - c_0b_1}{b_0}; \quad c_2 = \frac{a_2 - c_0b_2 - c_1b_1}{b_0}$$

and so on.

The crucial part of this theorem, though, is the third part: the uniform convergence of the series (according to Rudin's Theorem 7.17, which Ahlfors reproves in this setting) assures us that we can differentiate term by term; and since the radius of convergence of the series $f'(z) = \sum na_nz^{n-1}$ is the same as for $\sum a_nz^n$ this gives us the important

Corollary 2.2.2 *If $\sum a_nz^n$ is a power series with radius of convergence R , then the sum converges for all $|z| < R$ and defines a holomorphic function $f(z)$ in the disc $|z| < R$.*

Note that the series $\sum a_nz^n$ does *not* converge uniformly in the disc $|z| < R$; the theorem says only that it converges uniformly in $\Delta_\rho = \{z \mid |z| < \rho\}$ for any $\rho < R$; but since every point in the disc Δ_R lies in Δ_ρ for some $\rho < R$, this is enough to deduce that the function $f(z) = \sum a_nz^n$ is holomorphic in Δ_R , and that we can differentiate term by term.

Note also that the theorem does not say anything about the convergence of $\sum a_nz^n$ for $|z| = R$. In fact, the behavior can be quite varied. There is one theorem in this regard: even though it won't play a large role for us, *Abel's theorem* says that if $\sum a_nz_0^n$ converges for some z_0 with $|z_0| = R$, then indeed

$$\lim_{z \rightarrow z_0} f(z) = \sum a_nz_0^n,$$

as long as z approaches z_0 along an arc not tangent to the boundary of the disc $|z| < R$.

2.3 More holomorphic functions

So far, the only holomorphic functions we've found are the rational functions. As we said, power series give us a way to introduce many new ones. In fact, in

this lecture we'll describe all *elementary* functions, an elementary function being one that can be built up out of polynomials, exponentials, trigonometric functions and logarithm functions with the operations of addition, multiplication, division, composition and inversion.

2.3.1 The exponential function

We'll start by defining the exponential function

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

The first thing to check is its radius of convergence; since

$$\lim_{n \rightarrow \infty} \sqrt[n]{n!} = \infty$$

(a fun exercise to check!), the radius of convergence is ∞ , so e^z is defined and holomorphic for all z .

Next, we establish a basic functional equation for e^z , which is

$$e^a \cdot e^b = e^{a+b}.$$

Ahlfors does this by characterizing e^z as the unique solution of the differential equation $f'(z) = f(z)$ having value $f(0) = 1$, but we don't need to: since we know how to multiply power series (and presumably we know the binomial theorem), we can write

$$\begin{aligned} e^a e^b &= \left(\sum_{n=0}^{\infty} \frac{a^n}{n!} \right) \left(\sum_{n=0}^{\infty} \frac{b^n}{n!} \right) \\ &= \sum_{n=0}^{\infty} \left(\sum_{i+j=n} \frac{a^i b^j}{i! j!} \right) \\ &= \sum_{n=0}^{\infty} \frac{(a+b)^n}{n!} \\ &= e^{a+b}. \end{aligned}$$

A few things follow from this identity: to begin with, we have $e^z e^{-z} = 1$, from which in particular we conclude that e^z is never 0. From the power series,

we see directly that $e^x > 1$ if x is real and non-negative; since $e^x e^{-x} = 1$, it follows that $0 < e^x < 1$ if x is real and negative.

Next, since the coefficients of the power series for e^z are all real, we see that

$$e^{\bar{z}} = \overline{e^z}.$$

Thus, if $z = iy$ is purely imaginary, we have

$$|e^{iy}|^2 = e^{iy} \cdot \overline{e^{iy}} = e^{iy} \cdot e^{-iy} = 1;$$

that is, the function e^z takes the imaginary axis to the unit circle.

2.3.2 Trigonometric functions

Next, we define the sine and cosine functions in terms of the exponential: we set

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

From this, we can deduce that

$$e^{iz} = \cos z + i \sin z$$

and also that

$$\sin^2 z + \cos^2 z = 1.$$

Finally, we can just plug the power series for e^z into the definition of $\sin$ and $\cos$ to deduce their power series expansions

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots \quad \text{and} \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

A note for Ahlfors readers: you may recognize these as the power series expansions of the trigonometrically defined sine and cosine functions, at least for z real. *Ahlfors, however, does not.* In particular, one important fact about the exponential function—that it is periodic, with period $2\pi i$, or in other words

$$e^{z+2\pi i} = e^z \quad \text{for all } z \in \mathbb{C},$$

would follow immediately from this recognition. But Ahlfors is a purist: he wants to define the functions e^z , $\sin z$ and $\cos z$ purely in terms of the power series expansion for e^z , and deduce the periodicity of e^z from that without assuming any prior knowledge of anything. In fact, he winds up *defining* π to be one half of the smallest period of the function e^{iz} !

Chapter 3

More elementary functions

3.1 What's coming up

Today we're going to wrap up our discussion of Chapter 2 of Ahlfors; mainly, we want to finish our description of the exponential, trigonometric and logarithm functions; and then introduce the Riemann sphere, and describe rational functions $r(z)$ in those terms. This will finish our discussion of Chapter 2 in Ahlfors.

For the time being, we're going to skip over Chapter 3 (though we will almost certainly want to double back later and talk about the group of *linear fractional transformations* of the Riemann sphere, both because they are basic examples of conformal maps and because they arise in so many other areas of mathematics).

Starting next week, we'll take up the key notion of path integrals, which will lead us to the central theorems of complex analysis. This is the content of Chapter 4, which is the heart of the book and of this part of the course. We'll spend roughly three weeks on this chapter, and then go on to Chapter 5, in which we'll cover sections 1-4 (series, product and partial fraction developments of meromorphic functions).

3.2 Elementary functions

Since we're in the middle of a discussion of the exponential function and related functions, let's start today by wrapping up that topic, and then we'll go back to rational functions.

3.2.1 The exponential function

To start, let's recall the basic facts about the exponential function e^z . First, it's defined by the power series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

which has radius of convergence $R = \infty$. Next, it satisfies the functional equation

$$e^{a+b} = e^a \cdot e^b,$$

from which we deduce that e^z is never 0, and that the map $\mathbb{C} \rightarrow \mathbb{C}^*$ given by $z \mapsto e^z$ is a homomorphism from the group $\mathbb{C}$ of complex numbers, with group law given by addition, to the group $\mathbb{C}^*$ of nonzero complex numbers, with group law given by multiplication.

From the fact that the coefficients of the power series defining e^z are all real, we see that $e^{\bar{z}} = \overline{e^z}$, from which it follows that

$$|e^{iy}| = 1$$

from which we see that

$$|e^{x+iy}| = e^x \quad \text{and} \quad \arg(e^{x+iy}) = \arg(e^{iy}). \quad (3.1)$$

Geometrically, this means that the map $z \mapsto e^z$ takes the horizontal line $\text{Im}(z) = \theta$ to a ray through the origin. It also means that the restriction of this map to the imaginary axis gives a map

$$\mathbb{R} = \{z \mid \text{Re}(z) = 0\} \rightarrow S^1 = \{z \mid |z| = 1\}$$

which we'll see is the universal covering space of S^1 .

Finally, the trickiest aspect of the exponential function is the fact that it is *periodic*. In general, we say a function f on the complex plane is *periodic with period* ω if it satisfies the functional equation $f(z + \omega) = f(z)$ for all z . In fact, e^z is periodic with period $2\pi i$, which is to say simply that

$$e^{2\pi i} = 1.$$

Ahlfors proves this by a clever argument that avoids assuming any prior knowledge of trigonometric functions or their power series, but you don't have to feign ignorance.

3.2.2 Trigonometric functions

Next, we introduce the functions $\sin$ and $\cos$, defined by

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

From this, we can deduce that

$$e^{iz} = \cos z + i \sin z$$

and also that

$$\sin^2 z + \cos^2 z = 1.$$

We can just plug the power series for e^z into the definition of $\sin$ and $\cos$ to deduce their power series expansions

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots \quad \text{and} \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

Finally, we can use this to refine the equation 3.1: we have

$$|e^{x+iy}| = e^x \quad \text{and} \quad \arg(e^{x+iy}) = y$$

3.2.3 The logarithm

The function $\log(z)$ is defined to be the inverse of the exponential; that is, $w = \log(z)$ is defined to be a solution of the equation $e^w = z$. This is a little

tricky, since there are (by what we just saw) infinitely many such solutions, all differing by integral multiples of $2\pi i$. Specifically, we have

$$\operatorname{Re}(\log z) = \log |z| \quad \text{and} \quad \operatorname{Im}(\log z) = \arg z.$$

The situation here is very similar to what we saw when we considered the square root (or more generally the n th root) function: for any given z , there are many possible values of $\log z$, but *no way to choose one consistently throughout the domain $\mathbb{C}^*$ of the function.*

There are two ways of dealing with this. One is to simply think of $\log z$ as a multi-valued function, or (in this case) as a function that takes values in the quotient $\mathbb{C}/2\pi i\mathbb{Z}$. The other is to restrict the domain of the function to a subset of $\mathbb{C}^*$ where it can be single-valued: for example, if we take

$$\Omega = \{z \in \mathbb{C} \mid \operatorname{Re}(z) > 0\},$$

we can define a single valued function $\log z$ on Ω simply by requiring that $-\pi/2 < \operatorname{Im}(\log z) < \pi/2$. (We could go further: if we take Ω to be the complement of the negative real axis, we can define $\log z$ unambiguously in Ω by requiring that $-\pi < \operatorname{Im}(\log z) < \pi$.) In fact, as we'll see in a moment, we can choose a single-valued branch of the function $\log z$ in any simply connected open subset $\Omega \subset \mathbb{C}^*$.

One convention we'll adopt in regard to the logarithm: if z is a positive real number, by $\log z$ we'll mean the (unique) real value of $\log z$.

Finally, we should point out that a consequence of the formula $e^{a+b} = e^a e^b$ is the functional equation

$$\log zw = \log z + \log w.$$

Given the multi-valued nature of the $\log$, this should be viewed as an equation of sets, or equivalently as an equality in $\mathbb{C}/2\pi i$.

3.2.4 What do we mean by a^b ?

A priori, the answer to this is clear: we mean

$$a^b = e^{b \log a}$$

The issue is, this is again a multi-valued function. In general, there's not much we can do about that, but there are some cases in which the ambiguity is less:

1. If a is real and positive, we take $\log a$ to be real as well, so a^b is unambiguous.
2. If $b \in \mathbb{Z}$, then the ambiguity of $\log a$ doesn't matter, since $e^{2\pi ib} = 1$; in this case, of course, we can just say $a^b = a \times a \times \cdots \times a$ (b times) if b is positive, or $1/(a \times a \times \cdots \times a)$ ($-b$ times) if b is negative.
3. If $b = m/n \in \mathbb{Q}$, with $(m, n) = 1$, then we see that a^b has n possible values, like the n th root function.

3.2.5 Inverse trigonometric functions

Given that

$$w = \cos z = \frac{e^{iz} + e^{-iz}}{2},$$

we can solve this equation to express z in terms of w . First, we can multiply the equation $2w = e^{iz} + e^{-iz}$ through by e^{iz} to arrive at

$$(e^{iz})^2 - 2we^{iz} + 1 = 0;$$

we can solve this quadratic polynomial to arrive at

$$e^{iz} = w \pm \sqrt{w^2 - 1}.$$

Thus

$$z = \frac{1}{i} \log(w \pm \sqrt{w^2 - 1})$$

and so

$$\arccos w = -i \log(w \pm \sqrt{w^2 - 1}) = \pm i \log(w + \sqrt{w^2 - 1})$$

(since $(w + \sqrt{w^2 - 1})(w - \sqrt{w^2 - 1}) = 1$). The other inverse trig functions can similarly be defined in terms of logarithms.

We see from this that functions like $\arccos$ and $\arcsin$, as functions of a complex variable, are even more multi-valued than the logarithm. The graph

of $w = \cos z$ is much harder to visualize than that of $w = e^z$, but hopefully as we develop more intuition with holomorphic functions we'll get to a point where we have a functional understanding of what it looks like. For what it's worth, Ahlfors' attempt to describe the graph of $w = \cos z$ is in his section 3.4.3. (the term "Riemann surface," in this setting, means simply the graph $\Gamma \subset \mathbb{C} \times \mathbb{C}$ of the function).

3.2.6 Elementary functions

Consider the operations we can carry out to create new functions out of functions we know: we can add, subtract, multiply and divide them; we can compose them and we can invert them (meaning, take the inverse function). Here we have to allow possibly multi-valued functions; for example, if $p(z) = z^d + a_{d-1}z^{d-1} + \cdots + a_0$ is a polynomial, the inverse function would associate to any w a root of

$$z^d + a_{d-1}z^{d-1} + \cdots + (a_0 - w).$$

With that said, we define an *elementary* function to be one that can be constructed in this manner from the basic functions $1, z$ and e^z . This is a broad class of functions, including (as we've just seen) trigonometric and inverse trigonometric functions.

There are others, though. While the product, quotient and chain rules tell us in principle that the class of elementary functions is closed under differentiation, it was discovered by Abel and others in the 19th century that it is not closed under taking the indefinite integral. We will see some non-elementary functions arise later in this course, notably the gamma function $\Gamma(z)$, Riemann's zeta function $\zeta(z)$ and (maybe) the Weierstrass $\mathcal{P}$ function.

3.2.7 What does the graph of e^z look like?

The basic object I want to look at is

$$\Gamma = \{(z, w) \in \mathbb{C} \times \mathbb{C}^* \mid w = e^z\}$$

This lives inside $\mathbb{C} \times \mathbb{C}^* \subset \mathbb{C}^2 \cong \mathbb{R}^4$, so it's a little hard to visualize, but we can describe the projection on the second factor $\Gamma \rightarrow \mathbb{C}^*$.

To do this, recall the description of the map $z \mapsto e^z$ as sending the horizontal line $z = x + iy \mid y = y_0$ in $\mathbb{C}$ to the ray $\arg w = y_0$ in $\mathbb{C}^*$. As y_0 increases, of course, this ray rotates counterclockwise around the origin; of course when y_0 increases by 2π we return the same ray. So we can picture Γ as an infinite spiral staircase lying over the punctured plane $\mathbb{C}^*$ —in fact, this is the universal covering space of the punctured plane $\mathbb{C}^*$.

3.2.8 Rational functions

Rational functions are defined to be quotients of polynomials; that is

$$r(z) = \frac{p(z)}{q(z)} = \lambda \frac{(z - a_1) \cdots (z - a_m)}{(z - b_1) \cdots (z - b_n)}$$

with $a_i \neq b_j$ for all i and j . Here the a_i are again called the *zeroes* of r , and the b_j are called the *poles* of r .

In order to better understand rational functions, we should introduce a basic notion: the *Riemann sphere*. This is just the sphere $S = S^2$, viewed as the one-point compactification of the complex plane $\mathbb{C}$; the added point is called “the point at infinity” and denoted ∞ . If $\Omega \subset S$ is an open subset and $f : \Omega \rightarrow \mathbb{C}$ a function, we’ll say that f is holomorphic at a point $z \neq \infty \in \Omega$ if it’s holomorphic in the usual sense; we’ll say that f is holomorphic at ∞ if the function $g(z) = f(1/z)$ is holomorphic at 0. More generally, if $f : \Omega \rightarrow S$ is a map with ∞ a possible value, we’ll say that f is holomorphic at a point $z \in \Omega$ with $f(z) = \infty$ if the function $1/f$ is holomorphic at z .

With all that said, rational functions can be thought of as holomorphic maps $r : S \rightarrow S$. (Indeed, we will see in due course that they are *all* the holomorphic functions $S \rightarrow S$.) We can extend the notion of zero or pole to the point ∞ : if we set

$$s(z) = r\left(\frac{1}{z}\right),$$

we define the order of zero or pole of r at ∞ to be the order of zero or pole of $s(z)$ at 0. Explicitly, if we write our rational function r as

$$r(z) = \frac{p(z)}{q(z)} = \lambda \frac{(z - a_1) \cdots (z - a_m)}{(z - b_1) \cdots (z - b_n)}$$

as above, then if $m \geq n$, r will have a zero of order $m - n$ at ∞ ; and if $m \leq n$, it will have a pole of order $n - m$ at ∞ .

Exercise 3.2.1 Prove that a rational function $r(z)$ has the same number of zeroes as poles, counting multiplicity; and that similarly the functions $r(z) - \lambda$ all have the same number of zeroes.

3.2.9 Linear fractional transformations

An important special case of rational functions are rational functions of degree 1, often called *linear fractional transformations*. These are maps of the form

$$w = \frac{az + b}{cz + d}.$$

Note that the map is unchanged if we multiply all four coefficients a, b, c and d by a common scalar; and that in order for the function to be nonconstant we have to assume $ad - bc \neq 0$

These are actually automorphisms of the Riemann sphere (we will see that they are *all* the automorphisms of S), and as such they form a group G . (“Automorphism” here means a biholomorphic bijection; that is, a holomorphic map $S \rightarrow S$ with a holomorphic inverse) Explicitly, if

$$\phi : z \mapsto \frac{az + b}{cz + d} \quad \text{and} \quad \eta : z \mapsto \frac{ez + f}{gz + h}$$

then the composition $\eta \circ \phi$ sends z to

$$\begin{aligned} \frac{e\left(\frac{az+b}{cz+d}\right) + f}{g\left(\frac{az+b}{cz+d}\right) + h} &= \frac{e(az + b) + f(cz + d)}{g(az + b) + h(cz + d)} \\ &= \frac{(ea + fc)z + (eb + fd)}{(ga + hc)z + (gb + hd)}, \end{aligned}$$

another linear fractional transformation. In fact, if we associate to the linear fractional transformation $z \mapsto \frac{az+b}{cz+d}$ the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we see that composition of linear fractional transformation corresponds exactly to matrix multiplication. (Note that since we must have $ad - bc \neq 0$ as observed above, the linear fractional transformations correspond to invertible 2×2 matrices,

modulo scalar multiplication; in particular, this shows us that the inverse of a linear fractional transformation is again a linear fractional transformation.)

We have thus

$$G = PGL_2(\mathbb{C}).$$

Exercise 3.2.2 Prove that for any three distinct points $p, q, r \in S$, there is a unique linear fractional transformation carrying the points $0, 1, \infty$ to p, q, r in that order.

Linear fractional transformations are described in much greater detail in section 3.3 of Ahlfors, which you can read anytime; while we'll be skipping over Chapter 3 for now, I hope we can double back later and talk some more about them.

Chapter 4

Differential forms and path integrals

First off, I've been struggling with how to present the notions of differential forms and path integrals. The right way to do this would be to develop the notion of differentiable manifolds, their tangent and cotangent spaces; then define differential forms and the exterior derivative in general, and finally state Stokes' theorem. But to do this in anything like reasonable detail would take weeks that we don't have. So I've decided to go the opposite route: we're just going to define and describe the cases of immediate interest (integrals of 1-forms in the plane). If we have time later in the semester I'll devote a lecture to giving an overview of the general theory; if you want to see more in the meantime, I can suggest Chapter 10 of Rudin's book (though he avoids manifolds and cotangent spaces, plus I find the treatment very confusing).

One note about language: we will use the terms *smooth*, *differentiable* and $\mathcal{C}^\infty$ interchangeably: they all mean that the function in question has derivatives of all orders.

4.1 Integration of functions on $\mathbb{R}$ and $\mathbb{R}^2$

First, there is the notion of the Riemann integral, which is a linear map $I_b^a : \mathcal{C}[a, b] \rightarrow \mathbb{R}$ from the space of continuous real-valued functions on the interval $[a, b] \subset \mathbb{R}$ to $\mathbb{R}$. This is defined in the usual way: you break up the

interval $[a, b]$ into n subintervals of equal length $(b - a)/n$; choose a value x_i in the i th interval, take the sum

$$\frac{b - a}{n} \cdot \sum f(x_i)$$

and let $n \rightarrow \infty$.

There is a similar way of defining the integral of a bounded continuous function f on a bounded domain $\Omega \subset \mathbb{R}^2$, in which we overlay a grid of horizontal and vertical lines spaced $1/n$ apart (think of the sort of grate you use to make potatoes into french fries); choose a point p_i in each square contained in Ω , take the sum

$$\frac{1}{n^2} \cdot \sum f(p_i)$$

and let $n \rightarrow \infty$; we get in this way a linear map $I_\Omega : \mathcal{C}(\Omega) \rightarrow \mathbb{R}$.

We should point out that these constructions apply to complex-valued functions as well as real.

4.2 1-forms and their integrals

For this part, I want to you to try and forget you ever heard of complex numbers; everything will be real. Once we've stated the main definitions and theorems, we'll go back and re-introduce the complex numbers.

To start with, a definition:

Definition 4.2.1 Let $\Omega \subset \mathbb{R}^2$ be any open set. By a *1-form* ω on Ω , we will mean an expression of the form

$$\omega = f(x, y)dx + g(x, y)dy$$

where f and g are $\mathcal{C}^\infty$ functions on Ω .

We could think of this as a vector-valued function on Ω .

One related definition: if $h : \Omega \rightarrow \mathbb{R}$ is any smooth function, we can associate to it a 1-form called the *differential* of h and written dh ; this is the 1-form

$$dh := \frac{\partial h}{\partial x} dx + \frac{\partial h}{\partial y} dy$$

What do you do with 1-forms? The answer is, you integrate them along arcs in Ω : if $\gamma : [a, b] \rightarrow \Omega$ is any smooth map, written as

$$\gamma(t) = (\alpha(t), \beta(t))$$

we define the *integral of ω along γ* to be

$$\int_{\gamma} \omega := \int_{[a,b]} (\alpha'(t)f(\gamma(t)) + \beta'(t)g(\gamma(t))).$$

Concretely: at every point $p = \gamma(t)$ of the arc, we take the dot product of the tangent vector $\gamma'(t) = (\alpha'(t), \beta'(t))$ with the vector $\omega(p) = (f(p), g(p))$, and integrate the resulting function on $[a, b]$. In any event, such an integral is called a *path integral* or a *line integral*.

4.3 3 basic facts

There are three basic facts about path integrals:

- *The integral $\int_{\gamma} \omega$ is independent of the parametrization:* in other words, if $\phi : [c, d] \rightarrow [a, b]$ is any smooth map with $\phi(c) = a$ and $\phi(d) = b$, and we let $\delta(t) := \gamma(\phi(t))$, then

$$\int_{\delta} \omega = \int_{\gamma} \omega$$

In a sense, this is the whole *raison d'être* of 1-forms: you can integrate them along paths, independently of the choice of parametrization.

- *The fundamental theorem of calculus:* If h is any smooth function on Ω , and $\gamma : [a, b] \rightarrow \Omega$ any arc, we have

$$\int_{\gamma} dh = h(b) - h(a).$$

and finally

- *Green's theorem (a.k.a Stokes theorem)*: If γ is a simple closed curve in Ω , bounding a subregion $\Delta \subset \Omega$ (so in particular $\gamma(a) = \gamma(b)$), and $\omega = f(x, y)dx + g(x, y)dy$ is a 1-form on Ω , then we have

$$\int_{\gamma} \omega = I_{\Delta} \left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x} \right)$$

where I is as defined above.

This is all we're going to use about 1-forms and their path integrals; the remainder of this lecture will involve introducing the complex numbers and relating all this to holomorphic functions.

4.4 Working over $\mathbb{C}$

So far, everything we've done (with the exception of the observation that the Riemann integral makes sense for complex-valued functions) has involved real 1-forms on $\mathbb{R}^2$. We're now going to introduce the complex numbers and reformulate some of the above in that setting.

The first thing is truly negligible: we are going to identify the domain of our functions and 1-forms as $\mathbb{C}$ rather than as $\mathbb{R}^2$, which means that we'll write functions as $f(z)$ rather than $f(x, y)$ (though we will still want to write $z = x + iy$ and think of functions $f(z)$ as functions of two real variables on occasion.)

The second is also minor: we want to consider complex-valued 1-forms as well as real. What this means is simply that from here on out we'll define a 1-form on a domain $\Omega \subset \mathbb{C}$ to be an expression of the form

$$\omega = f(z)dx + g(z)dy$$

with f and g now complex-valued functions. We can carry out in this setting all the constructions we made above, but now all the functions involved will be complex-valued.

The third shift is the one that confuses people (myself included), though in fact it's pretty simple. We've defined the value of a 1-form at a point p to

be a linear combination of the symbols dx and dy ; now that we're allowing complex coefficients, this is just an element of the 2-dimensional complex vector space $\mathbb{C}\langle dx, dy \rangle$. But now we're going to introduce a different basis for that vector space: we set

$$dz := dx + idy \quad \text{and} \quad d\bar{z} := dx - idy$$

Note that we can also express our old basis $\{dx, dy\}$ in terms of the new: we have

$$dx = \frac{dz + d\bar{z}}{2} \quad \text{and} \quad dy = \frac{dz - d\bar{z}}{2i}.$$

Thus, the value of a complex-valued 1-form at a point can be expressed either as a complex linear combination of dx and dy , or as a complex linear combination of dz and $d\bar{z}$; we can go back and forth by using the equations above, though for our purposes it will turn out that expressing a 1-form in terms of dz and $d\bar{z}$ is more useful.

So, the objects we're dealing with haven't changed—they're still complex-valued $\mathcal{C}^\infty$ 1-forms; we're just writing them in terms of a different basis. Why is this confusing? Maybe because we can represent dx and dy by horizontal and vertical vectors¹, but it's not so clear how to represent dz or $d\bar{z}$.

So, why make this change of basis? One reason comes out if we write the exterior derivative of a complex-valued $\mathcal{C}^\infty$ function f in terms of our new basis. So, suppose that we have such a function, which we'll write as

$$f(x + iy) = u(x, y) + iv(x, y).$$

We can then write its exterior derivative in terms of the basis $\{dx, dy\}$ as

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right) dx + \left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) dy.$$

Now, if we plug in the relations

$$dx = \frac{dz + d\bar{z}}{2} \quad \text{and} \quad dy = \frac{dz - d\bar{z}}{2i}$$

¹We shouldn't—identifying the tangent and cotangent spaces via dot product ultimately makes everything more confusing—but we do.

we arrive at

$$df = \frac{1}{2} \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} - i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \right) dz + \frac{1}{2} \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} + i \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} \right) d\bar{z}.$$

Now, the key question: what happens if f is holomorphic? Immediately things simplify dramatically: first, the Cauchy-Riemann equations say exactly that the coefficient of $d\bar{z}$ is 0; and then we can likewise identify the coefficient of dz as the derivative f' of f ! Thus, *for a holomorphic function f we have*

$$df = f'(z)dz,$$

suggesting that the basis $\{dz, d\bar{z}\}$ really is a better basis for dealing with holomorphic functions.

In fact, this leads to a definition:

Definition 4.4.1 By a *holomorphic 1-form* on a region $\Omega \subset \mathbb{C}$ we mean a complex-valued 1-form of the form $\phi = f(z)dz$, where f is a holomorphic function.

In other words, for a complex-valued 1-form ϕ to be holomorphic means two things: first, that the values of ϕ are multiples of dz ; and then that the coefficient function be holomorphic. Thus the 1-form zdx is not a holomorphic 1-form, since its values are not multiples of dz ; and the 1-form xdz is not holomorphic because, while its values are multiples of dz , the coefficient function x is not holomorphic. But zdz is indeed holomorphic.

At this point, you should take some time out and work through some examples of 1-forms and their integrals. In particular, I'd suggest doing one example that is going to be important for us: this is the

Lemma 4.4.2 *Let γ be a parametrization of the unit circle; e.g., $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$ sends t to $(\cos t, \sin t)$. Then*

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i, & \text{if } n = -1; \text{ and} \\ 0, & \text{otherwise} \end{cases}$$

Proof This is just a matter of plugging in and remembering the standard tricks for integrating trigonometric functions. Here we have

$$z^n = \cos nt + i \sin nt; \quad dz = dx + idy \quad \text{and} \quad \gamma'(t) = (-\sin t, \cos t)$$

and plugging in we arrive at

$$\int_{\gamma} z^n dz = \int_0^{2\pi} (\cos nt + i \sin nt)(-\sin t + i \cos t) dt.$$

Collecting the real and imaginary parts of this expression we arrive at

$$\int_0^{2\pi} (-\cos nt \sin t - \sin nt \cos t) dt + i \int_0^{2\pi} (\cos nt \cos t - \sin nt \sin t) dt.$$

the first integral is always 0; the second is 0 unless $n = -1$, in which case the integrand is simply 1 and the integral equal to 2π .

4.5 Cauchy's theorem

We now come to the first major theorem. Suppose we have a holomorphic 1-form on a region $\Omega \subset \mathbb{C}$

$$\eta = f(z)dz$$

with coefficient function $f(z) = u(z) + iv(z)$. We can multiply out to express it as a linear combination of dx and dy :

$$f(z)dz = (u(z) + iv(z))(dx + idy) = (u + iv)dx + i(u + iv)dy.$$

Say now we want to integrate η over a simple closed curve γ in Ω . If we assume in addition that Ω is simply connected, so that γ traces out the boundary of a subregion $\Delta \subset \Omega$, we can apply Green's theorem as stated above, which says that

$$\int_{\gamma} \eta = I_{\Delta} \left(\frac{\partial i(u + iv)}{\partial x} - \frac{\partial (u + iv)}{\partial y} \right)$$

and the Cauchy-Riemann equations exactly say that this is 0!

The immediate consequence of this fact is

Theorem 4.5.1 (Cauchy's theorem) *If $f(z)$ is holomorphic in a simply connected region $\Omega \subset \mathbb{C}$, then*

$$\int_{\gamma} f dz = 0$$

for any closed loop γ in Ω .

This basic fact (slightly tweaked) is what drives the entire subject forward, as we'll see starting Wednesday. Interestingly, even though it may seem that this theorem states a fact about the global behavior of holomorphic functions, we'll see it also gives us tremendously strong insight into the local properties of these functions.

Chapter 5

The Cauchy integral formula

5.1 Cauchy's theorem

Today we're going to review the steps leading up to Cauchy's theorem, and then prove a slightly stronger form of the theorem. Once we have that, we can (finally!) start to prove some of the amazing theorems about holomorphic functions I've been telling you about.

5.1.1 Previously on Math 113

Let's start by recapping the basic definitions from last time:

First, a *complex-valued 1-form* on a region $\Omega \subset \mathbb{R}^2 = \mathbb{C}$ is an expression

$$\eta = p(x, y)dx + q(x, y)dy,$$

where p and q are complex-valued $\mathcal{C}^\infty$ functions. By a change of basis, the same form could be written as

$$\eta = r(x, y)dz + s(x, y)d\bar{z}$$

with r and s again complex-valued $\mathcal{C}^\infty$ functions; or, if we think of points in Ω as identified by a complex number, as

$$\eta = r(z)dz + s(z)d\bar{z}.$$

If $\gamma : [a, b] \rightarrow \Omega$ is any arc, and $\eta = p(x, y)dx + q(x, y)dy$ any 1-form as above, we define the path integral $\int_{\gamma} \eta$ by

$$\int_{\gamma} \eta := \int_a^b (p(\gamma(t))x'(t) + q(\gamma(t))y'(t)) dt$$

How to think of this: the cotangent space is defined to be the dual of the tangent space, and a 1-form η on Ω is a gadget that associated to every point in Ω an element of its cotangent space. Meanwhile, an arc $\gamma(t) = (x(t), y(t))$ has at every time t a tangent vector $(x'(t), y'(t))$. We simply feed the tangent vector to the arc at time t to the value of the 1-form η at the image point $\gamma(t)$, arriving at a scalar; we then take the Riemann integral of this scalar function on $[a, b]$.

It gets confusing because we are simultaneously identifying both the tangent and cotangent spaces to a point in the plane with $\mathbb{R}^2$, with the duality given by the “standard” dot product (thus, for example, in many calculus texts a 1-form is drawn as a vector field, which makes me crazy). Things will actually be clearer when we have the time to define tangent and cotangent spaces and keep the two straight.

Next, we define a *holomorphic 1-form* to be a complex-valued 1-form of the form

$$\eta = f(z)dz$$

with $f(z)$ holomorphic in Ω . This means two things: first, that the value of η at every point p lies in the subspace of the complex cotangent space $T_p^*\Omega \otimes \mathbb{C}$ spanned by dz ; and second, that the coefficient function is holomorphic.

Finally, we can apply Stokes’ theorem in the plane (a.k.a. Green’s theorem), which leads us to the fundamental

Theorem 5.1.1 (Cauchy’s theorem) *If η is a holomorphic 1-form in a simply connected region $\Omega \subset \mathbb{C}$, and $\gamma : [a, b] \rightarrow \Omega$ a closed loop in Ω , then the line integral*

$$\int_{\gamma} \eta = 0.$$

Note that, while we’re basing this on Stokes’ theorem, Ahlfors does not assume Stokes, but rather gives a self-contained proof of the Cauchy theorem in the relevant circumstances.

5.2 A slight tweaking of Cauchy's theorem

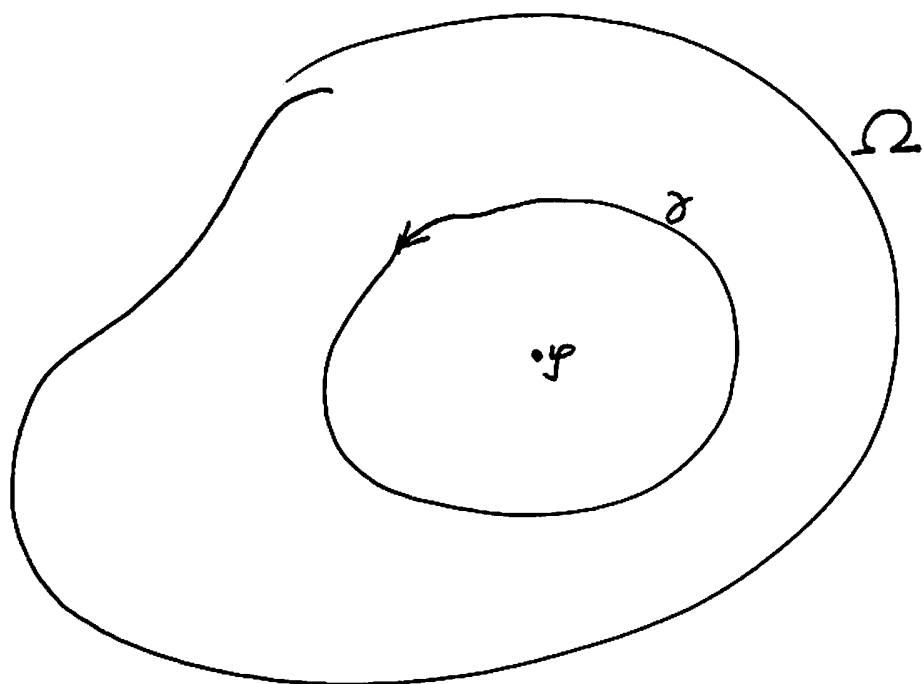
For many reasons, it will be useful to have a slightly stronger version of Cauchy's theorem, in which the 1-form η is allowed to have some singularities but the conclusion still holds. (This is Theorem 3 in Chapter 4.) Ahlfors does this in a rectangle, but in fact we can do it in any simply connected region.

Theorem 5.2.1 (tweaked Cauchy) *Let $\Omega \subset \mathbb{C}$ be a simply connected region and $\zeta \in \Omega$ a point. Suppose that η is a holomorphic 1-form in Ω with a singularity at ζ ; in other words, η is a holomorphic 1-form on $\Omega \setminus \{\zeta\}$. Assume that*

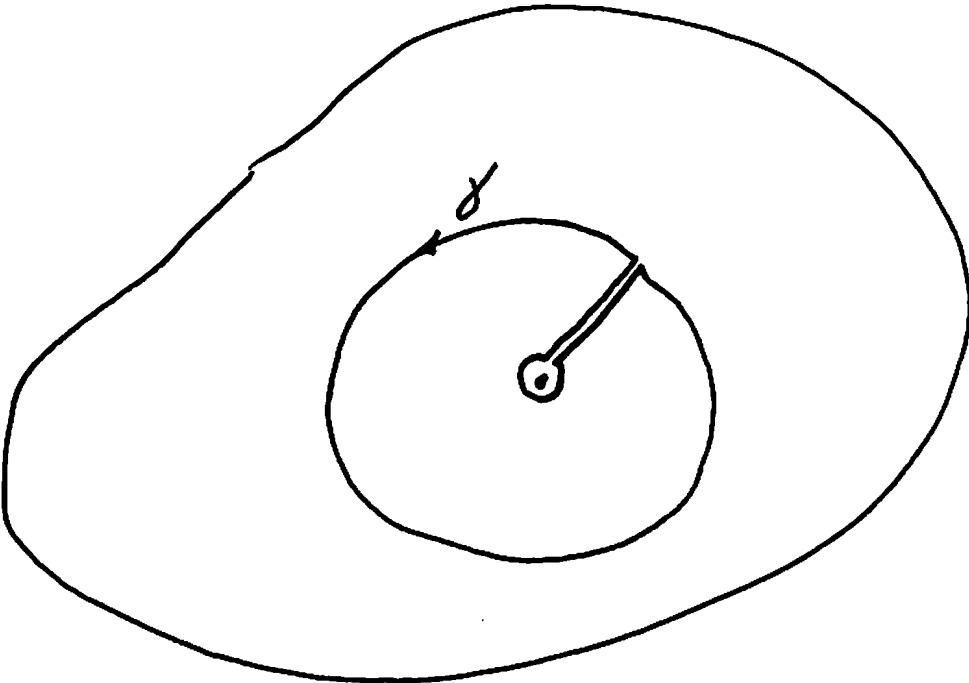
$$\lim_{z \rightarrow \zeta} (z - \zeta)f(z) = 0,$$

so the singularity isn't too bad. Then the integral of η along any closed loop in Ω is 0.

Proof: Suppose the region Ω , the point ζ and the loop γ are as pictured on the next page.



The trick is to replace the loop γ by another loop, as follows: we start by going around the loop γ as usual. But before we get back to the end, we take a detour: we veer off and head toward the point ζ along an arc α , stopping when we get within a distance ϵ of ζ . We then follow a circle of radius ϵ around ζ ; when we get all the way around the circle, we turn again and head back along the path α we took to go from the original loop to the circle around ζ (going the opposite way, of course). When we get back to γ , we resume our journey around γ until we get back to where we started. The resulting loop is pictured in the following diagram, on the next page.



There are now two points to be made. First, the integrals along α and α^{-1} exactly cancel; and, by virtue of the condition $\lim_{z \rightarrow \zeta} (z - \zeta)f(z) = 0$, the integral around the circle of radius ϵ around ζ goes to zero as $\epsilon \rightarrow 0$. Thus the limit, as $\epsilon \rightarrow 0$, of the integral around this modified path, is equal to the integral $\int_{\gamma} \eta$ along the original path. (That's one point, if you're counting.)

The second point is that *the modified path encloses a region in which η is holomorphic*—we've excised the singularity. Thus Cauchy's theorem applies; the integral over the modified path is 0, and hence so is the original integral.

Of course, as Ahlfors points out, the same conclusion holds if we allow η to have finitely many singularities, as long as each one satisfies the condition above that $\lim_{z \rightarrow \zeta} (z - \zeta)f(z) = 0$.

5.3 The Cauchy integral formula

5.3.1 Winding numbers

Before we derive the Cauchy integral formula, we want to give the definition of winding numbers and to establish some notations/facts about them.

The basic idea is simple, given our knowledge of the fundamental group. Say we have a point $a \in \mathbb{C}$ in the complex plane, and a closed loop $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{a\}$ in the plane that does not pass through a . We have

$$\pi_1(\mathbb{C} \setminus \{a\}) = \mathbb{Z},$$

with the generator the class of a path that simply transverses the boundary of a disc around a once in the counterclockwise direction. We define the *winding number* $n(\gamma, a)$ of the loop γ with respect to the point a to be simply the element of $\mathbb{Z}$ corresponding to the class $[\gamma] \in \pi_1(\mathbb{C} \setminus \{a\}) = \mathbb{Z}$.

As a sort of sanity check, we'll mention: if γ is a simple closed curve in the plane, the Jordan curve theorem says that it bounds a region $\Omega \subset \mathbb{C}$. In this situation, the winding number of γ with respect to the point a will be ± 1 for every point $a \in \Omega$ (depending on whether γ traverses the boundary of Ω in a clockwise or counterclockwise direction), and 0 for every $a \in \mathbb{C} \setminus \overline{\Omega}$.

The crucial fact about winding numbers is one that follows from the calculation we made in class: if γ traces out the unit circle in $\mathbb{C}$ in a counterclockwise direction (e.g., if $\gamma(t) = e^{it}$ for $0 \leq t \leq 2\pi$), then

$$\int_{\gamma} \frac{dz}{z} = 2\pi i.$$

From this, it follows that more generally that if $a \in \mathbb{C}$, and $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{a\}$ is a closed loop in the plane that does not pass through a , then

$$\int_{\gamma} \frac{dz}{z - a} = n(\gamma, a) \cdot 2\pi i.$$

Note that Ahlfors, lacking at this point knowledge of the fundamental group, adopts this as the *definition* of the winding number!

5.3.2 Finally, the Cauchy integral formula

Finally, we can use the slightly tweaked version of Cauchy's theorem to prove the Cauchy integral formula. The set-up is this: we have a holomorphic function $f(z)$ on a simply connected region $\Omega \subset \mathbb{C}$. We suppose we have a simple closed curve $\gamma : [a, b] \rightarrow \Omega$ which bounds a subregion $\Delta \subset \Omega$, and let $a \in \Delta$ be any point. (Equivalently, we could say we have a simple closed curve γ bounding a simply connected region $\Delta \subset \mathbb{C}$, and a holomorphic function f defined on the closure $\overline{\Delta}$, meaning it's defined on some open set $\Omega \subset \mathbb{C}$ containing $\overline{\Delta}$.)

We're going to apply the Cauchy theorem to the 1-form $F(z)dz$, where

$$F(z) = \frac{f(z) - f(a)}{z - a}.$$

Note that this is not necessarily a holomorphic 1-form in all of Δ , since it may have a singularity at a ; but since f is continuous in Δ , the 1-form $F(z)dz$ will satisfy the hypotheses of tweaked Cauchy. Thus, we have

$$\int_{\gamma} \frac{f(z) - f(a)}{z - a} dz = 0$$

or in other words,

$$\int_{\gamma} \frac{f(z)}{z - a} dz = f(a) \int_{\gamma} \frac{dz}{z - a} = 2\pi i f(a).$$

Rearranging, we arrive at the *Cauchy integral formula*

$$f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz.$$

Finally, I want to make a completely trivial alteration to the formula. In our applications of this formula, we'll see that its strength lies in the fact that it gives $f(a)$ as a function of a . Since a is not a letter we usually use for a variable, we're going to replace a by z and z by ζ ; we arrive at the

Theorem 5.3.1 (Cauchy integral formula) *Let $f(z)$ be a holomorphic function on a region $\Omega \subset \mathbb{C}$, and $\gamma : [a, b] \rightarrow \Omega$ a simple closed curve which bounds a subregion $\Delta \subset \Omega$. For any point $z \in \Delta$,*

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{\zeta - z}.$$

We'll be spending the next week or two investigating the consequences of this formula. Let me just point out now that it reflects one aspect of holomorphic functions I mentioned earlier: their rigidity. In particular, we are saying that *the value of f at any point in Δ is determined by its values on the boundary of Δ* —not something that could possibly be true for $\mathcal{C}^\infty$ functions in general.

5.4 Consequences

5.4.1 Derivatives!

One of the first consequences of Cauchy is a proof of the assertion I've made before: that even though the condition of being holomorphic requires only that the first derivative be defined (we don't even require it, a priori, to be continuous!), in fact a holomorphic function has derivatives of all orders.

To do this, we simply take the Cauchy integral formula and view both sides, as we suggested, as functions of z . We have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{\zeta - z}.$$

for all $z \in \Delta$. Since all the terms in the integral are bounded and continuous, we can differentiate both sides of this equality with respect to z , to arrive at

$$f'(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{(\zeta - z)^2},$$

and repeating this $n - 1$ more times we have

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{(\zeta - z)^{n+1}}$$

So, derivatives of all orders exist! Thus in particular *holomorphic functions are $\mathcal{C}^\infty$* .

5.4.2 Liouville's theorem

Finally (for now!) we have the always popular

Theorem 5.4.1 (Liouville's theorem) *Let f be a holomorphic function on the complex plane $\mathbb{C}$. If f is bounded, then f is constant.*

So, bounded $\mathcal{C}^\infty$ functions $\mathbb{R} \rightarrow \mathbb{R}$, like e^{-x^2} , $\sin x$ or $1/(1+x^2)$, don't have analogues in the world of holomorphic functions. (The functions e^{-z^2} and $\sin z$ are defined and holomorphic on all of $\mathbb{C}$, but not bounded; the function $1/(1+z^2)$ is neither defined for all $z \in \mathbb{C}$ nor bounded.)

Proof. Suppose we have $|f(z)| < M$ for all $z \in \mathbb{C}$. Let C_r be the circle of radius r in $\mathbb{C}$. According to the formula above, we have

$$f'(0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta) d\zeta}{\zeta^2}.$$

But now look at the integrand: by hypothesis the numerator is bounded by M ; and the denominator has absolute value r^2 . Since the length of the path C_r is $2\pi r$, it follows that the integral goes to 0 as $r \rightarrow \infty$; since it's independent of r , this means simply that $f'(0) = 0$. Finally, we could apply this argument to any value of z to deduce that $f'(z) = 0$ for all z , and hence that f is constant.

Corollary 5.4.2 (The fundamental theorem of algebra) *Any polynomial $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0 \in \mathbb{C}[z]$ of positive degree $n > 0$ has a root.*

Proof. This is an immediate consequence of Liouville: if p had no zeroes, the reciprocal $q(z) = 1/p(z)$ would be defined and holomorphic on the entire complex plane $\mathbb{C}$. Since $n > 0$, we have $\lim_{z \rightarrow \infty} q(z) = 0$, so q is bounded and hence constant, a contradiction.

Chapter 6

Singularities of holomorphic functions

6.1 Clarifications from last time

I wanted to start by clarifying three points that came up in Wednesday's lecture. In two of the three cases, what was lacking was simply a reference to Ahlfors or Rudin; in the third, I was simply wrong and I'll have to ask you to take on faith an assertion that you probably won't see proved until you take a course like 231a.

6.1.1 The proof of tweaked Cauchy

In the proof of the “tweaked” version of the Cauchy theorem—in which we relax the hypothesis to allow the function $f(z)$ to have a mild singularity at a point $a \in \Omega$ —we replaced the original path of integration γ by a different path, in which we veer off from the path γ at one point, follow an arc α to the circle C_ϵ of radius ϵ around a , go once around C_ϵ , then back out along the path α^{-1} to where we left γ along the path α^{-1} ; finally we resume the loop γ . Let's call this modified loop γ_ϵ .

The argument then proceeded in two steps. First, we observe that the path γ_ϵ encloses a region in which the integrand $f(z)dz$ is holomorphic ev-

everywhere, and so the original Cauchy theorem applies to tell us

$$\int_{\gamma_\epsilon} f(z)dz = 0.$$

The second point is to argue that

$$\lim_{\epsilon \rightarrow 0} \int_{\gamma_\epsilon} f(z)dz = \int_\gamma f(z)dz;$$

given this, tweaked Cauchy follows.

Now, the argument for the second point was that since the modified arc γ_ϵ consists of four segments, we have

$$\int_{\gamma_\epsilon} f(z)dz = \int_\gamma f(z)dz + \int_\alpha f(z)dz + \int_{C_\epsilon} f(z)dz - \int_\alpha f(z)dz.$$

The terms involving the integral over α cancel, and so we simply have to show

$$\lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} f(z)dz = 0.$$

Here, the missing ingredient is an observation that Ahlfors makes in section 4.1.1 (specifically, (9) on p. 108), and that I neglected to state in lecture: that for path integrals in general,

$$\left| \int_\delta f(z)dz \right| \leq \int_\delta |f(z)||dz|;$$

and in particular, if $|f(z)| < M$ for all z along the path, then the absolute value of the integral is less than M times the arc length of the path. Given this, what we need to establish is that

$$\lim_{\epsilon \rightarrow 0} \left(2\pi\epsilon \cdot \max_{|z-a|=\epsilon} |f(z)| \right) = 0;$$

and this is exactly what is implied by the condition $\lim_{z \rightarrow a} (z - a)f(z) = 0$.

6.1.2 The existence of derivatives

The second issue arose in the proof that a holomorphic function f on a region $\Omega \subset \mathbb{C}$ has derivatives of all orders. The idea was straightforward: we want to use the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{\zeta - z}$$

where we can take γ to be a small circle around z . The formula applies to every point z inside the circle, and so we can differentiate both sides of the Cauchy formula to arrive at

$$f'(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{(\zeta - z)^2}$$

and more generally

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(\zeta) d\zeta}{(\zeta - z)^{n+1}}.$$

The issue is, we need to justify taking the derivative of the integral by differentiating the integrand. There are two ways we could do this. One, we could cite the standard real analysis theorem asserting we can do this under relatively mild conditions: if $\phi(x, t)$ is a bounded, continuous real-valued function of two real variables, we have

$$\frac{d}{dt} \int_a^b \phi(x, t) dx = \int_a^b \frac{\partial \phi}{\partial t}(x, t) dx.$$

See, for example, Chapter 9 (page 236) of Rudin (the statement in Rudin is maximally confusing, but basically, this holds as long as all the terms are defined). Note that we are applying this statement not to the line integral $\int_{\gamma} \eta$ per se, but to the ordinary Riemann integral $\int_a^b \gamma^* \eta$ to which it refers.

Alternatively, Ahlfors gives a self-contained argument, based on his Lemma 3. This is somewhat more complicated, and if you're comfortable with the application of Rudin's theorem you don't need to go through it all; but in fact Lemma 3 will come up again in Chapter 5 when we talk about power series expansions, so it wouldn't be wasted effort.

There is also another reason for looking at Lemma 3: it answers a very basic question. We've seen that a holomorphic function $f(z)$ on a region Ω bounded by a simple closed curve C is determined by its values on C (assuming that f extends to a continuous function on $\overline{\Omega}$). The natural follow-up question is, can every continuous complex-valued function ϕ on C be the boundary values of a holomorphic function f on Ω ? The answer, according to Ahlfors' Lemma 3, is yes: we simply define f by

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{\phi(\zeta) d\zeta}{\zeta - z}$$

Of course, we still have to show that f has the boundary values we want, but we'll do that when this question comes up later.

6.1.3 Dependence of a path integral on the homotopy class of the path

On this one I just flat out screwed up. The statement I wanted to make, to simplify the treatment of winding numbers, is this: assume the 1-form η is closed in a region $\Omega \subset \mathbb{C}$. If $\gamma : [0, 1] \rightarrow \Omega$ is a closed loop, starting and ending at p , then *the integral $\int_{\gamma} \eta$ depends only on the class $[\gamma] \in \pi_1(\Omega, p)$.*

This statement is certainly true: in fact, the integral $\int_{\gamma} \eta$ depends only on the homology class of γ , which is a coarser invariant than the homotopy class. But we lack the tools to establish it, and so I think this is one of those things you take on faith until you acquire the technique to prove it.

6.2 Liouville and the FTA

I want to get back to deriving immediate consequences of the Cauchy integral formula. To begin with, we have

Theorem 6.2.1 (Liouville's theorem) *A bounded holomorphic function on $\mathbb{C}$ is constant.*

The proof is simple, relying on the expression of the derivatives of f as path integrals. Specifically, say $|f(z)| < M$ for all $z \in \mathbb{C}$. If we denote by C_r

the circle $\{z : |z| = r\}$ of radius r around the origin, then we've seen that

$$f^{(n)}(0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta) d\zeta}{\zeta^{n+1}}.$$

Now, the denominator of the integral has absolute value r^{n+1} along the path of integration, while the numerator is bounded by M . The path length is $2\pi r$, and it follows that

$$|f^{(n)}(0)| < n!Mr^{-n};$$

since the left hand side is independent of r we conclude that $f^{(n)}(0) = 0$. Finally, we could apply the same argument to any point in the plane, so we may deduce that the derivative f' of f is identically 0, and hence that f is constant.

Finally, we can give, as a direct consequence of Liouville's theorem, yet another proof of the fundamental theorem of algebra. This is completely straightforward: let

$$p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0 \in \mathbb{C}[z]$$

be a polynomial with complex coefficients of degree $n > 0$, and assume that $p(z)$ is never 0. Then the reciprocal $q(z) = 1/p(z)$ is a holomorphic function in the entire plane. Moreover, since $n > 0$ we have $\lim_{z \rightarrow \infty} p(z) = \infty$, and correspondingly $\lim_{z \rightarrow \infty} q(z) = 0$. Thus q is bounded, and hence by Liouville constant, a contradiction.

6.3 Singularities of holomorphic functions I: removable singularities

One of the striking features of holomorphic functions is that their singularities are very prescribed: in fact, there are only three kinds of singularities of holomorphic functions, one of which isn't even a singularity at all.

The set-up here is straightforward: we have a region $\Omega \subset \mathbb{C}$ and a point $a \in \Omega$ (for the purposes of this discussion, nothing is lost if we just take Ω to be a disc). We are given a holomorphic function $f(z)$ on the complement $\Omega \setminus \{a\}$; our goal is to describe the behavior of $f(z)$ when z approaches a .

The first case to consider is the mildest: suppose to begin with that

$$\lim_{z \rightarrow a} (z - a)f(z) = 0.$$

In this case a is called a *removable singularity* of f ; examples are functions like

$$f(z) = \frac{\sin z}{z} \quad \text{and} \quad f(z) = \frac{e^z - 1}{z}$$

(with $a = 0$ and Ω any neighborhood of 0).

In this case, we can draw a circle C around a , and the tweaked Cauchy theorem tells us that

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta) d\zeta}{\zeta - z}$$

for $z \neq a$. But the right hand side is a function analytic in the whole disc; so we see that f —a priori only defined in $\Omega \setminus \{a\}$ —can be extended to a holomorphic function in all of Ω . In other words, a removable singularity is no singularity at all.

There is an important consequence of this fact: given any function $f(z)$ that is holomorphic in Ω with $f(a) = 0$, we can apply this to the function $f(z)/(z - a)$ to deduce the

Corollary 6.3.1 *Let f be a holomorphic function in the region Ω , and suppose that $f(a) = 0$ for some $a \in \Omega$. Then there exists a holomorphic function g in Ω such that*

$$f(z) = (z - a)g(z)$$

In other words, holomorphic functions are like polynomials in this respect. Note that this is in marked contrast to the case of continuous functions, where a function like $f(x) = \sqrt[3]{x}$ is continuous and takes the value 0 at $x = 0$, but the function $f(x)/x$ cannot be extended continuously to the whole line.

This corollary has some pretty amazing corollaries of its own! To set this up, suppose now that f is actually holomorphic in all of Ω . We can apply the corollary to the function

$$F(z) = \frac{f(z) - f(a)}{z - a},$$

which has visibly a removable singularity; it follows that we can write

$$f(z) = f(a) + (z - a)f_1$$

for some holomorphic function f_1 on Ω . Note that

$$f_1(a) = \lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} = f'(a).$$

We can now apply the same construction to the new function f_1 , to find a holomorphic function f_2 in Ω with

$$f_1(z) = f_1(a) + (z - a)f_2$$

and we can continue in this way forever; we get a sequence of functions f_n , all defined and holomorphic in Ω , satisfying

$$f_{n-1}(z) = f_{n-1}(a) + (z - a)f_n(z) \quad \text{and} \quad f^{(n)}(a) = n!f_n(a).$$

Adding these all up, we get for any n

$$f(z) = f(a) + \frac{f'(a)}{1!}(z-a) + \frac{f^{(2)}(a)}{2!}(z-a)^2 + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(z-a)^{n-1} + f_n(z)(z-a)^n$$

for some holomorphic function f_n on Ω . In addition, we have an integral expression for the function f_n : if C is a circle centered at a (and contained in Ω), we can write

$$f_n(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta) d\zeta}{(\zeta - a)^n (\zeta - z)}$$

6.3.1 That theorem I've been telling you about

Let's start with a basic theorem:

Theorem 6.3.2 *Let Ω be a connected region in the plane $\mathbb{C}$, $a \in \Omega$ and f a holomorphic function on Ω . If $f^{(n)}(a) = 0$ for all n , then f is identically zero.*

Before proving this, let's remark that this is once more in marked contrast to the C^∞ case. For example, consider the function

$$f(x) = \begin{cases} e^{-1/x^2}, & \text{if } x > 0; \text{ and} \\ 0, & \text{if } x \leq 0. \end{cases}$$

It's not hard to see that all the derivatives of e^{-1/x^2} limit to 0 as $x \rightarrow 0$, so this function is $\mathcal{C}^\infty$ with all derivatives 0 at 0, contrary to the statement of the theorem. This in turn allows us to construct $\mathcal{C}^\infty$ function with "local support:" for example, the function

$$g(x) = \begin{cases} e^{-1/(1-x^2)}, & \text{if } |x| < 1; \text{ and} \\ 0, & \text{if } |x| \geq 1. \end{cases}$$

is $\mathcal{C}^\infty$ on all of $\mathbb{R}$, with support the interval $(-1, 1)$; scaling, we can find such a function with support $(-\epsilon, \epsilon)$ for any ϵ . Adding such functions to a given function allows us to alter the values of the function locally without affecting the values at a distance.

To prove the theorem, we just observe that if all derivatives of f vanish at a , then we can write

$$f(z) = (z - a)^n f_n(z)$$

for any n . At the same time, we can use the integral expression for $f_n(z)$ above to deduce that the limit $\lim_{n \rightarrow \infty} (z - a)^n f_n(z) = 0$ for any fixed z inside the circle C , and hence that $f \equiv 0$ inside the circle C .

Finally, we use the connectedness of Ω to deduce that f vanishes identically in Ω . This is straightforward: if $\Gamma \subset \Omega$ is the locus of points where all derivatives of f vanish, we see that Γ is closed, since it's the intersection of the closed subsets $f^{(n)}(z) = 0$; but by the above argument it's open as well. Since the hypothesis rules out $\Gamma = \emptyset$, we see that we must have $\Gamma = \Omega$.

The consequences of this are enormous. To start, assume now that f is holomorphic in Ω and *not* identically zero. Then at every point $a \in \Omega$, there is a smallest integer h such that $f^{(h)}(a) \neq 0$; we say in this case that f has a *zero of order h at a* . That means in turn that in a neighborhood of a we can write

$$f(z) = (z - a)^h g(z), \quad \text{with } g(a) \neq 0.$$

Think about this: if $g(a) \neq 0$, then by continuity there will be a neighborhood U of a such that $g \neq 0$ in U ; it follows that *a is the only zero of the function f in U* . In other words, *the zeroes of a holomorphic function other*

than 0 are isolated; every one is alone in its own neighborhood. Thus, for example, if f is a holomorphic function on the connected region Ω , then

- If f vanishes on any open subset of Ω it vanishes identically in Ω ;
- If f vanishes on any arc in Ω it vanishes identically in Ω ; and
- If f vanishes on any subset of Ω with an accumulation point in Ω it vanishes identically in Ω .

These in turn translate into the sort of theorems I was telling you about: if two holomorphic functions on Ω agree on any of the three types of subsets above, they agree everywhere in Ω .

Chapter 7

Singularities, and the local picture of a holomorphic map

7.1 Poles

We come now to the second possible type of singularity of a holomorphic function: poles. To recall the setup, we have a domain $\Omega \subset \mathbb{C}$, a point $a \in \Omega$, and a function f on $\Omega \setminus \{a\}$ defined and holomorphic everywhere in Ω except for a . We just discussed the case where

$$\lim_{z \rightarrow a} (z - a)f(z) = 0;$$

the next case we'll consider is when

$$\lim_{z \rightarrow a} f(z) = \infty,$$

meaning that for any M there exists a $\delta > 0$ such that

$$0 < |z - a| < \delta \implies |f(z)| > M.$$

In this case, the singularity of f is called a *pole*.

In this case we have already done most of the work: if f is holomorphic on $\Omega \setminus \{a\}$ and $\lim_{z \rightarrow a} f(z) = \infty$, then there is a neighborhood Ω' of a such that f is nonzero in $\Omega' \setminus \{a\}$; and in $\Omega' \setminus \{a\}$ we can set

$$g(z) := \frac{1}{f(z)}.$$

g is then a holomorphic function in $\Omega' \setminus \{a\}$, and we have $\lim_{z \rightarrow a} g(z) = 0$, so by the removable singularity theorem g extends to a holomorphic function in all of Ω' .

Now, as we established in the last lecture, the function g will vanish to some finite, integral order h at a , meaning that

$$g(z) = (z - a)^h j(z)$$

where j is holomorphic in Ω' and $j(a) \neq 0$. We have, correspondingly,

$$f(z) = (z - a)^{-h} k(z),$$

where $k(z) := 1/j(z)$ is again holomorphic and nonzero in Ω' . In other words, a pole is very much like a zero, except that the factor $z - a$ appears to a negative power rather than a positive one!

A function which is analytic in Ω except for poles is called *meromorphic*. Note the following:

- The zeroes *and* poles of a meromorphic function f are isolated; that is, if $a \in \Omega$ is any point, there is a neighborhood U of a in which f has no other zeroes or poles.
- The quotient $h = f/g$ of two holomorphic functions f and g is meromorphic (as long as g doesn't vanish identically!). Actually, the converse is true in certain circumstances: for example, we'll see a little later that any meromorphic function on $\mathbb{C}$ is the quotient of two holomorphic functions on $\mathbb{C}$.
- From the definition, we can think of a meromorphic function as a holomorphic map to the Riemann sphere $S = \mathbb{C} \cup \{\infty\}$.
- We haven't yet established that a holomorphic function f has a power series expansion in powers of $z - a$ for any point a in its domain. But we will, and the consequence is that *meromorphic functions have Laurent series expansions*: if we can write

$$f(z) = (z - a)^{-h} k(z),$$

with k holomorphic and $k(a) \neq 0$, we can write

$$k(z) = a_0 + a_1(z - a) + a_2(z - a)^2 + \dots$$

and hence

$$f(z) = a_0(z - a)^{-h} + a_1(z - a)^{-h+1} + \dots$$

The sum of the terms with negative powers of $z - a$ is called the *polar part*, or *principal part* of the function.

In general, meromorphic functions are every bit as well-behaved as holomorphic ones; indeed, as we said above, they're simply holomorphic functions with target the Riemann sphere S rather than $\mathbb{C}$.

7.2 Essential singularities

We come now to the third and last class of singularity, what are called essential singularities. This is a priori a sort of “none of the above” class; but as we'll see we can still prove some striking theorems about the behavior of functions with essential singularities.

To reiterate the set-up: Ω will be an open subset of $\mathbb{C}$, $a \in \Omega$ any point and f a holomorphic function on $\Omega \setminus \{a\}$. We have so far seen what happens if either

$$\lim_{z \rightarrow a} (z - a)f(z) = 0 \quad \text{or} \quad \lim_{z \rightarrow a} f(z) = \infty.$$

We observe now that if

$$\lim_{z \rightarrow a} (z - a)^c f(z) = 0 \text{ or } \infty$$

for any c , then one of those two conditions must hold. If the singularity is neither removable or a pole, then we must have

$$\lim_{z \rightarrow a} (z - a)^c f(z) \neq 0 \text{ or } \infty \quad \text{for all } c.$$

In this case, it is called an *essential singularity*. An example of a function with an essential singularity would be the function $f(z) = e^{1/z}$, on the punctured plane $\mathbb{C} \setminus \{0\}$. (It's a good exercise to try to see directly that $\lim_{z \rightarrow 0} z^c e^{1/z}$ does not exist for any c , even if we allow ∞ as a possible limit.)

To get a sense of how deeply weird essential singularities are, we have the

Theorem 7.2.1 *Let f be holomorphic in $\Omega \setminus \{a\}$ with an essential singularity at a . Then in any neighborhood of a , the values of the function $f(z)$ come arbitrarily close to every complex number: precisely, for any $\epsilon > 0$, any $\delta > 0$ and any complex number w , there exists a $z \in \Omega \setminus \{a\}$ such that*

$$|z - a| < \epsilon \quad \text{and} \quad |f(z) - w| < \delta.$$

Without the epsilons and deltas, what this is saying is that *in any neighborhood of an essential singularity the values of the function come arbitrarily close to every complex number*. In particular, this says that for any complex number w , I can find a sequence $z_1, z_2, \dots \in \Omega \setminus \{a\}$ tending to a , and such that $\lim_{n \rightarrow \infty} f(z_n) = w$. In other words, when a holomorphic function fails to have a limit, it *really* fails to have a limit!

To prove the theorem, suppose the conclusion fails: that is, for some δ and $\epsilon > 0$ and $w \in \mathbb{C}$, we have $|f(z) - w| > \delta$ for all z with $|z - a| < \epsilon$. Then the function

$$g(z) = \frac{1}{f(z) - w}$$

is bounded in a neighborhood of a and so extends to a holomorphic function in all of Ω . We can then write

$$f(z) = \frac{1}{g(z)} + w,$$

from which we see that f is meromorphic, and does not have an essential singularity.

7.3 The local picture of a holomorphic map

We are not out of applications of the Cauchy integral formula! Another important consequence is what's called the *argument principle*, which we'll describe next. (There's also a version that applies to meromorphic functions, which we'll discuss a little later.)

The starting point is a familiar set-up: we have a function f holomorphic in a region Ω , which Ahlfors takes to be a disc (though that doesn't seem to be necessary); and we have a closed curve γ in Ω . On the first pass, you should keep it simple: assume γ is in fact a simple closed curve, bounding

a subregion $\Delta \subset \Omega$, so that the winding number of γ with respect to any point $a \in \Omega$ not on γ is either 1 or 0, according to whether or not a lies in the region Δ enclosed by γ .

We now ask a basic question: *how many zeroes does f have in Δ ?* Three points: first, we have to assume that no zeroes of f actually lie *on* the path γ . Second, since the zeroes of f are isolated, and the closure of Δ is compact, there can be only finitely many. And third, to clarify the question, we want to count the zeroes *with multiplicity*: for each $a \in \Delta$, we've seen that we can write $f(z) = (z - a)^h g(z)$, with h a nonnegative integer and g holomorphic with $g(a) \neq 0$. We call h the *order* of the zero, denoted $\text{ord}_a(f)$; and the precise form of the question is to find

$$\sum_{a \in \Delta} \text{ord}_a(f).$$

All that said, the problem is pretty straightforward. Applying multiple times the basic principle that a function f with a zero of order h at a can be written in the form $f(z) = (z - a)^h g(z)$ with $g(a) \neq 0$, we see that if $z_1, \dots, z_k$ are the zeroes of f in Δ , and $\text{ord}_{z_\alpha}(f) = m_\alpha$, then we can write

$$f(z) = (z - z_1)^{m_1} (z - z_2)^{m_2} \dots (z - z_k)^{m_k} \cdot g(z)$$

where g is holomorphic and nowhere zero in Δ . Now, consider the function

$$\frac{f'(z)}{f(z)} = \frac{d}{dz} \log f(z).$$

(Note that while $\log f(z)$ is not well-defined, it is locally well-defined up to a constant integral multiple of $2\pi i$, so its derivative *is* well-defined.) Either directly or via the interpretation as the derivative of $\log f(z)$, we see that we have

$$\frac{f'(z)}{f(z)} = \frac{m_1}{z - z_1} + \dots + \frac{m_k}{z - z_k} + \frac{g'(z)}{g(z)}.$$

Now, since g is nowhere 0, the quotient $\frac{g'(z)}{g(z)}$ is holomorphic throughout Δ , and so we have

$$\int_{\gamma} \frac{g'(z)}{g(z)} dz = 0.$$

At the same time, by our previous calculation we know that

$$\int_{\gamma} \frac{m_{\alpha}}{z - z_{\alpha}} dz = m_{\alpha} \cdot 2\pi i.$$

Thus we have the answer to our question, once more in the form of an integral:

Theorem 7.3.1 *Under the hypotheses above, the number of zeroes of f inside the simple closed curve γ is given by*

$$\sum_{z_{\alpha} \in \Delta} \text{ord}_{z_{\alpha}}(f) = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz$$

This is Theorem 10 in Ahlfors, with two slight mods: we are assuming the path γ is a simple closed curve, so all the relevant winding numbers $n(\gamma, z_{\alpha})$ are 1; and where Ahlfors in the sum repeats each zero as many times as its order, we simply include it once with coefficient $\text{ord}_{z_{\alpha}}(f)$.

There is another, very useful way to view this formula, if we think of f as giving a map from Δ to the complex plane $\mathbb{C}$ with coordinate w via $w = f(z)$. We can then consider the arc Γ in the w -plane given as the composition $f \circ \gamma$. We have then

$$\int_{\gamma} \frac{f'(z)}{f(z)} dz = \int_{\Gamma} \frac{dw}{w};$$

or in other words, *the number of zeroes of f inside the loop γ is equal to the winding number of the composed path Γ around the origin in the w -plane.*

The integral representation of the number of zeroes of f in the region Δ has a major consequence. Suppose we deform the function f , specifically replacing $f(z)$ by $f(z) - a$ for small values of a . Given that f doesn't vanish on the arc γ , it has a minimum absolute value ϵ on γ , and for values of a with $|a| < \epsilon$ we see that the integral

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - a} dz$$

varies continuously with a . But at the same time, Theorem 10 tells us that the value of the integral is an integer! The conclusion is that the integral is constant (again, for $|a| < \epsilon$); in other words,

Proposition 7.3.2 *Under the hypotheses above, there exists an $\epsilon > 0$ such that for $|a| < \epsilon$, the number of times f assumes the value a in Ω is the same as the number of times it assumes the value 0.*

Let me point out, once more, how different this is from the behavior of $\mathcal{C}^\infty$ functions. A function of a real variable x like $f(x) = x^2$ has a zero (of order 2) at $x = 0$; but if we deform the function just a little to $f(x) = x^2 + c$, with $c > 0$, the zero simply disappears. We're saying that doesn't happen with holomorphic functions (and indeed if x were a complex variable, the two zeroes would still be there!).

Once more, the reason for the difference is topology. For example, if you prove the following topological lemma, you can deduce the proposition above:

Exercise 7.3.3 Let γ be a closed loop in $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, and let $r = \min_{z \in \gamma} |z|$. Prove that for $w \in \mathbb{C}$ with $|w| < r$, we have

$$n(\gamma, w) = n(\gamma, 0)$$

Chapter 8

More consequences of the argument principle

8.1 Local behavior of holomorphic functions

On Monday, we established an integral formula for the number of zeroes of a holomorphic function f in a region Ω , in terms of an integral around the boundary of Ω . This formula has a number of consequences, both local and global; here we want to focus on what the formula tells us about the local behavior of holomorphic functions. (For an example of a global consequence, see Rouché's theorem (page 153 in Ahlfors), which we'll get to next lecture.)

To emphasize the local implications of the formula, let's set it up a little differently. Suppose f is holomorphic in a region Ω , $z_0 \in \Omega$ is any point and say $f(z_0) = w_0$. Finally, suppose that z_0 is a zero of order n of the function $f(z) - w_0$.

Choose an $\epsilon > 0$ such that the functions $f(z_0) - w_0$ and $f'(z)$ have no zeroes in the punctured disc $0 < |z - z_0| \leq \epsilon$, and set

$$\delta = \min_{z: |z - z_0| = \epsilon} |f(z) - w_0|.$$

By the argument we gave last time, then, we see that for all w with $|w - w_0| < \delta$, the function $f(z) - w$ also has exactly n zeroes in the disc $|z - z_0| < \epsilon$, counting multiplicity. But the condition that $f'(z) \neq 0$ for all z with $0 <$

$|z - z_0| < \epsilon$ tells us that these zeroes are all simple, so that there must be exactly n of them. In sum, then, we have proved

Proposition 8.1.1 *Let f be a holomorphic function in a neighborhood of z_0 , with value $f(z_0) = w_0$, and assume that the function $f(z) - w_0$ has a zero of order n at z_0 . For all sufficiently small $\epsilon > 0$, there exists $\delta > 0$ such that for all w with $0 < |w - w_0| < \delta$, the equation $f(z) = w$ has exactly n solutions with $|z - z_0| < \epsilon$.*

In other words, a punctured neighborhood of z_0 maps n -to-1 onto a punctured neighborhood of w_0 . Note that this is saying two things: first, that for w near w_0 , there are n solutions to the equation $f(z) = w$ near z_0 counting multiplicity; and second, that for w close to (but not equal to) w_0 , the function $f(z) - w$ has only simple zeroes, so that there are actually n distinct solutions. The latter follows just from the fact that the zeroes of the derivative f' are isolated, so that for ϵ small the derivative f' will have no zeroes in the punctured disc $0 < |z - z_0| < \epsilon$.

Note that in case $\text{ord}_{z_0}(f - w_0) = 1$, this says that f maps a neighborhood of z_0 one-to-one to a neighborhood of w_0 . More generally, it's saying that if $\text{ord}_{z_0}(f - w_0) = n$, then locally, the map given by f looks like the map $z \mapsto z^n$ around the origin. Precisely, we're saying that if we let

$$\Gamma = \{(z, w) : f(z) = w\}$$

be the graph of f , then for $\delta > 0$ small, there exists a neighborhood U of $(z_0, w_0) \in \Gamma$ such that the map

$$\pi_2 : U \setminus \{(z_0, w_0)\} \rightarrow B_\delta(w_0) \setminus \{w_0\}$$

is the connected n -sheeted covering space of the punctured disc $B_\delta(w_0) \setminus \{w_0\}$.

So, if $\text{ord}_a(f - f(a)) = 1$, then the map f is a local homeomorphism: there are neighborhoods U of $a \in \mathbb{C}$ and V of $f(a) \in \mathbb{C}$ such that f maps U homeomorphically to V . The points a such that $\text{ord}_a(f - f(a)) > 1$ are called *ramification points* of the map; note that since these are exactly the zeroes of f' they are isolated. Finally, the images of the ramification points are called the *branch points* of the map.

8.2 The maximum principle

One immediate consequence of the general proposition above is that if f is a nonconstant holomorphic function on $\Omega \subset \mathbb{C}$, then the map $\Omega \rightarrow \mathbb{C}$ given by f is *open*. This means the image of an open subset is open; equivalently, it's enough to check that if p is any point in the domain, the image of any neighborhood of p contains a neighborhood of the image point $f(p)$ in the target, which we have just done.

Once more: even well-behaved $\mathcal{C}^\infty$ maps are not in general open. More typical is a map like $(x, y) \mapsto (x^2, y)$, which folds the plane over along the y -axis; at a point $(0, y)$ on the y -axis, the map is not open. In general, a $\mathcal{C}^\infty$ map $(u, v) = f(x, y)$ from $\mathbb{R}^2$ to $\mathbb{R}^2$ is said to be *smooth* at a point (x, y) if it maps a small neighborhood of (x, y) homeomorphically onto its image in the target $\mathbb{R}^2$; by the inverse function theorem, this will be the case if and only if the Jacobian determinant

$$J(f) = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} \neq 0.$$

If this is not the case, (x, y) is called a *singular point* of f . Typically, a $\mathcal{C}^\infty$ map from $\mathbb{R}^2$ to $\mathbb{R}^2$ will be singular along a curve, while as we've seen with a nonconstant holomorphic map $\Omega \rightarrow \mathbb{C}$ the singular locus will consist of isolated points.

One corollary of the fact that nonconstant holomorphic maps are open is the *maximum principle*, which says that if f is a nonconstant holomorphic function on a region $\Omega \subset \mathbb{C}$, then *the absolute value $|f(z)|$ does not have any local maxima in Ω* ; in other words, given any $z \in \Omega$ and any neighborhood U of z in Ω , there is another value $z' \in U$ such that $|f(z')| > |f(z)|$. This follows directly from the openness of the map: after all, the image of U will contain an open neighborhood of $f(z)$, and any neighborhood of $f(z)$ in $\mathbb{C}$ contains points of absolute value strictly greater than $|f(z)|$.

Here, for fun, is another, more directly analytic argument for the maximum principle. Choose an $r > 0$ such that the circle C of radius r around z

is contained in Ω . We then have by the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta) d\zeta}{\zeta - z}.$$

We can parametrize C by $\theta \mapsto z + re^{i\theta}$ for $0 \leq \theta \leq 2\pi$, upon which the integral reduces to

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) d\theta;$$

in other words, *the value of f at z is the average of its values on a circle of radius r around z* . Thus the only way we could have $|f(z)| \geq |f(z + re^{i\theta})|$ for all θ would be if we had equality throughout; if that's the case for all sufficiently small r it would follow that f is a constant.

8.3 Non-local applications

So far we've been focussed on applications of the Cauchy formula to the local geometry of a holomorphic map. We're now going to shift gears and consider “global” consequences—corollaries that involve, either as hypotheses or conclusions, the behavior of a function throughout its domain of definition.

8.3.1 Schwarz' lemma

As an example of the application of the maximum principle, we have

Theorem 8.3.1 (Schwarz' Lemma) *Let $\Delta = \{z \in \mathbb{C} : |z| < 1\}$ be the unit disc in the complex plane, and suppose that $f : \Delta \rightarrow \Delta$ is a holomorphic map (that is, f is a holomorphic function on Δ such that $|f(z)| < 1$ for all $z \in \Delta$). Suppose in addition that $f(0) = 0$. Then we must have*

$$|f(z)| \leq |z| \text{ for all } z \in \Delta; \quad \text{and} \quad |f'(0)| \leq 1.$$

Moreover, if equality holds in the second inequality or in the first for any $z \neq 0$, then we must have $f(z) = cz$ with $c \in \mathbb{C}$ of modulus 1.

Proof. The proof is almost shorter than the statement! Briefly, since $f(0) = 0$, the function

$$g(z) := \frac{f(z)}{z}$$

has a removable singularity at the origin, and so extends to a holomorphic function in all of Δ with $g(0) = f'(0)$. By hypothesis, $|g(z)| \leq 1/r$ on the circle $C_r = \{z : |z| = r\}$, and so by the maximum principle we see that $|g(z)| \leq 1/r$ for all z with $|z| \leq r$; letting $r \rightarrow 1$, we arrive at $|g(z)| \leq 1$ for all $z \in \Delta$, which is the first conclusion. Finally, if $|g(z)| = 1$ for any $z \in \Delta$, then by the maximum principle g must be constant; this is the second statement.

There are three remarks to be made here. First, the statement of Schwarz's lemma might not strike you as remarkable. To see how truly weird it is, though, consider the following equivalent formulation:

Theorem 8.3.2 *Let f be a holomorphic function on the unit disc $\Delta = \{z : |z| < 1\}$, with $f(0) = 0$. If $|f'(0)| > 1$, then $\exists z \in \Delta$ with $|f(z)| > 1$.*

I mean, come on! If a beginning calculus student asserted something like that about differentiable functions on $(-1, 1) \subset \mathbb{R}$, you'd flunk them, and rightly.

Second, Schwarz' lemma might strike you as hyperspecialized, and unlikely to arise often in practice. As Ahlfors points out, though, it can be applied much more broadly by normalizing. For example, if instead of requiring $f(0) = 0$, suppose $f(0) = w \in \Delta$. We can then normalize by composing f with a linear fractional transformation $S : \Delta \rightarrow \Delta$ that carries w to 0; applying Schwarz to the composite $S \circ f$, we arrive at a version of Schwarz in this more general case.

More generally still, we will see in Section 1 of Chapter 6 (well, we won't get there in this course, but it's actually pretty readable if you want to take a look) that given any simply connected region $\Omega \subsetneq \mathbb{C}$, there is a one-to-one holomorphic map $g : \Omega \rightarrow \Delta$ —in other words, any two simply connected regions in the plane are conformally equivalent. Given any holomorphic map $f : \Omega \rightarrow \Omega$, then, we can conjugate by g to arrive at a holomorphic map $g \circ f \circ g^{-1} : \Delta \rightarrow \Delta$; we can apply Schwarz to that map and derive a Schwarz inequality for holomorphic maps $\Omega \rightarrow \Omega$.

A third point, which we will not pursue here but which underlies much of the ubiquity of Schwarz' lemma, involves distances. To set this up, we can

introduce a metric on Δ in which the length of an arc γ in Δ is

$$\int_{\gamma} \frac{|dz|}{1 - |z|^2}$$

(in particular, this means the points on the boundary are infinitely far away!). This is called the *Poincaré metric*, or the *hyperbolic metric* on Δ ; it has the property that every holomorphic bijection $f : \Delta \rightarrow \Delta$ preserves distances in the hyperbolic metric. More generally, as we said in the last paragraph, every simply connected region $\Omega \subsetneq \mathbb{C}$ is conformally equivalent to Δ ; if we port over the Poincaré metric from Δ , we get what is also called the Poincaré metric on Ω .¹

We're not going to get into metrics here. But in these terms, there is a very simple interpretation of the Schwarz lemma: it says that *any holomorphic map $f : \Delta \rightarrow \Delta$ is a contraction*, meaning that it is distance-decreasing. And, of course, there's a corresponding statement for an arbitrary simply connected region $\Omega \subsetneq \mathbb{C}$: in terms of the Poincaré metric on Ω , it says that *any holomorphic map $f : \Omega \rightarrow \Omega$ is a contraction*. In this form, the Schwarz lemma has extensions to other Riemann surfaces, and to complex manifolds of higher dimensions; it truly is surprisingly ubiquitous.

You may have noticed we're writing $\Omega \subsetneq \mathbb{C}$ rather than $\Omega \subset \mathbb{C}$. There's a reason for that: alone among simply connected open subsets of $\mathbb{C}$, the complex plane $\mathbb{C}$ itself is *not* conformally equivalent to the disc Δ .

Exercise 8.3.3 Prove that there does not exist a holomorphic bijection $f : \Delta \rightarrow \mathbb{C}$.

8.3.2 Rouché's theorem

Our next theorem is another application of the argument principle. We've already seen this in one setting: if f is holomorphic in a simply connected region $\Omega \subset \mathbb{C}$, and if $\gamma : (a, b) \rightarrow \Omega$ is a simple closed curve in Ω bounding a region Δ such that f has no zeroes on γ , then the number of zeroes of f in

¹Of course, there are many different conformal bijections from Ω to Δ , but by the last remark they all give rise to the same Poincaré metric on Ω .

Δ is given by an integral:

$$\sum_{a \in \Delta} \text{ord}_a(f) = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz.$$

In fact, the same formula applies verbatim if f is *meromorphic*, as long as it has no zeroes or poles along γ . What's more, the same proof works: if the zeroes of f in Δ are $a_1, \dots, a_k$, with a_α a zero of order m_α ; and the poles of f in Δ are $b_1, \dots, b_l$, with b_α a pole of order n_α , we can write

$$f(z) = \frac{(z - a_1)^{m_1} \dots (z - a_k)^{m_k}}{(z - b_1)^{n_1} \dots (z - b_l)^{n_l}} g(z),$$

where g is holomorphic and nowhere zero in Δ . The same argument as before then yields

$$\int_{\gamma} \frac{f'(z)}{f(z)} dz = 2\pi i \left(\sum m_\alpha - \sum n_\beta \right).$$

There was also another interpretation of this argument: if we think of f as giving a map $\Omega \rightarrow \mathbb{C}$, and we let $\Gamma = f \circ \gamma$ be the image in $\mathbb{C}$ of the arc γ , we can interpret the integral

$$\int_{\gamma} \frac{f'(z)}{f(z)} dz = \int_{\Gamma} \frac{dw}{w}$$

so we conclude that the *number of zeroes of f in Δ minus the number of poles of f in Δ is the winding number of Γ around 0*. In this form, we have as an almost immediate corollary

Theorem 8.3.4 (Rouché's theorem) *Let Ω be a simply connected region in $\mathbb{C}$, γ a simple closed curve in Ω and $\Delta \subset \Omega$ the region enclosed by γ . If f and g are holomorphic functions on Ω satisfying the inequality*

$$|f(z) - g(z)| < |f(z)|$$

for all z on γ , then f and g have the same number of zeroes in Δ .

(Note that the hypotheses imply that neither f nor g have any zeroes on γ .)

Proof This follows by applying the argument principle to the meromorphic function $h(z) = g(z)/f(z)$. The point is, the hypotheses of the theorem tell us that

$$\left| \frac{g(z)}{f(z)} - 1 \right| < 1$$

on γ ; or in other words, the image Γ of γ under the map h lies entirely in the open disc $B_1(1)$ of radius 1 around the point $w = 1$. But this disc does not include 0, and so any closed loop contained in $B_1(1)$ has winding number $n(\Gamma, 0) = 0$ with respect to 0. Applying the argument principle, we see that h must have the same number of zeroes and poles in Δ .

Rouché's theorem is often used to locate the zeroes of a holomorphic function. To use the example in Ahlfors, consider for example the polynomial

$$g(z) = z^7 - 2z^5 + 6z^3 - z + 1.$$

We know from the fundamental theorem of algebra that g has 7 roots, but can we say roughly where those roots are located? For example, how many of the roots are inside the unit disc?

Rouché gives us the answer. To apply it, we set $f(z) = 6z^3$, so that $f(z) - g(z) = -z^7 + 2z^5 + z - 1$. We have then $|f(z)| = 6$ for all z on the unit circle $|z| = 1$, while $|f(z) - g(z)| \leq 5$ when $|z| = 1$. Rouché then tells us that g has exactly three zeroes in the unit disc.

Chapter 9

Residue calculus

9.1 Residues

Up to now, we've been using the Cauchy integral formula to give a representation of a holomorphic function f as a path integral, which we could manipulate to prove results about the behavior of f . We haven't actually developed any new techniques for evaluating integrals.

That is about to change. Our last application of the Cauchy theorem will be to use it to evaluate definite integrals, including many that could not be evaluated with standard calculus techniques. The body of techniques we'll describe here is broadly called the *residue calculus*.

To set it up, suppose that we want to evaluate a path integral $\int_{\gamma} f(z)dz$ around a simple closed curve γ in a region $\Omega \subset \mathbb{C}$ in the complex plane. (Of course, most definite integrals are not a priori of this form; we'll see in what follows a number of techniques for converting an ordinary integral $\int_a^b f(x)dx$ into one of this form.) In the simplest setting, if f is holomorphic in Ω then Cauchy's theorem tells us right off the bat that $\int_{\gamma} f(z)dz = 0$. We'll consider here the next simplest case, where f may have a finite number of singular points in Ω ; in other words, f is defined and holomorphic on $\Omega \setminus \{a_1, \dots, a_k\}$.

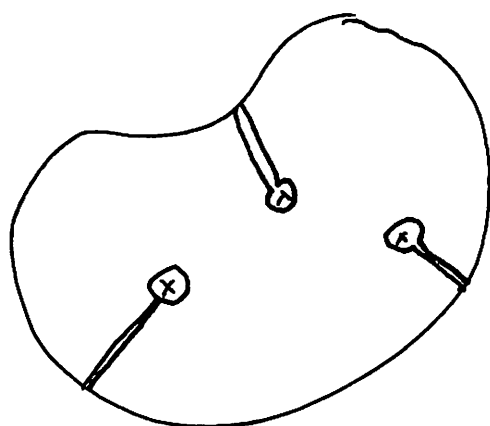
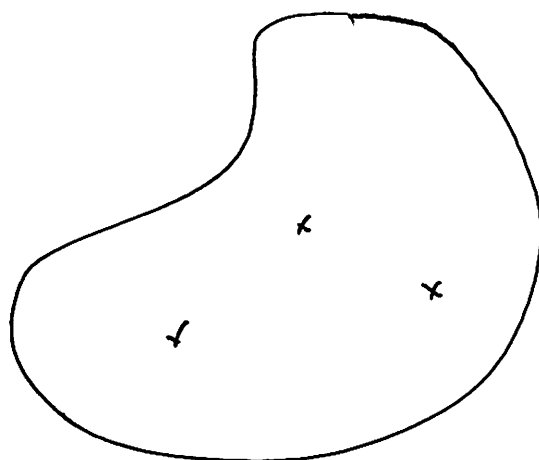
Let's first consider f in a neighborhood of each of the singular points. Suppose in general that we have a function f holomorphic in a punctured

disc; that is, f is defined and holomorphic on $\{z : 0 < |z - a| < r\}$ for some $r > 0$. Let ϵ be any number with $0 < \epsilon < r$, and let C_ϵ be the circle of radius ϵ around a . We define the *residue* of f at a to be the integral

$$\operatorname{Res}_a(f) := \frac{1}{2\pi i} \int_{C_\epsilon} f(z) dz,$$

noting that this is independent of $\epsilon < r$, since f is holomorphic in the punctured disc $\{z : 0 < |z - a| < r\}$.

Given this, we can go back to our original integral $\int_\gamma f(z) dz$, and replace the path γ of integration with a path γ' that excises the singularities, the way we did when we were proving tweaked Cauchy: for each $\alpha = 1, \dots, k$, we stop at one point in the path, take a left turn and head for a_α ; when we get to within a distance ϵ_α of a we go around the circle C_α of radius ϵ_α around a_α and then head back to our original path, as in the picture on the next page.



Since the new path γ' encloses a region in which f is everywhere holomorphic, we have

$$\int_{\gamma'} f(z)dz = 0,$$

and we may conclude that

$$\int_{\gamma} f(z)dz = \sum_{\alpha=1}^k \int_{C_{\alpha}} f(z)dz = 2\pi i \sum_{\alpha=1}^k \text{Res}_{a_{\alpha}}(f).$$

In other words, we've reduced the problem of evaluating the original integral to one of locating the singularities of f and finding the residue at each one. Of course, this leaves two questions: how do we calculate the residues in practice, and how do we convert an ordinary integral into a path integral around a closed loop? Happily, the answer to the first question is completely straightforward and algorithmic, and we'll describe that next. The answer to the second question is more ad hoc; we'll illustrate by example various techniques that can be used.

9.2 Calculating residues

This is a somewhat different question in the two cases of a pole and an essential singularity, the former being as you might expect the easier. So: to start, let's assume that f is defined and holomorphic in the punctured disc $\{z : 0 < |z - a| < r\}$, and that f has a pole at a , of order h .

In this case, we have seen that we can write

$$f(z) = a_{-h}(z - a)^{-h} + \cdots + a_{-1}(z - a)^{-1} + j(z),$$

with $j(z)$ holomorphic. We can integrate term by term, and all the terms except one are immediately seen to be 0. The one term that does contribute is the term $a_{-1}(z - a)^{-1}$, whose integral around the circle C_{ϵ} we have seen is $2\pi i \cdot a_{-1}$; we have accordingly

$$\text{Res}_a(f) = a_{-1}.$$

OK, then, how do we find a_{-1} ? The ease of the answer depends on the order h of the pole. By far the easiest case is that of a simple pole ($h = 1$), in which the function $g(z) = (z - a)f(z)$ has a removable singularity at a ; in other words, g extends to a holomorphic function at a , and we have simply

$$\operatorname{Res}_a(f) = g(a).$$

To illustrate, let's use Ahlfors' example: take

$$f(z) = \frac{e^z}{(z - a)(z - b)}$$

with $a \neq b$, so f will have simple poles at a and b . To find the residue at a , then, we consider the function

$$g(z) = (z - a)f(z) = \frac{e^z}{z - b}.$$

We have then

$$\operatorname{Res}_a(f) = g(a) = \frac{e^a}{a - b}.$$

What about the case of multiple poles? This is slightly harder, but we can use the same idea: if $\operatorname{ord}_a(f) = -h$, we can write as before

$$f(z) = a_{-h}(z - a)^{-h} + \cdots + a_{-1}(z - a)^{-1} + j(z),$$

and this time we consider the function $g(z) = (z - a)^h f(z)$, which is holomorphic. Now we can write

$$g(z) = a_{-h} + a_{-h+1}(z - a) + \cdots + a_{-1}(z - a)^{h-1} + j(z)(z - a)^h$$

Now, to isolate the relevant term a_{-1} , we take the $(h - 1)$ st derivative on both sides and evaluate at a ; we arrive at

$$\operatorname{Res}_a(f) = \frac{1}{(h - 1)!} g^{(h-1)}(a).$$

To illustrate, we can use a variant of the previous example; we'll take

$$f(z) = \frac{e^z}{(z - a)^2}$$

Now $g(z) = (z - a)^2 f(z) = e^z$, so $g'(z) = e^z$ and we have

$$\operatorname{Res}_a(f) = g'(a) = e^a.$$

What about the case of essential singularities? The situation there is messier, in part because *essential singularities need not be isolated singularities*: for example, the function

$$f(z) = \frac{1}{e^{1/z} - 1}$$

has poles at all the points $1/2\pi ni$ with n a nonzero integer, and of course an essential singularity at $z = 0$. In this situation, the residue $\operatorname{Res}_0(f)$ is not even defined (the definition of the residue $\operatorname{Res}_a(f)$ required that f be holomorphic in a punctured disc around a).

Nonetheless, if we have a function f with an essential singularity that *is* isolated, we can define the residue as before, and we even have a tool for calculating it: we'll see in Chapter 5, Section 1.3 that a function f holomorphic in a punctured disc $0 < |z - a| < R$ has a *Laurent series expansion*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - a)^n.$$

In this case, we can again integrate term by term to arrive at

$$\operatorname{Res}_a(f) = a_{-1}.$$

In the next lecture, we'll see how to use this set-up to evaluate definite integrals explicitly.

Chapter 10

Evaluating definite integrals

As we said, the second issue with applying residue calculus is converting a given definite integral into a path integral over a closed loop. Here there is no algorithm, or even broadly applicable technique; rather, there is a grab-bag of tricks that we can use on an ad-hoc basis. We will illustrate some of these techniques by working out examples.

10.1 Trigonometric/exponential integrals

Here we are integrating a rational function $R(\sin z, \cos z)$, or equivalently a rational function $S(e^{iz})$, over a period of the function; as an example, consider

$$\int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$$

where $a > 1$. Here a natural thing to do would be to make the substitution $z = e^{i\theta}$, which converts the integral over $[0, 2\pi]$ into an integral over the unit circle $|z| = 1$. We have then

$$d\theta = \frac{dz}{iz} \quad \text{and} \quad a + \cos \theta = \frac{z + 2a + z^{-1}}{2}$$

so we arrive at

$$\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} = \frac{2}{i} \int_{|z|=1} \frac{dz}{z^2 + 2az + 1}.$$

Now we can get to work. The integrand has simple poles at the (two distinct) roots of $z^2 + 2az + 1$, which are located at the points $-a \pm \sqrt{a^2 - 1}$. Of these, exactly one ($z = -a + \sqrt{a^2 - 1}$) is located inside the unit circle. To find the residue at this point, we set

$$g(z) = (z + a - \sqrt{a^2 - 1}) \frac{1}{z^2 + 2az + 1} = \frac{1}{z + a + \sqrt{a^2 - 1}};$$

evaluating this at $z = -a + \sqrt{a^2 - 1}$, we get that

$$\text{Res}_{-a+\sqrt{a^2-1}} \left(\frac{1}{z^2 + 2az + 1} \right) = \frac{1}{2\sqrt{a^2 - 1}}$$

and so the value of the original integral is

$$\frac{2}{i} \cdot 2\pi i \cdot \frac{1}{2\sqrt{a^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - 1}}$$

10.2 Rational functions

We already know how to integrate rational functions; the method of partial fractions (which is actually simpler in the complex setting than the real!) will always work. But to illustrate a technique for converting an integral over the real line into a path integral over a closed loop, let's consider integrals of the form

$$\int_{-\infty}^{\infty} R(z) dz$$

where $R(z)$ is a rational function. Note that in order for the integral to converge, the degree of the denominator has to be at least 2 greater than the degree of the numerator (and of course R can't have any poles along the real axis).

The trick here is simple: by definition,

$$\int_{-\infty}^{\infty} R(z) dz = \lim_{\rho \rightarrow \infty} \int_{-\rho}^{\rho} R(z) dz.$$

Now we can complete the line segment from $-\rho$ to ρ to a closed curve in $\mathbb{C}$ by adding on the semicircle C_ρ of radius ρ in the upper half plane. We have then

$$\int_{-\rho}^{\rho} R(z) dz + \int_{C_\rho} R(z) dz = 2\pi i \sum_a \text{Res}_a(R(z)),$$

where the sum is taken over all poles of $R(z)$ lying in the upper half disc $\{z : |z| < \rho \text{ and } \text{Im}(z) > 0\}$.

Now, by the hypothesis that the degree of the denominator of R is at least two greater than the degree of the numerator, the integral

$$\int_{C_\rho} R(z) dz \rightarrow 0$$

as $\rho \rightarrow \infty$: we have, for large ρ and a suitable constant C , that $|R(z)| < C/\rho^2$, while the length of the path of integration is $\pi\rho$. Taking the limit, we arrive at

$$\int_{-\infty}^{\infty} R(x) dx = 2\pi i \sum_a \text{Res}_a(R(z)),$$

where the sum is over all poles of R in the upper half plane.

10.3 Mixed rational and exponential functions

As we said, we already knew how to evaluate definite integrals of rational functions. But our new methods allow us to go further and evaluate integrals of the form

$$\int_{-\infty}^{\infty} R(z) e^{iz} dz,$$

with R again a rational function with denominator of degree at least 2 larger than the numerator. For example, let's do

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^2} dz.$$

The one part of the argument above that is problematic in this case is the fact that the integral over the semicircle C_ρ goes to 0 as $\rho \rightarrow \infty$. But this follows as before, given the fact that $|e^{iz}| < 1$ whenever $\text{Im}(z) > 0$. In this case, the function $\frac{e^{iz}}{1+z^2}$ has two simple poles, at $\pm i$; the residue at i is given as $g(i)$, where

$$g(z) = (z - i) \frac{e^{iz}}{1 + z^2} = \frac{e^{iz}}{z + i}.$$

and we conclude that

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^2} dz = 2\pi i \cdot \text{Res}_i\left(\frac{e^{iz}}{1+z^2}\right) = \frac{\pi}{e}.$$

(If you're wondering why the integral of a complex-valued function is a real number, the answer is simple: we can write

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^2} dz = \int_{-\infty}^{\infty} \frac{\cos z}{1+z^2} dz + i \int_{-\infty}^{\infty} \frac{\sin z}{1+z^2} dz$$

and the last integral is 0 by symmetry.)

10.4 Rational functions of degree -1

There is a variant of this method that allows us to extend the technique to cases where the denominator of the rational function R has degree only one greater than the degree of the numerator. In this case, the periodicity of the function e^{iz} on the real line ensures that the integral $\int_0^\infty e^{iz} R(z) dz$ converges, though not absolutely: it's like $\sum_1^\infty (-1)^n \frac{1}{n}$ converges, even though $\sum_1^\infty \frac{1}{n}$ does not.

There is, however, one thing that definitely does go wrong in this case: it's not true that the integral $\int_{C_\rho} e^{iz} R(z) dz$ goes to 0 as $\rho \rightarrow \infty$; so we have to find a different way of completing the integral to a path integral around a closed curve.

The solution here is simple: instead of using a semicircle, we consider the path integral around the perimeter of a rectangle. Specifically, let $\Gamma_{X_1, X_2, Y}$ be the rectangle of height Y whose bottom edge is the interval $[-X_1, X_2]$ in the real axis. We can then estimate the value of the integral of $e^{iz} R(z) dz$ along the top and vertical sides of the rectangle, and see that they go to 0 as Y goes to ∞ and then $X_1, X_2 \rightarrow \infty$. The conclusion is that

$$\begin{aligned} \int_{-\infty}^{\infty} e^{iz} R(z) dz &= \lim_{X_1, X_2 \rightarrow \infty} \lim_{Y \rightarrow \infty} \int_{\Gamma_{X_1, X_2, Y}} e^{iz} R(z) dz \\ &= 2\pi i \sum_a \text{Res}_a(e^{iz} R(z)), \end{aligned}$$

where the sum is again over all poles of R in the upper half plane.

10.5 Principal values

We have one more way to extend the range of these techniques, which involves the notion of *Cauchy principal value*. The idea is straightforward: up to now, we've always assumed that the function we're integrating does not have any singularities along the path of integration. But in fact, there is a way of dealing with the case where it does.

To set this up, suppose we have a function $f(z)$ with a simple pole at z_0 with residue B , so that we can write

$$f(z) = \frac{B}{z - z_0} + g(z)$$

with g holomorphic. If C_δ denotes the circle of radius δ around z_0 , we know that

$$\int_{C_\delta} f(z)dz = \int_{C_\delta} \frac{Bdz}{z - z_0} + \int_{C_\delta} g(z)dz = 2\pi iB$$

where the second equality follows from Cauchy's theorem, which says that the integral $\int_{C_\delta} g(z)dz = 0$.

Now we ask: suppose that instead of integrating all the way around C_δ , we only go a fraction α of the way around: in other words, we integrate along the arc $\gamma_{\delta,\alpha}$ given by

$$\begin{aligned} \gamma_{\delta,\alpha} : [\theta_0, \theta_0 + 2\pi\alpha] &\rightarrow \mathbb{C} \\ \theta &\mapsto z_0 + \delta e^{i\theta}. \end{aligned}$$

Of course, it's no longer true that the integral of $g(z)dz$ along this arc is 0. But it is in the limit, as $\delta \rightarrow 0$! We thus have

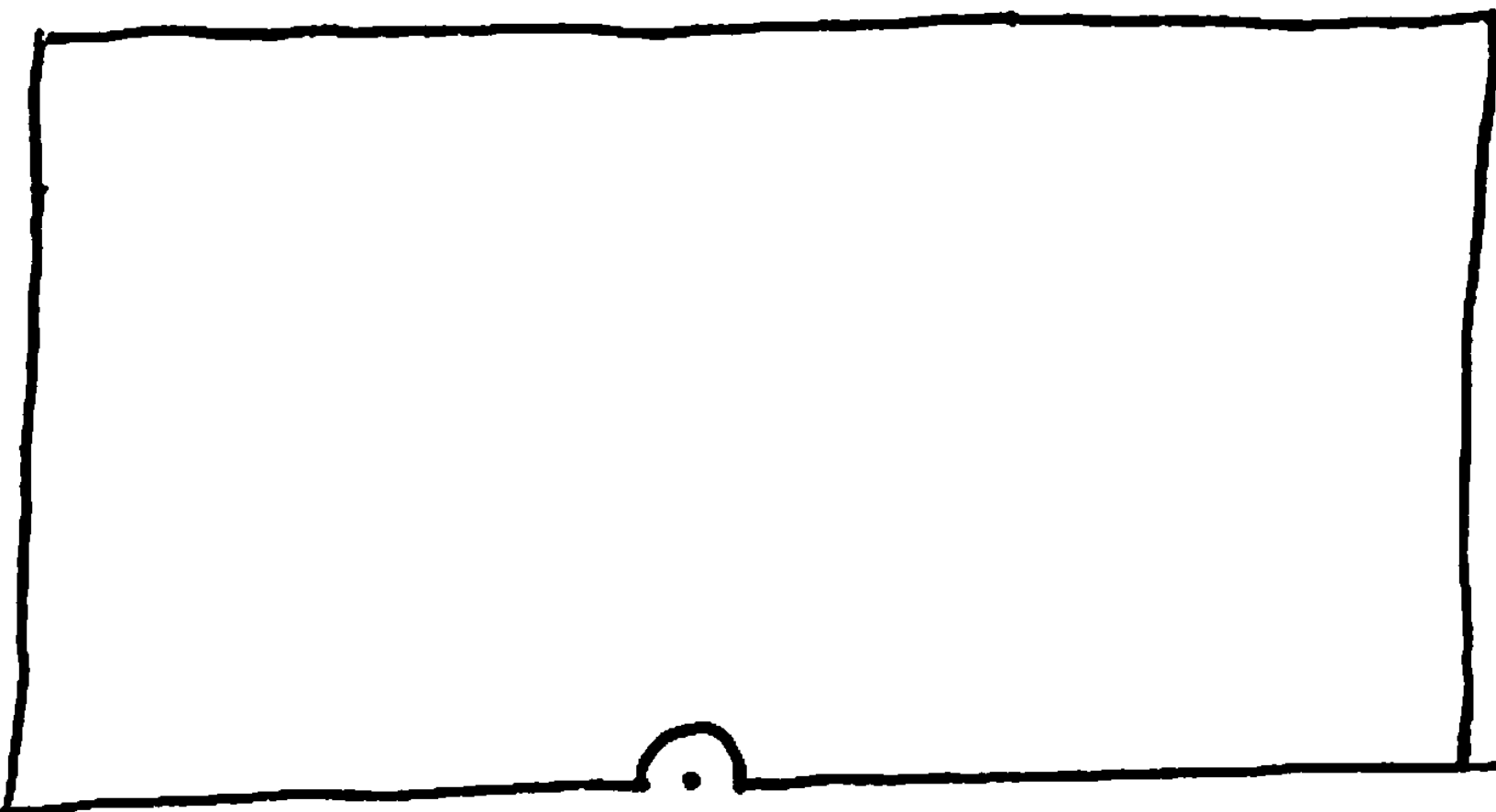
$$\lim_{\delta \rightarrow 0} \int_{\gamma_{\delta,\alpha}} f(z)dz = \lim_{\delta \rightarrow 0} \int_{\gamma_{\delta,\alpha}} \frac{Bdz}{z - z_0} = 2\pi iB\alpha$$

where in fact the second limit is unnecessary, since by direct calculation the second integral doesn't depend on δ .

To see how this can be applied, let's consider the integral

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \operatorname{Im} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx.$$

The trick here is to modify our path slightly to avoid going through the origin. Specifically, when we get the point $-\delta$ on the real axis we take a detour, and go around the upper half of the circle C_δ to the point $+\delta$; we then resume moving rightward along the real axis. (Ahlfors, for reasons I can't see, chooses to go around the lower half of the circle C_δ ; as far as I can tell you wind up in the same place.) Adding the other three sides of the rectangle $\Gamma_{X_1, X_2, Y}$ as above, we arrive at a closed path of integration, as pictured on the next page.



Now, the integrand $\frac{e^{iz}}{z}$ is holomorphic everywhere in the region enclosed by this path, and so Cauchy's theorem tells us immediately that the value of the integral is 0. At the same time, the integral over the other three sides of the rectangle $\Gamma_{X_1, X_2, Y}$ go to 0 as before. Letting C'_δ denote the upper half of the circle C_δ around the origin, then, we arrive at

$$\begin{aligned} \lim_{\delta \rightarrow 0} \left(\int_{-\infty}^{-\delta} \frac{e^{iz}}{z} dz + \int_{\delta}^{\infty} \frac{e^{iz}}{z} dz \right) &= \lim_{\delta \rightarrow 0} \left(- \int_{C'_\delta} \frac{e^{iz}}{z} dz \right) \\ &= \frac{1}{2} \cdot 2\pi i \cdot \text{Res}_0\left(\frac{e^{iz}}{z}\right) \\ &= \pi i. \end{aligned}$$

(The extra minus sign is because we are going around C'_δ in the clockwise direction, rather than the counterclockwise.)

Taking the imaginary part, we arrive at the conclusion

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi.$$

10.6 Non-integral powers of z

Our last example is the cutest. It involves a rational function $R(z)$ times a non-integer power of z , which may seem like a problem: as I may have mentioned once or twice, non-integer powers of z are not well-defined functions on $\mathbb{C}$. But, as it turns out, this is not a bug, it's a feature! It's exactly the multi-valued nature of z^α for $\alpha \notin \mathbb{Z}$ that makes it possible to evaluate these integrals by residues.

By way of an example, let's consider the integral

$$\int_0^\infty \frac{z^\alpha dz}{1+z^2}$$

where $\alpha \in (0, 1)$ is a real number strictly between 0 and 1, and z^α means the real value. (The condition $\alpha < 1$ ensures that the integral converges.) Note that if $\alpha = m/n$ is rational, we could do this by using the substitution $z = w^n$ to convert the integral to a rational function of w (though of course

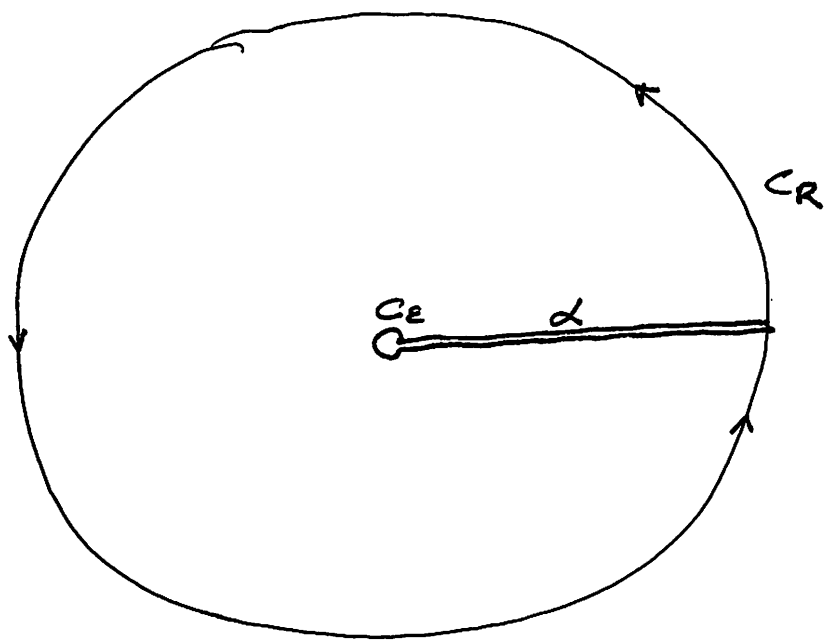
if n is large this could get complicated). The approach we'll describe here, by contrast, works equally well for any $\alpha \in (0, 1)$

The trick here is to use what's called a *keyhole integral*: we want to integrate over a closed loop Γ_R as pictured on the next page, in which you go out along the real axis from $1/R$ to R , then around the circle C_R of radius R , then back along the real axis from R back to $1/R$ and finally around the circle $C_{1/R}$ of radius $1/R$. Note that since the region Ω enclosed by our path is simply connected and does not include the origin, there exists a well-defined branch of $\frac{z^\alpha}{1+z^2}$, which we will take to be our function f . In fact, we can be explicit: the value of $z^\alpha = e^{\alpha \log z}$ we'll take will be the one corresponding to the value of $\log z$ with imaginary part between 0 and 2π .

We have then

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z) dz = 2\pi i \sum_{a \in \mathbb{C}} \text{Res}_a(f),$$

and we presumably know how to evaluate the right hand side.



But what about the left hand side? The integral has four parts:

$$\int_{\Gamma_R} f(z)dz = \int_{\alpha} f(z)dz + \int_{C_R} f(z)dz + \int_{\alpha^{-1}} f(z)dz + \int_{C_{1/R}} f(z)dz$$

and the two parts $\int_{C_R}$ and $\int_{C_{1/R}}$ can readily be seen to go to 0 as $R \rightarrow \infty$. As for the remaining two parts, it may seem at first that they will cancel, and we will be left with a value of 0 for the total, which would be of no use to us. But here we are saved exactly by the fact that z^α is not well-defined on the whole plane: *the values of f as we come back in along the real axis along α^{-1} are not the same as they were when we went out along α* ; rather, they are multiplied by $e^{2\pi i\alpha}$. In sum, then, we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z)dz = (1 - e^{2\pi i\alpha}) \int_0^\infty \frac{z^\alpha dz}{1 + z^2}.$$

It remains to evaluate the residues of $f(z)$ at the two poles $\pm i$. Since these are simple poles, it's pretty straightforward, as long as we're careful about assigning the right value to z^α at the poles. For example, the pole at $+i$ has residue the value at i of the function

$$g(z) = (z - i)f(z) = \frac{z^\alpha}{z + i}.$$

Now here by z^α we mean $e^{\alpha \log z}$, where the value of $\log z$ at $z = i$ is $\pi i/2$; thus the residue at i is

$$\text{Res}_i(f) = \frac{e^{\pi i\alpha/2}}{2i}.$$

Similarly, continuing counterclockwise around the unit circle, we arrive at $-i$ with the value $\log(-i) = 3\pi i/2$; we get correspondingly that

$$\text{Res}_{-i}(f) = \frac{e^{3\pi i\alpha/2}}{-2i}.$$

The sum of the residues, times $2\pi i$, is thus

$$2\pi i \sum_{a \in \mathbb{C}} \text{Res}_a(f) = \pi (e^{\pi i\alpha/2} - e^{3\pi i\alpha/2}),$$

and we conclude

$$\int_0^\infty \frac{z^\alpha dz}{1 + z^2} = \pi \frac{e^{\pi i\alpha/2} - e^{3\pi i\alpha/2}}{1 - e^{2\pi i\alpha}}$$

We can simplify this a bit: if we set $w = e^{\pi i \alpha / 2}$, this becomes

$$\pi \frac{w - w^3}{1 - w^4},$$

and cancelling the factor of $1 - w^2$ in both numerator and denominator, we arrive at

$$\int_0^\infty \frac{z^\alpha dz}{1 + z^2} = \pi \frac{w}{1 + w^2}.$$

I'll leave it to you to check that this is always real!

Chapter 11

Lecture 30: Power series expansions

Today we want to start Chapter 5, on series and product developments, by describing the power series and Laurent series expansions of holomorphic functions.

11.1 Recall on uniform convergence

Before we get underway, I want to recall the basic theorem from the chapter in Rudin we read: that if f_n is a sequence of differentiable functions converging *uniformly* to a function f , then f is differentiable and $\lim_{n \rightarrow \infty} f'_n(z) = f'(z)$. As an immediate consequence, we see that if f_n is holomorphic for all n , then f is as well. As we've already observed, this means in particular that a power series $\sum a_n z^n$ defines a holomorphic function within its radius of convergence. We will also apply this to infinite sums of other functions, and to infinite products as well.

There is one tweak we should mention: since the conclusion of the theorem is local, we can weaken the hypotheses slightly to require only that the convergence is *uniform on compact sets*—for example, if the domain of the functions f_n is the entire complex plane, it's sufficient to check uniformity on discs $|z| \leq R$.

Finally, it's also true that if $f_n \rightarrow f$ uniformly, then the integrals $\int_a^b f_n(z)dz$ converge to $\int_a^b f(z)dz$, which brings us to our weekly “that can't be right!” theorem:

Theorem 11.1.1 *Suppose f_n is a sequence of holomorphic functions on a domain Ω , converging uniformly on compact sets to f . If the functions f_n are nonzero in Ω , then either f is nonzero or f is identically zero.*

As always, we'll take a moment and point out how absurd this would be if we applied it to ordinary $\mathcal{C}^\infty$ functions: for example, the functions $f_n(x) = x^2 + \frac{1}{n}$ have no zeroes on $\mathbb{R}$, but the uniform limit $f(x) = x^2$ does.

Proof. Suppose $f(z)$ was not identically zero, but that $f(a) = 0$ for some $a \in \Omega$; say the order of f at a is m . We know that a is an isolated zero of f ; let Δ_ϵ be a disc around a in which f has no other zeroes, and let C_ϵ be the boundary of Δ_ϵ . Since f_n has no zeroes, we have

$$\int_{C_\epsilon} \frac{f'_n(\zeta)d\zeta}{f_n(\zeta)} = 0;$$

but by what we just said,

$$\int_{C_\epsilon} \frac{f'(\zeta)d\zeta}{f(\zeta)} = 2\pi im;$$

a contradiction.

11.2 Taylor series

Let's start by recalling the basic set-up. We have a function f holomorphic in a region Ω , and a point $a \in \Omega$. Our goal will be to show that, in some disc of positive radius around a in Ω , the function f is equal to a power series

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n.$$

We start with the basic construction of Ahlfors, page 125. We start by repeatedly invoking the fact that *if a holomorphic function in Ω has a zero at a , it is divisible by $z - a$* , in the sense that the ratio $f(z)/(z - a)$ has a

removable singularity at a ; that is, it is defined and holomorphic throughout Ω . We apply this first to the function $f(z) - f(a)$, which by this observation we can write as

$$f(z) - f(a) = (z - a)f_1(z).$$

In other words, we have

$$f(z) = f(a) + (z - a)f_1(z);$$

taking the derivative and evaluating at a we see also that $f_1(a) = f'(a)$.

Now we apply the same logic to the new function f_1 , and we see that we have

$$f(z) = f(a) + f'(a)(z - a) + (z - a)^2 f_2(z)$$

with $f_2(z)$ holomorphic in Ω ; doing this a total on n times we arrive at the expression

$$f(z) = f(a) + f'(a)(z - a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(z - a)^{n-1} + (z - a)^n f_n(z).$$

This is beginning to look like the Taylor series expansion of f around a . But there is one ingredient missing: if we want to use this to say that $f(z)$ is equal to the power series

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n$$

in a region Δ we have to show that for every $z \in \Delta$ the “remainder term” $(z - a)^n f_n(z)$ goes to 0 as $n \rightarrow \infty$.

The way to do this is to represent the remainder term as an integral. Specifically, suppose C is a circle, centered at the point a and entirely contained in Ω ; let Δ be the disc enclosed by C . Then Ahlfors derives the expression for f_n

$$f_n(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta) d\zeta}{(\zeta - a)^n (\zeta - z)}$$

which is valid for $z \in \Delta$ (see (29) on page 126).

Let's let R be the radius of the circle C , and let M denote the maximum of $|f(z)|$ for $z \in C$. Looking at the terms in the integral above, we see that $|\zeta - z| \geq R - |z - a|$ by the triangle inequality, and so

$$|f_n(z)| \leq R \frac{M}{R^n(R - |z - a|)}.$$

Thus, the remainder term $(z - a)^n f_n(z)$ has absolute value

$$|(z - a)^n f_n(z)| \leq \left(\frac{|z - a|}{R} \right)^n \frac{MR}{R - |z - a|}$$

and since $|z - a| < R$, this does indeed go to 0 as $n \rightarrow \infty$.

11.3 Operations on power series

We've said most of these before, but I wanted to record here the operations we can carry out explicitly on power series:

1. Addition: since power series converge absolutely within their radius of convergence, we can add power series term by term:

$$\sum a_n z^n + \sum b_n z^n = \sum (a_n + b_n) z^n$$

2. Multiplication: we have

$$\left(\sum a_n z^n \right) \left(\sum b_n z^n \right) = \sum_n (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0) z^n$$

3. Reciprocals: if $\sum a_n z^n$ is a power series with $a_0 \neq 0$, we can find the inverse power series by solving the equation

$$\left(\sum a_n z^n \right) \left(\sum b_n z^n \right) = 1$$

successively for $n = 0, 1, 2, \dots$: we have

$$\begin{aligned} b_0 &= \frac{1}{a_0} \\ b_1 &= \frac{-a_1 b_0}{a_0} \\ b_2 &= \frac{-a_2 b_0 - a_1 b_1}{a_0} \end{aligned}$$

and so on.

4. Composition: given two power series, say $\sum a_n z^n$ and $\sum b_n z^n$, representing functions f and g with $f(0) = 0$ (that is, with $a_0 = 0$), we can find the power series of the composition $g \circ f$ simply by plugging in and collecting terms: we write

$$\begin{aligned} g(f(z)) &= \sum b_n \left(\sum a_m z^m \right)^n \\ &= b_0 + b_1 a_1 z + (b_1 a_2 + b_2 a_1^2) z^2 + \dots \end{aligned}$$

5. Inversion: Given a power series $f(z) = \sum a_n z^n$ with $a_0 = 0$ and $a_1 \neq 0$ (that is, $f(0) = 0$ and $f'(0) \neq 0$), we know from section 4.3.3 that the map f takes a neighborhood of 0 homeomorphically onto a neighborhood of 0; that is, a holomorphic inverse function $g(z)$ exists, at least locally. We can find the power series expansion $g(z) = \sum b_n z^n$ of this inverse function by solving for successive coefficients: for example, using the formulas we found for the coefficients of the composition $g \circ f$, we see that if we want $g \circ f = z$, we must have

$$b_0 = 0; \quad b_1 = \frac{1}{a_1}; \quad b_2 = \frac{-b_1 a_2}{a_1^2};$$

and so on.

6. Differentiation: A power series $f(z) = \sum a_n z^n$ can be differentiated term by term within its radius of convergence, from which we see that

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

By the same token, if f is holomorphic in a disc (or any simply connected domain) there is a holomorphic function on the same domain with $g' = f$; the function g is unique up to the addition of a constant, and if $f(z) = \sum a_n z^n$ we can write

$$g(z) = c + \sum_{n=0}^{\infty} \frac{a_n}{n+1} z^{n+1}.$$

Chapter 12

Lecture 31: More power series

12.1 Recall

I want to recall what we proved last time: that if f is a function holomorphic in a region $\Omega \subset \mathbb{C}$ and $a \in \Omega$ any point, and the disc $\Delta = \{z : |z - a| < R\}$ of radius R around a is contained in Ω , then for all $z \in \Delta$ we have

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n;$$

in other words, every holomorphic function is given by a power series in any disc contained in its domain of definition. Note that it actually tells us a little more: we see from this that the radius of convergence of the power series with coefficients $\frac{f^{(n)}(a)}{n!}$ is at least R , where R is the distance from a to the boundary of Ω .

12.2 Power series of multivalued functions

We can use the operations above to build up a vocabulary of power series representations of elementary functions: starting with the power series for e^z , we get in this way descriptions of the basic trig, inverse trig and log functions. One area where we have to be careful, however is with multi-valued functions such as $\log z$ or z^α with $\alpha \notin \mathbb{Z}$.

Here, first of all, we have to bear in mind that since a convergent power series is necessarily a single-valued function, *the domain of convergence of*

the power series of a multivalued function can never include a branch point. Thus we can never express the function $\log z$, for example, as a power series in z ; rather, we have to expand it in powers of $z - a$; or, what is the same, expand a function like $\log(a + z)$ in powers of z . Typically, what's done is to expand $\log(1 + z)$ in powers of z .

Of course, if we want to represent any function f by a power series, we first have to specify a domain of definition and the values of f in that domain. In the case of the function $f(z) = \log(1 + z)$, for example, we can take our region Ω to be $\mathbb{C} \setminus \mathbb{R}^{\leq -1}$, where $\mathbb{R}^{\leq -1} = \{x \in \mathbb{R} : x \leq -1\}$; and we can specify which value of $\log(1 + z)$ to take for $z \in \Omega$ by requiring that

$$-\pi < \operatorname{Im} \log(1 + z) < \pi;$$

this is the unique branch of $\log(1 + z)$ in Ω with value 0 at $z = 0$. We can derive the power series in any one of a number of ways; we arrive at

$$\log(1 + z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n}.$$

Note that the radius of convergence is exactly 1: it must be at least 1, that being the distance from the origin to the boundary $\mathbb{R}^{\leq -1}$ of Ω ; and it can't be strictly greater than 1, because the function can't be defined in any open set containing -1 .

We have a similar situation with non-integer powers z^α of z : we can't define it in any neighborhood of the origin, and since we like to write power series in z rather than $z - a$, we instead represent the function $(1 + z)^\alpha$. Again, we have to specify a domain and the values of the function in that domain; for the former, we'll take the same domain Ω as last time, and we'll define the value of $(1 + z)^\alpha$ at a point $z \in \Omega$ to be $e^{\alpha \log(1+z)}$, where the value of $\log(1 + z)$ is specified as before: with

$$-\pi < \operatorname{Im} \log(1 + z) < \pi.$$

With all that said, we can again derive the power series expansion of $(1 + z)^\alpha$ by taking derivatives and evaluating at 0. (Note that once more we know in advance the radius of convergence of this power series will be 1: it's at least 1, since the disc of radius 1 around $z = 0$ is contained in Ω ; and it

can't be greater than 1 because the function $(1+z)^\alpha$ can't be defined in any neighborhood of $z = -1$.

To express the coefficients of this power series, it'll be helpful to introduce some new notation—or, rather, extend some old notation. Specifically, for $n \in \mathbb{N}$ and any $\alpha \in \mathbb{C}$ we define

$$\binom{\alpha}{n} := \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}.$$

With that convention, the power series expansion of $(1+z)^\alpha$ is given by

$$(1+z)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} z^n.$$

12.3 Laurent series

A Laurent series in z is like a power series, but involving negative as well as positive powers of z : something like

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

If we consider only the terms involving negative powers of z , this is a power series in the variable z^{-1} , as so it converges for all z *outside* a disc around the origin. The sum of the terms of positive degree in z will converge inside a disc; so that the domain of convergence of a Laurent series in general is an annulus

$$\Omega_{R_1, R_2} = \{z : R_1 < |z| < R_2\}.$$

Note that R_1 and R_2 can be 0, or ∞ , or anything in between; in particular, the domain of convergence can be empty (as indeed it can be with ordinary power series).

It is a basic theorem (Ahlfors p.185) that conversely, *any function $f(z)$ defined and holomorphic in an annulus Ω_{R_1, R_2} can be represented by a Laurent series*. The proof in Ahlfors is cute, but we won't go through it here.

One thing that confused me at first (and, truth be told, still does) is that the same function can have different Laurent series expansions, depending on where we want them to converge. Thus, for example, the function

$$f(z) = \frac{1}{1-z}$$

is given inside the unit disc $|z| < 1$ by the power series

$$f(z) = 1 + z + z^2 + \dots$$

But we can also write

$$\frac{1}{1-z} = \frac{1/z}{1/z-1} = \frac{-1}{z} \cdot \frac{1}{1-1/z}$$

from which we see that we can also write

$$f(z) = -z^{-1} - z^{-2} - z^{-3} - \dots,$$

which converges in the annulus $|z| > 1$.

One special case of Laurent series is the case where we have only finitely many terms of negative degree in z . This is of course the case of meromorphic functions: if f is meromorphic in a neighborhood of 0, with a pole of order m at 0, then we can write

$$f(z) = \frac{g(z)}{z^m},$$

with g holomorphic. The function g then has a Taylor series expansion

$$g(z) = \sum_{n=0}^{\infty} a_n z^n$$

valid in a disc $|z| < R$; and the Laurent series expansion of f in the annulus $0 < |z| < R$ is simply

$$g(z) = \sum_{n=-m}^{\infty} a_{n+m} z^n.$$

Note that the converse is not the case: a Laurent series with infinitely many terms of negative degree need not have an essential singularity (as in the example of $1/(1-z)$). But it *will* have an essential singularity if the inner radius of convergence is 0, that is, if the series converges on a punctured disc around 0.

Chapter 13

Partial fraction expansions

We want to move on from power series and consider other ways of expressing a function as an infinite sum or product of simpler functions. Today, we'll describe *partial fractions expansions* of meromorphic functions on the plane; in the next lecture we'll consider infinite product expansions of entire holomorphic functions.

To set up partial fractions, consider rational functions $R(z) = P(z)/Q(z)$. To start, we have two ways of representing such a function: first of all, we can factor P and Q and write

$$R(z) = \lambda \frac{(z - a_1)(z - a_2) \dots (z - a_k)}{(z - b_1)(z - b_2) \dots (z - b_l)};$$

or, if we collect like terms instead of just repeating them,

$$R(z) = \lambda \frac{(z - a_1)^{m_1}(z - a_2)^{m_2} \dots (z - a_e)^{m_e}}{(z - b_1)^{n_1}(z - b_2)^{n_2} \dots (z - b_f)^{n_f}}.$$

But there's another way to write such a function: by partial fractions. If the poles are all simple, we can write

$$R(z) = \frac{c_1}{z - b_1} + \frac{c_2}{z - b_2} + \dots + \frac{c_n}{z - b_n} + S(z)$$

where S is a polynomial; or, in the general case (allowing multiple poles) we can write

$$R(z) = \frac{C_1(z)}{(z - b_1)^{n_1}} + \frac{C_2(z)}{(z - b_2)^{n_2}} + \dots + \frac{C_f(z)}{(z - b_f)^{n_f}} + S(z)$$

where $C_i(z)$ is a polynomial of degree $\leq n_i - 1$ and S is again a polynomial.

We now ask: if we think of meromorphic functions on the whole plane $\mathbb{C}$ as analogues of rational functions, *are there analogues of either of these representations for meromorphic functions f on $\mathbb{C}$?* The answer is “yes” in both cases, and we’ll spend the next day or two seeing how to carry this out.

13.1 Setting up the problem

Our goal here is to do for meromorphic functions on $\mathbb{C}$ what the tool of partial fractions does for rational functions. To set this up, let’s describe in more generalizable terms what partial fractions does in the case of rational functions.

Briefly, suppose f is a rational function given as a quotient of polynomials; that is

$$f(z) = \lambda \frac{(z - a_1)^{m_1} (z - a_2)^{m_2} \dots (z - a_e)^{m_e}}{(z - b_1)^{n_1} (z - b_2)^{n_2} \dots (z - b_f)^{n_f}}.$$

Partial fractions gives us an alternative expression for f : we can write

$$f(z) = \frac{C_1(z)}{(z - b_1)^{n_1}} + \frac{C_2(z)}{(z - b_2)^{n_2}} + \dots + \frac{C_f(z)}{(z - b_f)^{n_f}} + S(z). \quad (13.1)$$

where S is a polynomial.

How should we interpret this last expression if f is meromorphic? Well, recall that if f is meromorphic, with a pole of order m at a point $b \in \mathbb{C}$, we can write $f(z) = g(z)/(z - b)^m$ with g holomorphic. If the power series of g around b is

$$g(z) = \sum_{n=0}^{\infty} a_n (z - b)^n,$$

then, we can write

$$f(z) = \frac{a_0}{(z - b)^m} + \frac{a_1}{(z - b)^{m-1}} + \dots + \frac{a_{m-1}}{z - b} + \sum_{n=0}^{\infty} a_{m+n} (z - b)^n.$$

The sum of the first m terms (the ones involving negative powers of $z - b$) is called the *polar part* or *singular part* of the meromorphic function f at the point b . Note that it’s a polynomial of degree m in $1/(z - b)$.

This is exactly what we're getting when we apply the method of partial fractions to a rational function: if f is a rational function, then in the expression (13.1) the term $C_i(z)/((z-b_i)^{n_i})$ is precisely the polar part of f at b_i . So that establishes our goal: suppose f is now an entire meromorphic function, with poles at $b_1, b_2, b_3, \dots$ (we'll assume infinitely many, just to avoid notational issues), and suppose that the polar part of f at b_ν is

$$P_\nu\left(\frac{1}{z-b_\nu}\right).$$

What we'd like to do is to write

$$f(z) = \sum_{\nu=1}^{\infty} P_\nu\left(\frac{1}{z-b_\nu}\right) + g(z)$$

with g an entire holomorphic function. Can we do it? Or could convergence issues doom us?

There are two sorts of questions we can ask in this regard. One is an abstract existence question: given a collection of points $\{b_\nu\} \subset \mathbb{C}$ with no limit points, does there exist an entire meromorphic function with poles exactly at the b_ν ? If we also specify a collection m_ν of positive integers, is there a function with a pole of order m_ν at b_ν ? And, to go whole hog, if we also specify a polar part at each point b_ν , is there an entire meromorphic function with exactly those polar parts?

The second question is not about constructing new functions but representing old ones: if we're given an entire meromorphic function—like, for example, $\pi^2 \csc^2(\pi z) = \pi^2 / \sin^2(\pi z)$, which has poles exactly at the integers—can we represent it in this way?

13.2 The partial fraction expansion of $\pi^2 \csc^2 \pi z$

Let's start with the second question, which is more fun. To begin with, from the power series expansion of $\sin z$ around 0

$$\sin \pi z = \pi z - \frac{\pi^3 z^3}{6} + \frac{\pi^5 z^5}{120} - \dots$$

we get

$$\sin^2 \pi z = \pi^2 z^2 - \frac{\pi^4 z^4}{3} + \cdots = \pi^2 z^2 \left(1 - \frac{\pi^2 z^2}{3} + \cdots \right).$$

Next, we see that

$$\pi^2 \csc^2 \pi z = \frac{1}{z^2} \left(1 + \frac{\pi^2 z^2}{3} + \cdots \right)$$

so the polar part of $\pi^2 \csc^2 \pi z$ at 0 is simply $1/z^2$. By the periodicity of the function $\pi^2 \csc^2 \pi z$, it follows that the polar part of $\pi^2 \csc^2 \pi z$ at the integer n is $1/(z - n)^2$.

So, can we write $\pi^2 \csc^2 \pi z$ as a sum of its polar parts plus an entire holomorphic function? Absolutely—the only thing standing in our way is the convergence of the sum

$$\sum_{n=-\infty}^{\infty} \frac{1}{(z - n)^2},$$

and it's easy to see that this converges for any value of $z \in \mathbb{C} \setminus \mathbb{Z}$. (Of course, we need more than this if we want the sum to converge to a holomorphic function: we need to show that it converges uniformly on compact subsets of $\mathbb{C} \setminus \mathbb{Z}$; but this is also straightforward.) We thus have

$$\pi^2 \csc^2 \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{(z - n)^2} + g(z), \quad (13.2)$$

where g is an entire holomorphic function.

This may seem like not all that big a deal if we don't know what g is! But in fact we know enough about g to determine it explicitly. To begin with, it is an entire holomorphic function; and it's *periodic with period 1*, that is, we have

$$g(z + 1) = g(z) \quad \forall z$$

(this follows from the defining equation (13.2) for g above, and the fact that the other two terms in the equation are visibly periodic with period 1). Thus

the function g just repeats its values in the vertical strip $0 \leq \operatorname{Re}(z) \leq 1$ over and over.

We can also estimate $|g(z)|$ as the imaginary part of z goes to $\pm\infty$: we have

$$\sin \pi z = \frac{e^{\pi iz} - e^{-\pi iz}}{2i}.$$

Now, when $\operatorname{Im} z \rightarrow \infty$, the first term in the numerator goes to 0 and the absolute value of the second goes to ∞ ; thus we can bound $|\sin \pi z|$ from below in the region $\operatorname{Im} z > y_0$, and correspondingly $|\pi^2 \csc^2 \pi z|$ is bounded from above in that region. Likewise, when $\operatorname{Im} z \rightarrow -\infty$, the second term in the numerator goes to 0 and the absolute value of the first goes to ∞ ; thus as before we can bound $|\sin \pi z|$ from below in the region $\operatorname{Im} z < -y_0$, and correspondingly $|\pi^2 \csc^2 \pi z|$ is bounded from above in that region.

It's also pretty straightforward to see that the other term in (13.2)—the infinite sum—is bounded from above in such a region.

Finally, we note that $|g(z)|$ (being continuous) is bounded on the compact set $\{z : 0 \leq \operatorname{Re} z \leq 1; |\operatorname{Im} z| \leq y_0\}$ and so, by periodicity, is bounded in the strip $\{z : |\operatorname{Im} z| \leq y_0\}$. Thus, in sum, g is bounded in the entire plane, hence constant by Liouville; and since it goes to 0 when $\operatorname{Im} z \rightarrow \infty$, *it must be identically zero*. We arrive at a somewhat unexpected formula:

$$\pi^2 \csc^2 \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{(z-n)^2}.$$

13.3 Existence questions

Let's go back now to the other question we raised: can we use partial fraction expansions to prove the existence of entire meromorphic function with arbitrarily specified poles, order and polar parts?

Before we give the answer, let's do an illustrative example. This will be the question I asked in class: is there an entire meromorphic function $f(z)$ with a simple pole with residue 1 at each of the integers and holomorphic on $\mathbb{C} \setminus \mathbb{Z}$?

We have specified that f should have polar part $1/(z - n)$ at $n \in \mathbb{Z}$, and so the natural thing to try would be simply

$$f_0(z) = \sum_{n=-\infty}^{\infty} \frac{1}{z - n}.$$

But now we do run into convergence issues: the sum simply doesn't converge anywhere. But there is a solution: we can add to each term in the sum a holomorphic function, which will not affect the pole but will help the sum converge. In this case, all we have to do is to subtract from each term $1/(z - n)$ with $n \neq 0$ its value at 0, which is $-1/n$; we arrive at the modified sum

$$f(z) = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z - n} + \frac{1}{n} \right) = \frac{1}{z} + \sum_{n \neq 0} \frac{z}{n(z - n)}$$

which, happily, does converge, and has exactly the desired polar parts.

We'll come back and say more about this particular function, but first let's give the answer to our general existence question. Suppose we are given an arbitrary collection of points $b_\nu \in \mathbb{C}$ with no limit points, and for each ν an arbitrary polynomial P_ν . We have the

Theorem 13.3.1 *There exists an entire meromorphic function $f(z)$, holomorphic on $\mathbb{C} \setminus \{b_\nu\}$ and having polar part $P_\nu \left(\frac{1}{z - b_\nu} \right)$ at b_ν .*

The proof will be along the lines of what we did in the last example: we consider first the sum

$$f_0(z) = \sum_{\nu} P_\nu \left(\frac{1}{z - b_\nu} \right).$$

If that converges, we're done. If not, we can subtract from each term in the sum a polynomial, which will make the sum converge without affecting the polar parts. Specifically, we choose a sequence of nonnegative integers m_ν , and from the ν th term in the sum we subtract the sum of the first m_ν terms in the Taylor series of the function $P_\nu \left(\frac{1}{z - b_\nu} \right)$ at the origin. Denoting this sum by $p_\nu(z)$, Ahlfors shows that *for a suitable choice of m_ν , the sum*

$$f(z) = \sum_{\nu} \left(P_\nu \left(\frac{1}{z - b_\nu} \right) - p_\nu(z) \right)$$

converges uniformly on compact sets in $\mathbb{C} \setminus \{b_\nu\}$, proving the existence of an entire meromorphic function with the desired polar parts.

13.4 Back to the function with simple poles at the integers

Let's return to our function

$$f(z) = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z-n} + \frac{1}{n} \right) = \frac{1}{z} + \sum_{n \neq 0} \frac{z}{n(z-n)}.$$

Since the convergence is uniform on compact subsets of $\mathbb{C} \setminus \mathbb{Z}$, it is permissible to take the derivative term-by-term; we arrive at

$$f'(z) = \sum_{n=-\infty}^{\infty} \frac{-1}{(z-n)^2}$$

and now we recognize the sum on the right: it's $-\pi^2 \csc^2 \pi z$! Pulling out our old calculus textbooks, we deduce that

$$f(z) = \pi \cot \pi z + C$$

for some constant C . Since both sides are odd functions of z (that is, $f(-z) = -f(z)$), the constant C must be 0, and we have another identity

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z-n} + \frac{1}{n} \right).$$

Finally, we should point out that another remedy for the failure of the sum $\sum_{n \in \mathbb{Z}} 1/(z-n)$ to converge is to pair off the terms corresponding to n and $-n$; since

$$\frac{1}{z-n} + \frac{1}{z+n} = \frac{2z}{z^2 - n^2}$$

we can also write

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2},$$

which does converge.

Chapter 14

Infinite product expansions

Just as our last topic, partial fractions expansions, was motivated by an analogy between meromorphic functions and rational functions, our starting point here is the parallel analogy between holomorphic functions and polynomials. To start, a polynomial is by definition a function of the form

$$p(z) = a_n z^n + \cdots + a_1 z + a_0.$$

But as you know, there's another way to write a polynomial: we can factor it and write

$$p(z) = \lambda \prod (z - b_i)$$

A natural question to ask is then, can we do something analogous for entire holomorphic functions, that is, holomorphic functions on the entire complex plane? Of course, a holomorphic function may well have infinitely many zeroes, meaning that the finite product above will have to be replaced by an infinite product, so we have to start by saying what it means for an infinite product to converge.

14.1 Preliminary: convergence of infinite products

What does it mean for an infinite product $\prod p_i$ to converge? This may seem obvious: as in the case of infinite sums, our first instinct might be to say that

an infinite product $\prod p_i$ converges if the sequence of partial products

$$P_n = \prod_{i=1}^n p_i$$

converges, and of course we'd set the infinite product equal to the limit $\lim_{n \rightarrow \infty} P_n$.

This is almost right, but there's one problem. In analogy with the case of infinite sums, the convergence of an infinite product should depend on the whole sequence; it shouldn't be affected if, for example, we change one term. But here we have a problem with the number 0: if any one term $p_i = 0$, then the partial sums from that point on are all zero, meaning that under our naive definition it would automatically converge.

The solution to this problem is simply to make a special rule regarding 0 in an infinite product: we say that $\prod p_i$ converges if

1. at most finitely many terms $p_i = 0$; and
2. the partial products of the nonzero terms

$$P_n = \prod_{i \in \{1, 2, \dots, n\} : p_i \neq 0} p_i$$

converge to a nonzero limit.

This may seem like an awkward definition, and it is. But it works well with what we want to do.

Having defined convergence, let's fix the notation. It's pretty clear that in order for an infinite product $\prod p_i$ to converge, the terms have to tend to 1 (this is one of the reasons we excluded the case where the limit of the partial products was 0). We therefore write infinite products in general as $\prod (1 + a_n)$; the convergence or divergence of the product will reflect the rate at which a_n goes to 0.

We can be more precise about that: not unreasonably, assuming no factors are 0, *the infinite product $\prod (1 + a_i)$ will converge if and only if the infinite*

sum $\sum \log(1 + a_i)$ *does*. Here we have to specify which value of $\log(1 + a_n)$ to take; we choose the one with imaginary part between $-\pi$ and π . This causes some complication in the proof, which would otherwise be, “well, duh.”

In fact, we can go further: the series $\sum \log(1 + a_n)$ converges *absolutely* if and only if the series $\sum a_n$ does. This follows from the general fact that if c_n is any sequence tending to 1, then the series $\sum a_n$ converges absolutely if and only if the series $\sum c_n a_n$ does. Note that we need to specify absolute convergence: the conditional convergence of a series $\sum a_n$ may depend on cancellation between the positive and negative terms, which multiplication by c_n can disrupt.

Thus, the infinite product $\prod(1 + a_i)$ converges absolutely if and only if the series $\sum a_i$ does.

14.2 Product expansions: the basic set-up

Given all the above, let's formulate our basic construction and questions. The goal is to express a given entire holomorphic function $f(z)$ as a product, to highlight its zeroes, the way we write a polynomial as a product $\lambda \prod (z - b_\nu)$ of linear factors, so we'll suppose $f(z)$ is an entire holomorphic function with zeroes at $b_1, b_2, b_3, \dots$. Of course an infinite product of the form $\prod (z - b_\nu)$ can't possibly converge, so instead we consider an infinite product of the form

$$\prod_{\nu=1}^{\infty} \left(1 - \frac{z}{b_\nu}\right).$$

(Of course, if the function $f(z)$ happens to have a zero at $z = 0$, we have to treat that separately, so we'll throw in a factor of z^m if this is the case.)

Assuming the infinite product does converge, the ratio of f to this infinite product is then an entire holomorphic function with no zeroes, and by at least one homework problem any such function can be written as $e^{g(z)}$ for some entire holomorphic function g . In total, then, we have

$$f(z) = z^m e^{g(z)} \prod_{\nu=1}^{\infty} \left(1 - \frac{z}{b_\nu}\right).$$

(Note that we are adopting Ahlfors' notational convention that if the function $f(z)$ has a zero of order m , the corresponding factor $(1 - \frac{z}{b})$ is repeated m times.)

We then have, in analogy with the situation with partial fraction expansions, two questions:

1. The abstract existence question: given any sequence $b_1, b_2, \dots \in \mathbb{C}$ with $\lim_{\nu \rightarrow \infty} b_\nu = \infty$, does there exist an entire holomorphic function with zeroes exactly at the points b_ν ?
2. The representation question: can we represent known functions, like $\sin$ and $\cos$, in this way?

In fact, the answers to both are exactly analogous to what we found with partial fractions. Let's start as we did with partial fractions, by considering a case of the second question: the representation of the function $\sin(\pi z)$ as an infinite product.

14.3 $\sin \pi z$ as an infinite product

The function $\sin(\pi z)$ has zeroes exactly at the integers, so naively the infinite product would be

$$z \prod_{n \neq 0} \left(1 - \frac{z}{n}\right).$$

Unfortunately, this doesn't converge, since $\sum 1/n$ doesn't. Happily, though, there is a remedy for this: very much in analogy with what we did in the case of partial fractions, we can try multiplying each factor by a nonzero holomorphic function to make it smaller (that is, closer to 1) and help convergence without altering the zeroes.

What factor should we choose? Well, we're trying to make the sum

$$\sum \log \left(1 - \frac{z}{n}\right)$$

converge, so by analogy with the partial fractions case, one approach would be to try subtracting from each term in the sum the first few terms in the

Taylor series expansion of the function around the origin. In this case, the Taylor series of the term $\log(1 - \frac{z}{n})$ is

$$\log\left(1 - \frac{z}{n}\right) = -\frac{z}{n} - \frac{z^2}{2n^2} - \dots$$

so we can try adding z/n to the term $\log(1 - \frac{z}{n})$; or, in other words, multiplying the original factor of $(1 - z/n)$ by $e^{z/n}$. This produces the series

$$z \prod_{n \neq 0} \left(1 - \frac{z}{n}\right) e^{z/n},$$

and in fact this does converge, as we see by summing the logs.

We'll see in a moment how to carry this out in general, but first let's finish this example. Given the convergence of the modified series, we know we can write

$$\sin \pi z = z e^{g(z)} \prod_{n \neq 0} \left(1 - \frac{z}{n}\right) e^{z/n} \quad (14.1)$$

for some entire holomorphic function $g(z)$. Again, this doesn't do us much good if we can't identify the missing factor $e^{g(z)}$, but in fact we can!

The trick here is to take the logarithmic derivative $f'(z)/f(z) = \frac{d}{dz} \log f(z)$ on both sides of (14.1), since this converts products into sums. (We have to check that, according to our definition of convergence of an infinite product, we can take the logarithmic derivative term-by-term; this is another reason for defining convergence as we did.) Going through (14.1), we have

- the logarithmic derivative of $\sin \pi z$ is $\pi \cos \pi z / \sin \pi z = \pi \cot \pi z$;
- the logarithmic derivative of z is $1/z$;
- the logarithmic derivative of $e^{g(z)}$ is $g'(z)$; and
- the logarithmic derivative of the infinite product is

$$\sum_{n \neq 0} \left(\frac{1}{z - n} - \frac{1}{n} \right).$$

Adding these all up, we have

$$\pi \cot \pi z = \frac{1}{z} + g'(z) + \sum_{n \neq 0} \left(\frac{1}{z-n} - \frac{1}{n} \right).$$

But we have already seen that

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z-n} - \frac{1}{n} \right),$$

so we conclude that $g'(z) = 0$; that is, $g(z)$ is a constant!

Finally, to determine the constant $g(z)$, just divide both sides of (14.1) by z and take the limit as $z \rightarrow 0$: on the left we get π and on the right we get $e^{g(z)}$, so $g(z) = \log \pi$ and we have altogether

$$\sin \pi z = \pi z \prod_{n \neq 0} \left(1 - \frac{z}{n} \right) e^{z/n};$$

or, combining terms corresponding to n and $-n$,

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2} \right). \quad (14.2)$$

If I could editorialize for a moment, this is one of the things I love about complex analysis. Of course, it introduces you to the world of holomorphic functions, a beautiful and exotic place—a secret garden in the middle of the city. But you also get to see some old friends in a new light. Once you’ve seen the exponential function, for example, as a function of a complex variable—seen that its graph is the universal covering space of the punctured plane $\mathbb{C}^*$, and understood correspondingly the multivalued nature of the logarithm function—you can’t go back. And I thought I knew all about the function $\sin x$ in high school—but I never saw the formula (14.2) until I read it in Ahlfors.

14.4 The general existence theorem

Let’s return to the abstract existence question: given any sequence $b_1, b_2, \dots \in \mathbb{C}$ with $\lim_{\nu \rightarrow \infty} b_\nu = \infty$, does there exist an entire holomorphic function with

zeroes exactly at the points b_ν ? Again, if the infinite product

$$\prod \left(1 - \frac{z}{b_\nu}\right)$$

converges, we're done; that's our function. If it diverges, on the other hand, we can do as we did in the case of $\sin \pi z$ and throw in an additional factor in each term that will ensure convergence without changing the location of the zeroes.

The choice of extra factor is suggested by the examples we've seen. We want the sum

$$\sum \log \left(1 - \frac{z}{b_\nu}\right)$$

to converge. We can write out the Taylor series of the factors: we have

$$\log \left(1 - \frac{z}{b_\nu}\right) = -\frac{z}{b_\nu} - \frac{z^2}{2b_\nu^2} - \frac{z^3}{3b_\nu^3} - \dots$$

and the idea would be to subtract off the first m_ν terms of this series; that is, replace the factor

$$\left(1 - \frac{z}{b_\nu}\right)$$

in the original product with

$$\left(1 - \frac{z}{b_\nu}\right) e^{\frac{z}{b_\nu} + \frac{1}{2}\left(\frac{z}{b_\nu}\right)^2 + \dots + \frac{1}{m_\nu}\left(\frac{z}{b_\nu}\right)^{m_\nu}}$$

At the end of the day, we can see—exactly as in the case of partial fractions—that *for a suitable choice of m_ν , the infinite product*

$$\prod \left(1 - \frac{z}{b_\nu}\right) e^{\frac{z}{b_\nu} + \frac{1}{2}\left(\frac{z}{b_\nu}\right)^2 + \dots + \frac{1}{m_\nu}\left(\frac{z}{b_\nu}\right)^{m_\nu}}$$

converges. This establishes the abstract existence theorem

Theorem 14.4.1 *If $b_1, b_2, \dots \in \mathbb{C}$ is any sequence with $\lim b_\nu = \infty$, then there exists an entire holomorphic function $f(z)$ with zeroes exactly at the points b_ν .*

Note that here we're allowing repeats of the b_ν , the convention being that if b appears n times in the sequence then f should have a zero of order n at b . (Note that no $b \in \mathbb{C}$ can appear infinitely many times in the sequence, since that would contradict the hypothesis that $\lim_{\nu \rightarrow \infty} b_\nu = \infty$.)

As an immediate corollary of this theorem:

Corollary 14.4.2 *Any meromorphic function on $\mathbb{C}$ is the quotient of two entire holomorphic functions.*

Proof Suppose f has poles at $b_1, b_2, \dots$. There is an entire holomorphic function $g(z)$ with zeroes at exactly those points; and the product $h(z) = f(z)g(z)$ is everywhere holomorphic, so $f = h/g$.

Chapter 15

Nevanlinna theory

There's a beautiful branch of complex analysis called *value distribution theory*, or *Nevanlinna theory*. We won't have time to develop it, but we can at least explain the content of the two main theorems of Nevanlinna theory.

The starting point is our old friend the fundamental theorem of algebra. In one of its many forms, it says that a polynomial of degree d takes every value the same number of times (counting multiplicity). We've said a number of times that entire holomorphic functions behave like polynomials; is there an analogue of the FTA for them?

For the remainder of this lecture, we'll consider an entire meromorphic function f on $\mathbb{C}$, which we may think of as a holomorphic map from $\mathbb{C}$ to the Riemann sphere $S = \mathbb{C} \cup \{\infty\}$.

15.1 The counting function and the proximity function

To begin with, we need an analogue of the notion of “the number of times f takes the value a ,” since of course that number will in general be infinite. We introduce a function that measures the frequency of zeroes of $f(z) - a$, as follows: we let

$$n_f(r, a) = \#\{z \in \mathbb{C} : |z| < r \text{ and } f(z) = a\}$$

that is, the number of zeroes of $f(z) - a$ in the disc $|z| < r$; and to smooth out this function we take

$$N_f(a, r) = \int_0^r n(a, t) \frac{dt}{t}.$$

The rate of growth of $N(r)$ is our measure of how frequently f assumes the value a . Note that since f is assumed meromorphic, $a = \infty$ is also a possibility; in that case $N_f(\infty, r)$ measures the frequency of the poles of f . Note also that we have to assume $f(0) \neq a$, a minor annoyance.

We'd like to say, by analogy with the fundamental theorem of algebra, that the functions $N_f(a, r)$ all have the same order of growth as $r \rightarrow \infty$. But this is visibly false: the function e^z takes every value $a \neq 0, \infty$ infinitely many times, but is never 0 or ∞ . What we need is an error term, which comes from what is called the *proximity function* $m_f(a, r)$. To begin with, we define, for positive real numbers x ,

$$\log^+(x) = \max(0, \log x).$$

We then set

$$m_f(a, r) = \frac{1}{2\pi} \int_0^{2\pi} \log^+ \left| \frac{1}{f(re^{i\theta}) - a} \right| d\theta.$$

This is called a proximity function because it measures how close the values of f on the circle of radius r get to a . If $a = \infty$, we take instead

$$m_f(\infty, r) = \frac{1}{2\pi} \int_0^{2\pi} \log^+ |f(re^{i\theta})| d\theta,$$

which is a measure of how large $|f(z)|$ is (that is, how close to ∞ $f(z)$ is) on the circle of radius r .

We now set

$$T_f(a, r) := N_f(a, r) + m_f(a, r),$$

called the *characteristic function* of f for the value a .

15.2 The first main theorem

In these terms, we have the First Main Theorem of Nevanlinna theory:

Theorem 15.2.1 *For any two values $a, b \in S$, we have*

$$T_f(a, r) = T_f(b, r) + O(1)$$

In other words, an entire meromorphic function takes all values with equal frequency, if you count (via the proximity function) the amount of time the function spends “close” to a .

The function e^z gives a good example. Look at the image, under the map $e^z : \mathbb{C} \rightarrow S$, of the circle of radius r around the origin in $\mathbb{C}$. The left half of the circle will spiral around the value 0, spending most of its time within a small circle around 0. (More precisely: if Δ_ϵ is the disc of radius ϵ around 0, then as $r \rightarrow \infty$, the fraction of the circle $|z| = r$ that maps into Δ_ϵ approaches $1/2$, regardless of how small ϵ is.) Likewise, most of the right hand half of the circle of radius r is mapped very close to the point $\infty \in S$ —again, for any neighborhood U of $\infty \in S$, the fraction of the circle $|z| = r$ that is mapped into U goes to $1/2$ as $r \rightarrow \infty$.

For any other value $a \in S \setminus \{0, \infty\}$, by contrast, the portion of the circle $|z| = r$ that’s mapped into a neighborhood of a goes to 0 as the neighborhood shrinks. According to the FMT, this exactly compensates for the fact that e^z has no zeroes or poles, but assumes every other possible value infinitely many times.

15.3 The second main theorem

Now, what we we’re really interested in, if we’re trying to extend the Fundamental Theorem of Algebra, is the function $N_f(a, r)$; we think of the function $m_f(a, r)$ as an “error term.” But that only works if indeed m_f accounts for only a small fraction of T_f . We make, accordingly, one more definition: for a given entire meromorphic function f , and a given value $a \in S$, we define the *defect* of a to be

$$\delta_f(a) = \liminf_{r \rightarrow \infty} \frac{m_f(a, r)}{T_f(a, r)} = \limsup_{r \rightarrow \infty} 1 - \frac{N_f(a, r)}{T_f(a, r)}.$$

Essentially, this measures the failure of f to assume a given value a as frequently as the characteristic function would lead you to expect; we’ll say a value a is *defective* for the function f if $\delta_f(a) \neq 0$.

This said, we have the amazing Second Main Theorem of Nevanlinna Theory:

Theorem 15.3.1 *For any entire meromorphic function f ,*

$$\sum_{a \in S} \delta_f(a) \leq 2.$$

In particular, this means f has only countably many defective values; in other words, with at most countably many exceptions, *an entire meromorphic function assumes all values with the same frequency.*

Note that the SMT is sharp, as shown by the example of $f(z) = e^z$: for this function, we have $\delta_f(0) = \delta_f(\infty) = 1$, and $\delta_f(a) = 0$ for all other values.

Finally, we mention another special case of the SMT:

Theorem 15.3.2 (Little Picard theorem) *An entire meromorphic function that omits three values $a \in S$ is constant; in particular, an entire holomorphic function that omits two values $a \in \mathbb{C}$ is constant.*

Chapter 16

Special functions

So far, we have used infinite series and product expansions to do two things: one, to prove abstract existence theorems (there is an entire holomorphic function with given zeroes $b_1, b_2, \dots \in \mathbb{C}$; there is an entire meromorphic function with given polar parts, etc.); and two, to give new ways of representing known functions like $\csc \pi z$ and $\sin \pi z$.

There is another important use of these expansions, which is to introduce new functions into our vocabulary of functions. These arise in a number of ways; I'll start by describing these in general with examples; and then we'll study two examples in more detail.

16.1 Encoding arithmetic information

One thing we can do with infinite expansions is to encode arithmetic sequences. By way of an example, consider the *partition function* $p(n)$, defined to be the number of ways of expressing the integer n as a sum of positive integers. Thus, for example, we have

$$2 = 1 + 1; \text{ so } p(2) = 2;$$

$$3 = 2 + 1 = 1 + 1 + 1; \text{ so } p(3) = 3;$$

$$4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1; \text{ so } p(4) = 5.$$

The function $p(n)$ has many fascinating arithmetic properties—for example, there are all sorts of theorems about the congruence class of $p(n)$ mod

various primes. (An example: $p(5k+4) \equiv 0 \pmod{5}$, a theorem of Ramanujan.) But I want to focus here on a more quantitative question: how fast does $p(n)$ grow with n ? Does it have polynomial growth, for example, or exponential?

One way to approach a question like this is to encode the values $p(n)$ as the coefficients of an infinite series. For example, we can consider the holomorphic function defined by the power series

$$P(z) := \sum_{n=0}^{\infty} p(n)z^n.$$

The idea is that the analytic properties of the function $P(z)$ reflect the behavior of the sequence $p(n)$, and so we can use the methods of complex analysis to better understand the partition function.

But how do we analyze $P(z)$? One way is to give a second expansion of the function: we can show that

$$P(z) = \prod_{n=1}^{\infty} \frac{1}{1 - z^n}.$$

To see this, write out the infinite product as

$$(1 + z + z^2 + \dots)(1 + z^2 + z^4 + \dots)(1 + z^3 + z^6 + \dots) \dots$$

Then for any n , the partition of n as a sum of a_1 1s, a_2 2s, a_3 3s and so on corresponds to the product of the term z^{a_1} in the first factor, the term z^{2a_2} in the second factor, the term z^{3a_3} in the third factor, and so on. Thus if we multiply out the infinite product, we get one z^n for every partition of n .

From the representation of $P(z)$ as an infinite product, we can see that it is defined and holomorphic in the open unit disc. We can also see that it has a “natural boundary” on the unit circle: since the factors of the product have poles at the points $e^{2\pi i\alpha}$ for all $\alpha \in \mathbb{Q}$, we see that $P(z)$ blows up when z approaches any point on the circle $|z| = 1$ (like the function in the problem on your homework!).

This implies in particular that the radius of convergence of the power series $\sum p(z)z^n$ is 1, which already tells us something about the asymptotic behavior of the function $p(n)$. But in fact we can get a lot more out of it: by a more detailed analysis of the function $P(z)$, which we won't attempt here, it's possible to show that

$$p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}$$

(the symbol “ $\sim$ ” here means the ratio of the two sides approaches 1 as $n \rightarrow \infty$). In particular, the answer to our naive question, “does $p(n)$ have polynomial growth, or exponential?” is “neither:” it grows faster than any polynomial, but slower than any exponential function $e^{\lambda n}$ of n .

16.2 Finding new functions with special properties

Again, we're going to do this by example. We have introduced the notion of a periodic function: an entire meromorphic function f on $\mathbb{C}$ is said to be *periodic with period ω* if for all z we have

$$f(z + \omega) = f(z)$$

(and hence, of course, $f(z + n\omega) = f(z)$ for all $n \in \mathbb{Z}$).

There is also a notion of a *doubly periodic* function: for a pair of nonzero complex numbers ω_1, ω_2 linearly independent over $\mathbb{R}$ (that is, such that $\omega_1/\omega_2 \notin \mathbb{R}$), we say that an entire meromorphic function is *doubly periodic with periods ω_1, ω_2* if

$$f(z + \omega_1) = f(z + \omega_2) = f(z) \quad \forall z \in \mathbb{C}$$

(so that $f(z + n\omega_1 + m\omega_2) = f(z)$ for all $n, m \in \mathbb{Z}$).

Mathematicians in the 19th century came to realize that the integrals of certain algebraic functions necessarily involved doubly periodic functions. (I'll describe how this came about on Wednesday.) But there are no elementary functions that are doubly periodic! Thus, to give these integrals

explicitly we would have to introduce new functions, and so enlarge our basic vocabulary of functions.

In fact, it's very easy to construct doubly periodic functions using partial fraction expansions:¹ for example, consider for a positive integer k the sum

$$f(z) = \sum_{m,n \in \mathbb{Z}} \frac{1}{(z - m\omega_1 - n\omega_2)^k}.$$

For $k \geq 3$ this converges uniformly on compact subsets of $\mathbb{C} \setminus \{m\omega_1 + n\omega_2\}$ to give a doubly periodic function. In case $k = 2$, we can achieve convergence by throwing in an extra term in each summand; we arrive at

$$\mathcal{P}(z) = \frac{1}{z^2} + \sum_{\omega = m\omega_1 + n\omega_2; (m,n) \neq (0,0)} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

This is known as the *Weierstrass $\mathcal{P}$ -function*, and you can read all about it (and doubly periodic functions in general) in Chapter 7 of Ahlfors.

16.3 The Gamma function

Let's focus now on the two special functions of interest to us, starting with the Gamma function.

The starting point for the definition of Γ is a naive question: is there an entire meromorphic function that behaves like the function $n!$ on nonnegative integers? Of course, that's not a well-posed question; there are lots of entire functions with the same values at the integers. To be more precise, let's take the essential feature of the factorial function to be the property that $(n+1)! = (n+1)n!$, and ask if we can find/construct an entire meromorphic function satisfying the functional equation

$$F(z+1) = (z+1)F(z) \quad \text{or, equivalently,} \quad F(z) = zF(z-1)$$

Note that no polynomial can satisfy this, since $F(z+1)$ would have strictly more zeroes than $F(z)$. In fact, what this functional equation says is that

¹Note that by Liouville's theorem, a doubly periodic holomorphic function is constant.

the zeroes of $F(z+1)$ —which are exactly the zeroes of $F(z)$ shifted to the left by 1—are the zeroes of $F(z)$ plus an additional zero at -1 . It follows that F must have zeroes exactly at the non-negative integers.

Now, for various reasons I want to flip that, and ask for an entire function with zeroes exactly at the negative integers. We know how to write down a function with those zeroes: we just take the infinite product

$$G(z) := \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n},$$

where of course the factor $e^{-z/n}$ is thrown into the n th factor to promote convergence. Note that, comparing this to the product expansion of the sine function we derived earlier, we see that this function G satisfies the equation

$$zG(z)G(-z) = \frac{\sin \pi z}{\pi};$$

this will be useful later on.

So, does G satisfy a functional equation of the sort we want? Well, we know that $G(z-1)$ has exactly the same zeroes as G , plus one additional zero at $z=0$; so the ratio $G(z-1)/zG(z)$ is entire holomorphic and nowhere zero. That means we can write this quotient as $e^{\gamma(z)}$ for some entire holomorphic function γ ; in other words, we have

$$G(z-1) = zG(z)e^{\gamma(z)}.$$

Both sides being products, if we want to determine $\gamma(z)$ we should take the logarithmic derivative on both sides. Carrying this out, we see that in fact $\gamma(z)$ is a constant, denoted simply γ and called *Euler's constant*. It can be expressed as a limit:

$$\gamma = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right) - \log n;$$

its approximate value is .57722.

We are still two manipulations away from the Gamma function! First of all, to get rid of the factor of e^γ in the functional equation for G , we set

$$H(z) = e^{\gamma z} G(z)$$

so that $H(z)$ satisfies the simpler equation

$$H(z-1) = zH(z).$$

Finally, we can define the Gamma function! We set

$$\Gamma(z) = \frac{1}{zH(z)}.$$

This is called *Euler's Gamma function*. The immediate properties to note are:

- Γ is an entire meromorphic function, with poles at the nonpositive integers and no zeroes;
- We have the explicit representation

$$\Gamma(z) = \frac{e^{-\gamma z}}{z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right)^{-1} e^{z/n}$$

- Γ satisfies the functional equation

$$\Gamma(z+1) = z\Gamma(z)$$

- $\Gamma(1) = 1$ (by the explicit representation above), so by the functional equation we have

$$\Gamma(n) = (n-1)!$$

for positive integers n (why they couldn't arrange it to have $\Gamma(n) = n!$ I don't know).

- Finally, the relation $\pi z G(z) G(-z) = \sin \pi z$ above translates into the identity

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

There are many, many other identities satisfied by the Gamma function, and many things you can do with it. For example, if you're interested in the asymptotic behavior of $n!$ as $n \rightarrow \infty$, you can apply the techniques of complex analysis in the large to arrive (after four grueling pages) at *Stirling's formula*:

$$\Gamma(z) = \sqrt{2\pi} z^{z-\frac{1}{2}} e^{-z} e^{J(z)}$$

where $J(z) \rightarrow 0$ as $z \rightarrow \infty$ in a half-plane $\operatorname{Re}(z) \geq c > 0$. In particular, $n!$ has the same asymptotic growth as $n^{n+\frac{1}{2}}e^{-n}$ (meaning the ratio is bounded from above and below).

For us, the Gamma function will be simply a useful tool in defining and establishing basic properties of the *zeta function*, which we'll discuss next. Before we go on, though, I want to mention one other formula for the Gamma function which will be useful:

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$$

whenever $\operatorname{Re}(z) > 0$.

16.4 The zeta function

The Riemann zeta function is an example of a type of infinite expansion we haven't encountered yet: the *Dirichlet series*. These are sums of the form

$$\sum_{n=1}^{\infty} a_n n^{-z}$$

and they are very often used to encode arithmetic functions a_n . One notational point: for some reason, when dealing with Dirichlet series, people use the variable $s = \sigma + it$ instead of $z = x + iy$.

The Riemann zeta function is the simplest of all Dirichlet series: it is, simply,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

This converges absolutely when $\sigma = \operatorname{Re}(s) > 1$, and uniformly for $\sigma > \sigma_0 > 1$, so it defines a holomorphic function in the half plane $\operatorname{Re}(s) > 1$. One of the central facts to establish about ζ is that, even though the sum clearly fails to converge when $\sigma < 1$, the function $\zeta(s)$ can be extended to a meromorphic function in the whole complex plane, with a pole at $s = 1$; and indeed the key questions we have about ζ are about its values in the parts of the plane where the Dirichlet series does not converge.

The reason the Riemann zeta function plays such a central role in number theory is a second representation of $\zeta(s)$: we have

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

If we write

$$\frac{1}{1 - p^{-s}} = 1 + p^{-s} + p^{-2s} + \dots$$

(and ignore convergence issues), the content of this formula is simply the theorem of unique factorization: if we write a positive integer n as a product of primes $n = \prod p_i^{a_i}$, then the term n^{-s} in the Dirichlet sum corresponds to the product of the terms $p_i^{-a_i s}$ from the corresponding factors in the product.

The upshot of this representation of the zeta function is that the analytic behavior of $\zeta(z)$ reflects the distribution of the primes. The simplest example of this is the observation that the fact that the sum $\sum \frac{1}{n}$ diverges—so that $\zeta(z)$ has a singularity at $z = 1$ —implies that there are infinitely many primes: otherwise, the product above would be finite and the function ζ would be holomorphic at $z = 1$. (In fact, this tells us that the sum

$$\sum_p \frac{1}{p}$$

diverges, which is a stronger statement than just saying there are infinitely many primes; it says that they're not too sparse.) But the connection goes much, much deeper than this; the location of the zeroes of ζ , for example, has implications for the error term in classical approximations of the number $\pi(x)$ of primes $p \leq x$. This is the subject of the *Riemann Hypothesis*.

Leaving aside for now the connection between the zeta function and number theory, let's mention some of the basic facts about the function. One is elementary: we've asserted that

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt.$$

If we make a change of variable from t to nt , we deduce that

$$n^{-z} \Gamma(z) = \int_0^\infty e^{-nt} t^{z-1} dt$$

and now summing over n yields

$$\zeta(s)\Gamma(s) = \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx.$$

As a consequence, we can express $\zeta(s)$ as a path integral: we have

$$\zeta(s) = -\frac{\Gamma(1-s)}{2\pi i} \int_C \frac{(-z)^{s-1}}{e^z - 1} dz$$

where the path C is the “keyhole” path we’ve used before (Fig 5.1 on page 214 in Ahlfors). The significance of this last representation for $\zeta(s)$ is that *it is not restricted to the half plane $\operatorname{Re}(s) > 1$* : we see from this that *the zeta function extends to an entire meromorphic function*, with a simple pole at $s = 1$ with residue 1.

There is one more basic formula involving the zeta function, which is a functional equation satisfied by $\zeta(s)$. It relates the values of ζ at the points s and $1-s$; explicitly, we have

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

Using this equation and the fact that ζ has no zeroes in the half-plane $\sigma > 1$ (by the product expansion), we can locate all the zeroes of ζ outside the *critical strip* $0 < \sigma < 1$. The location of the zeroes in that critical strip, on the other hand, remains mysterious; it is the subject of the *Riemann hypothesis*, which asserts that *all zeroes of ζ in the critical strip lie on the line $\sigma = 1/2$* .

The Riemann hypothesis has implications for both the density of primes and your bank account: the Clay Institute is offering \$1,000,000 for a proof or disproof. I wouldn’t consider proving the Riemann hypothesis a sound financial strategy, however: it’s been verified for the first 3,000,000 zeroes of ζ (I’ll explain in class how they do this), but we’re still nowhere near a proof.

Chapter 17

Abelian integrals: how topology, complex analysis and algebraic geometry grew up together

17.1 A calculus problem we can solve

To start with, I want to go back to the 18th century and the early 19th century. Mathematicians had been applying the techniques of integral calculus to solve a range of problems arising in math and physics, and in so doing had built up a fairly large roster of functions whose integrals they could evaluate. These included most of the functions whose integrals you are taught in the first calculus course you take: exponential and logarithmic functions, and the basic trig and inverse trig functions.

It also included some simple *algebraic functions*, specifically functions involving the square root of a quadratic polynomial: for example,

$$\int_{z_0}^{z_1} \frac{dz}{\sqrt{z^2 + 1}}.$$

These days, students are taught to do these by trig substitutions; I don't know how they were done originally, but in any event I want to suggest a different approach.

What I want to suggest is that we think of this as a path integral on a Riemann surface. The issue is that, as complex analysis has taught us, a function like $\sqrt{z^2 + 1}$ is not well-defined on the complex plane: for every value of z other than $\pm i$, there are two possible values of $w = \sqrt{z^2 + 1}$, which form a two-sheeted covering space of $\mathbb{C} \setminus \{i, -i\}$; if we start out choosing one value of w for a given value of z , and then let z wander around in a circle around either $\pm i$, letting w vary continuously with z , we come back to the opposite value of w .

I'd propose, accordingly, introducing the locus

$$\Gamma = \{(z, w) \in \mathbb{C} \times \mathbb{C} : w^2 = z^2 + 1\}$$

and thinking of z and w as well-defined functions on Γ , rather than as multiple-valued functions on $\mathbb{C}$. In fact, Γ is a first example of a *complex manifold*: at every point $p = (z, w) \in \Gamma$, one of the two projection maps will be a local homeomorphism of a neighborhood of $p \in \Gamma$ with a neighborhood of either $z \in \mathbb{C}$ or $w \in \mathbb{C}$, and we can carry out all the constructions of complex analysis via these identifications. (This is indeed one of the ways in which manifolds first arose!) In particular, we can think of our original integral as a path integral

$$\int_{p_0}^{p_1} \frac{dz}{w}$$

between two given points $p_0 = (z_0, w_0)$ and $p_1 = (z_1, w_1) \in \Gamma$.

All this might seem like nothing more than pointless complication, if it were not for one fact: Γ *is biholomorphic to a region $\Omega \subset \mathbb{C}$ of the complex plane!* Explicitly, we have a biholomorphic bijection

$$S \setminus \{\pm 1\} \cong \Gamma$$

(where S is the Riemann sphere) given in the forward direction by

$$\lambda \mapsto \left(\frac{2\lambda}{1 - \lambda^2}, \frac{1 + \lambda^2}{1 - \lambda^2} \right)$$

and the other way by

$$(z, w) \mapsto \frac{w - 1}{z}.$$

(Check that these are in fact inverse bijections!) We can thus carry our original path integral over from Γ to $S \setminus \{\pm 1\}$; we have

$$w = \frac{1 + \lambda^2}{1 - \lambda^2} \quad \text{and} \quad dz = \frac{1 + \lambda^2}{(1 - \lambda^2)^2} d\lambda$$

so

$$\int_{p_0}^{p_1} \frac{dz}{w} = \int_{\lambda_0}^{\lambda_1} \frac{d\lambda}{1 - \lambda^2}$$

which we can do by partial fractions.

This may seem like a roundabout way to evaluate an integral that you could have just looked up in the back of your old calculus text. But in fact this approach applies much more broadly. Let me show you how I arrived at the maps between $S \setminus \{\pm 1\}$ and Γ . The idea is, you choose a point on the locus Γ , as pictured on the next page; I chose the point $(0, 1)$. Now, for any $\lambda \in \mathbb{C}$, we can consider the line L_λ through the point $(0, 1)$ with slope λ . We parametrize the line L_λ as $\{(z, \lambda z + 1)\}$; the equation $w^2 = z^2 + 1$ restricted to L_λ is then

$$(\lambda z + 1)^2 = z^2 + 1,$$

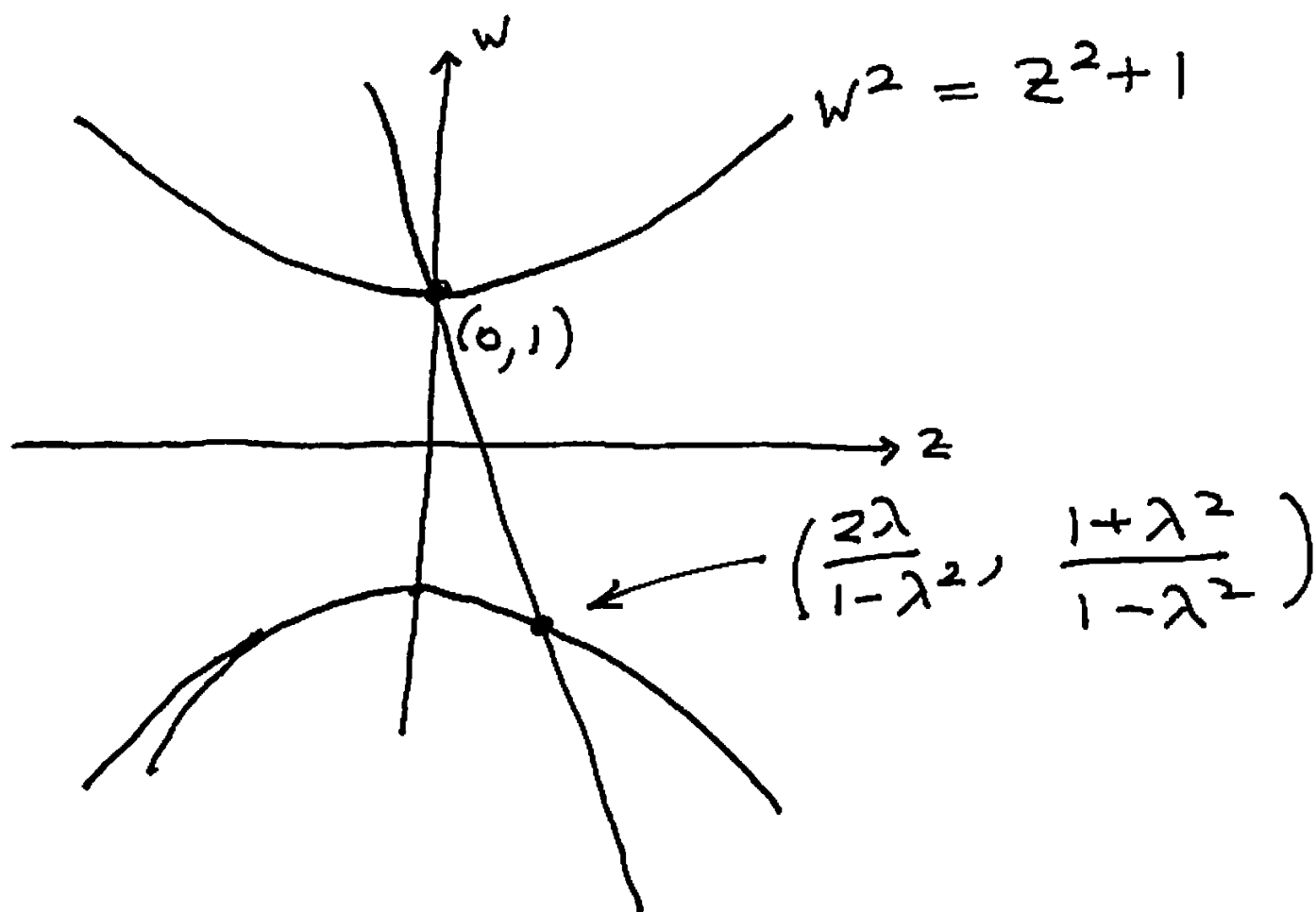
or in other words

$$(\lambda^2 - 1)z^2 + 2\lambda z = 0.$$

This is of course a quadratic polynomial, with two roots in general; one root ($z = 0$) coming from the point $(0, 1)$ and the other, at

$$z = \frac{2\lambda}{1 - \lambda^2}$$

coming from the remaining point of intersection of $L_\lambda \cap \Gamma$. This yields a biholomorphic map between Γ and the complement $S \setminus \{\pm 1\}$ of the two points ± 1 in the Riemann sphere S , and transforms the differential $\frac{dz}{w}$ into a rational differential on S , which we can always evaluate by the method of partial fractions.



line w /slope λ
thru $(0, 1)$

The point is, this process works on any path integral on a subset $\Gamma \subset \mathbb{C} \times \mathbb{C}$ defined by a quadratic polynomial $Q(z, w)$; and in particular *it works to evaluate the integral of any algebraic function involving a square root of a quadratic polynomial*. You don't have to memorize a long list of trig substitutions (or carry around that massive calculus textbook); this is truly a one-size-fits-all approach to integrating algebraic functions.

17.2 A calculus problem we can't

The methods above have their limitations, as 19th century mathematicians soon discovered. The occasion of their discovery was a simple problem: to calculate the arc length of a portion of the ellipse $ax^2 + by^2 = 1$ between two points (x_0, y_0) and (x_1, y_1) . Applying the standard integral formula for arc length (and massaging the results), they arrived at an integral of the form

$$\int_{x_0}^{x_1} \frac{dx}{\sqrt{1+x^4}}.$$

And there they stayed. Despite decades of attempts, they were unable to evaluate such integrals explicitly in terms of known functions. It wasn't until the second half of the 19th century that mathematicians, led by Riemann and others, began to see the reason for their failure; and that discovery ultimately transformed the subjects of complex analysis, topology and algebraic geometry.

The key to everything was to look at the graph of the function $\sqrt{1+x^4}$ in the complex plane; that is, the locus

$$\Phi = \{(z, w) \in \mathbb{C} \times \mathbb{C} : w^2 = z^4 + 1\}.$$

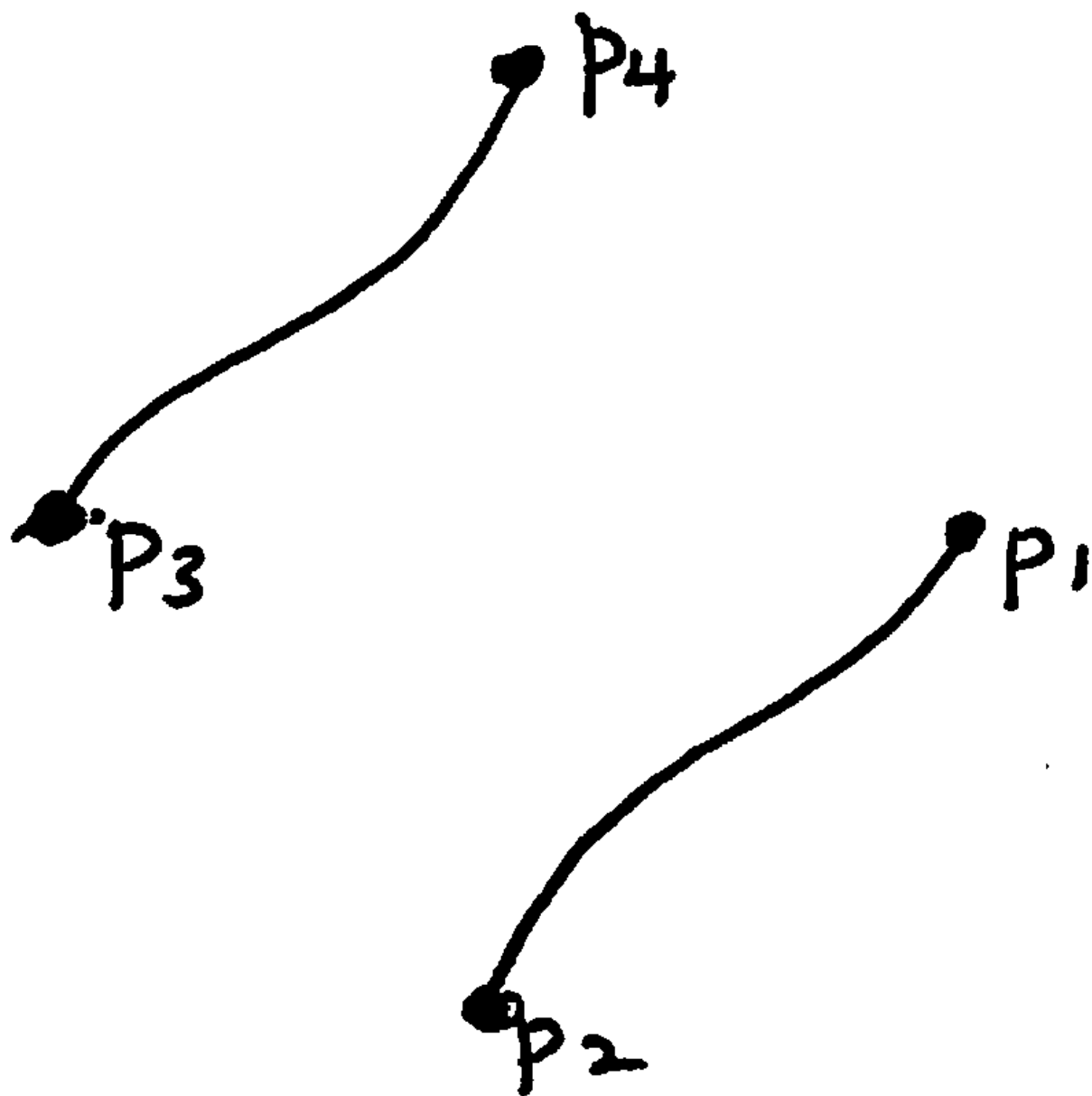
This is going to be hard to visualize, but we've been building up to this all semester, so give it a shot. I claim that Φ *is not an open subset of a sphere, as Γ was, but rather an open subset of a torus*, and this makes all the difference.

How do we see this? The key is to look at the projection map $\pi_1 : \Phi \rightarrow \mathbb{C}$ sending (z, w) to z . This map is two-to-one everywhere except over the points $z = \frac{1}{\sqrt{2}}(\pm 1 \pm i)$ where $z^4 = -1$. (For convenience we'll call these points $p_1, \dots, p_4$, and the points in Φ lying over them $q_1, \dots, q_4$). In fact,

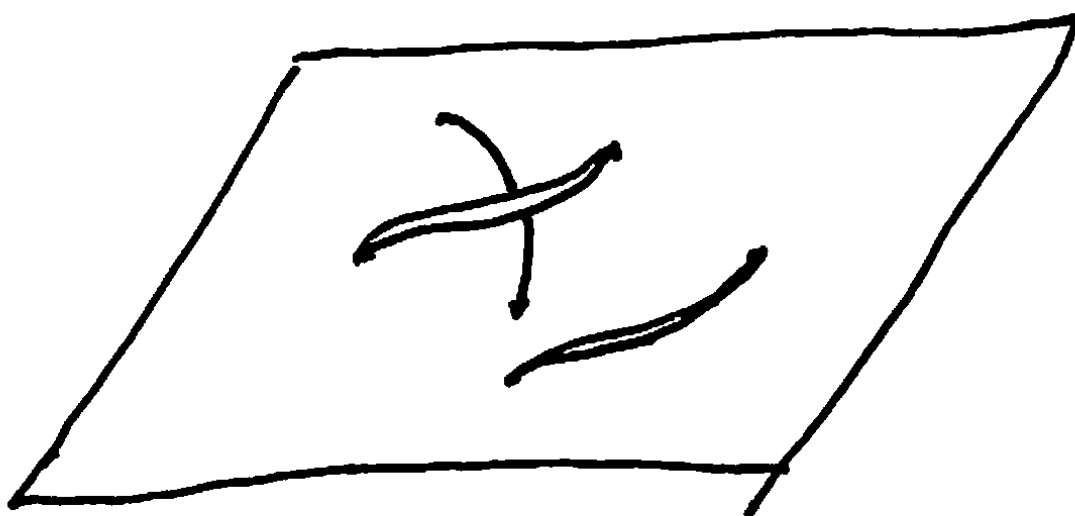
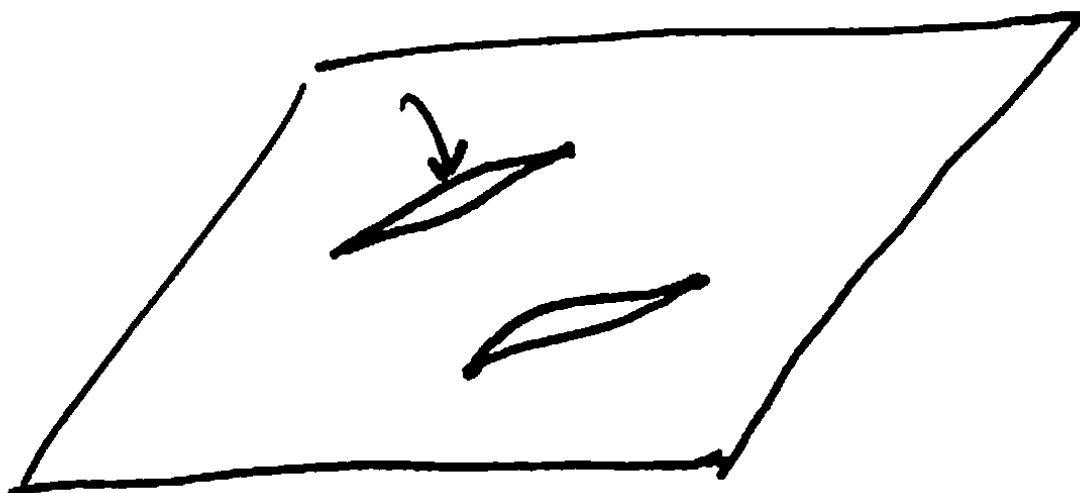
away from those four points, *the map $\Phi \setminus \{q_i\} \rightarrow \mathbb{C} \setminus \{p_i\}$ is a two-sheeted covering space*; and the points p_i are called *branch points*, meaning if you draw a small circle around any of the points p_i and lift it as a path to Φ , you wind up on the opposite sheet. More generally, if γ is a closed curve in $\mathbb{C} \setminus \{p_i\}$, and you lift γ to a path in $\Phi \setminus \{q_i\}$, you wind up on the same sheet (i.e., the lift is again a closed loop) if and only if the sum

$$\sum_i n(\gamma, p_i) \equiv 0 \pmod{2} \quad (2)$$

of the winding numbers of γ around the points p_i is even.

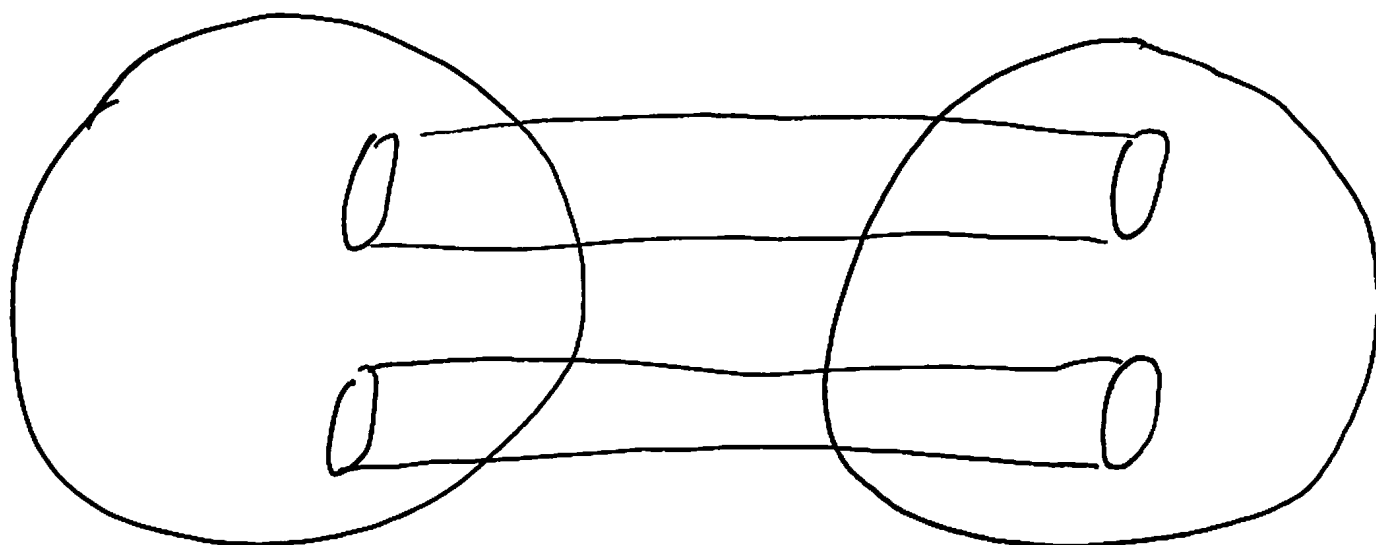


How do we visualize this? Here's one way: suppose we draw an arc γ joining the points p_1 and p_2 , and another arc δ joining p_3 and p_4 . Let $\mathbb{C}' = \mathbb{C} \setminus (\gamma \cup \delta) \subset \mathbb{C}$, and let $\Phi' = \pi_1^{-1}(\mathbb{C}') \subset \Phi$ be the preimage of $\mathbb{C}'$. The point is, according to the rule about getting from one sheet of the cover Φ to the other, Φ' is disconnected: it consists of two connected components Φ_1 and Φ_2 , each homeomorphic to $\mathbb{C}'$.



Now, what happens when we add the arcs γ and δ and their preimages in Φ back in? Well, basically we're identifying the component of the boundary S^1 of Φ_1 lying over γ with the corresponding component of the boundary of Φ_2 , and likewise for the components of the boundaries of Φ_1 and Φ_2 lying over δ . The topological space Φ can thus be constructed as follows: take two copies of the plane $\mathbb{C}$; cut two holes in each, and identify the boundaries of the holes on one copy with the boundaries of the holes in the other.

If we want to compactify, we can compactify the original plane $\mathbb{C}$ to the Riemann sphere S by adding a point at ∞ , and we can likewise compactify Φ by adding one point at ∞ to each copy of the plane $\mathbb{C}$; the resulting space $\bar{\Phi}$ can be described topologically as: take two spheres S^2 ; cut two holes in each and identify the boundaries of the holes in one with the boundaries of the holes in the other, and voilà: a torus!



This has many ramifications. In algebraic geometry, it means that unlike the case of the curve

$$\Gamma = \{(z, w) \in \mathbb{C} \times \mathbb{C} : w^2 = z^2 + 1\},$$

where as we saw we could give an isomorphism with a subset of the Riemann sphere (a birational isomorphism with $\mathbb{P}^1$, in the language of algebraic geometry), no such birational isomorphism can exist between Φ and $\mathbb{P}^1$. In the language developed as a result of this observation, we say that the curve given by $w^2 = z^2 + 1$ is *rational*, while the curve $w^2 = z^4 + 1$ has genus 1; and with that the study of algebraic curves got underway.

(It may seem jarring to call a locus like Φ a curve, when in fact it's obviously a surface! That's the algebraic geometer in me: what else would you call the zero locus of a single polynomial in two variables? By way of warning: we also call $\mathbb{C}$ the *complex line*; to us, the "complex plane" means $\mathbb{C} \times \mathbb{C}$)

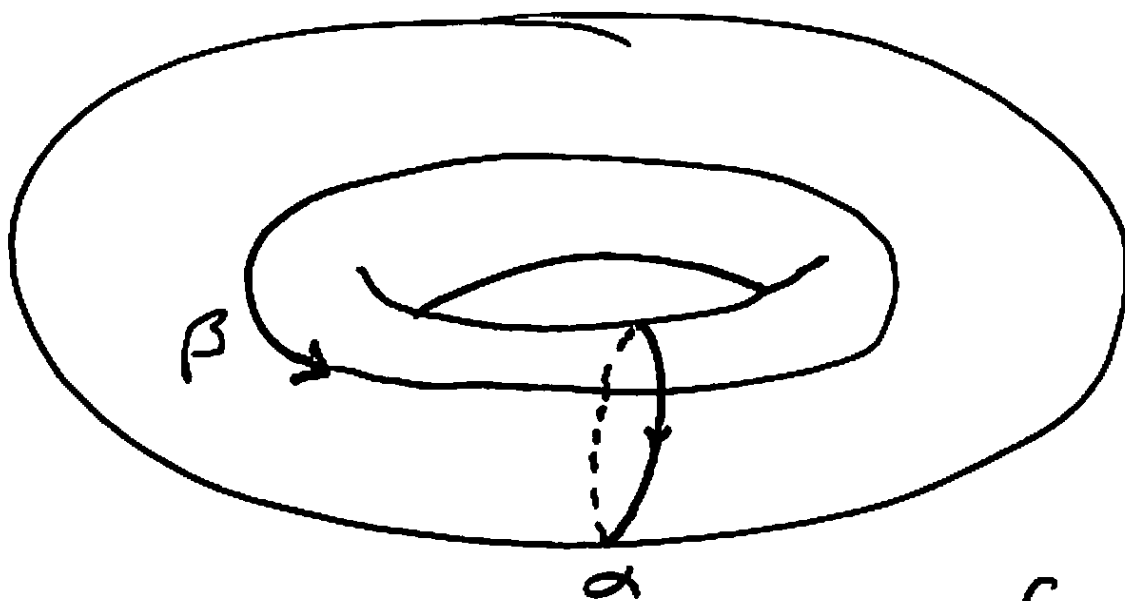
It also has ramifications for complex analysis. The point is, if we view an integral like

$$\int_{x_0}^x \frac{dx}{\sqrt{1+x^4}}$$

as a path integral on the Riemann surface $\overline{\Phi}$, then what we see is that, since a torus is not simply connected, *the integral depends on the path of integration*, even though the integrand is a holomorphic 1-form. Specifically, if we choose two loops α, β in Φ generating $\pi_1(\overline{\Phi})$, then the integral is defined only up to integral linear combinations of the *periods*

$$\omega_1 = \int_{\alpha} \frac{dz}{w} \quad \text{and} \quad \omega_2 = \int_{\beta} \frac{dz}{w}$$

What this means is that the inverse function of the integral is a doubly periodic function on the complex plane!



$$\omega_1 = \int_{\alpha} \gamma$$

$$\omega_2 = \int_{\beta} \gamma$$

That, finally, is when 19th century mathematicians saw that an integral of the form

$$F(x) = \int_{x_0}^x \frac{dx}{\sqrt{1-x^4}}$$

could not be expressed in terms of elementary functions. This led, of course, to a search for doubly periodic functions; Weierstrass, for example, introduced the Weierstrass $\mathcal{P}$ -function, and indeed we can express the integral $F(x)$ in terms of the inverse of $\mathcal{P}$.

(I can't help observing that this is a fascinating example of how mathematical objects get named. Today, integrals like $\int \frac{dx}{\sqrt{1+x^4}}$ are called *elliptic integrals*; curves like $w^2 = z^4 + 1$ are called *elliptic curves*, and there are generalizations to what are called *hyperelliptic curves*—all because the original integral that started it all arose in connection with arc length on an ellipse!)

Well, that's as far as I'm going to go with this. If you're interested to learn more about these questions and their generalizations, there are courses out there in complex analysis, algebraic geometry and topology that will hopefully satisfy your curiosity. I just wanted to show you how closely intertwined these subjects are, and why the choice of subject matter for 55b—a combination of topology and complex analysis—is not as odd a pairing as it might seem.